

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
			2012-05-09

SCHEM, MLB, KEPLER, 2PHASE, D2
FSB, 5/9/2012

D

Page	Contents	Sync	Date
1	Table of Contents	D2_KEPLER	01/13/2012
2	System Block Diagram	D2_KEPLER	01/13/2012
3	Power Block Diagram	D2_KEPLER	01/13/2012
4	Revision History	D2_KEPLER	01/13/2012
5	BOM Configuration	D2_KEPLER	01/13/2012
6	BOM Variants	D2_KEPLER	01/13/2012
7	Functional / ICT Test	D2_KEPLER	01/13/2012
8	Power Aliases	D2_KEPLER	01/13/2012
9	Signal Aliases	D2_KEPLER	01/13/2012
10	CPU DMI/PEG/FDI/RSVD	D2_KEPLER	01/13/2012
11	CPU CLOCK/MISC/JTAG	D2_KEPLER	01/13/2012
12	CPU DDR3 INTERFACES	D2_KEPLER	01/13/2012
13	CPU POWER	D2_KEPLER	01/13/2012
14	CPU POWER AND GND	D2_KEPLER	01/13/2012
15	CPU DECOUPLING-I	D2_SEAN	03/05/2012
16	CPU DECOUPLING-II	D2_SEAN	03/06/2012
17	PCH SATA/PCIe/CLK/LPC/SPI	D2_KEPLER	01/13/2012
18	PCH DMI/FDI/PM/Graphics	D2_KEPLER	01/13/2012
19	PCH PCI/USB/TP/RSVD	D2_KEPLER	01/13/2012
20	PCH GPIO/MISC/NCTF	D2_KEPLER	01/13/2012
21	PCH POWER	D2_CLEAN	03/19/2012
22	PCH GROUNDS	D2_KEPLER	01/13/2012
23	PCH DECOUPLING	D2_CLEAN	03/19/2012
24	CPU & PCH XDP	D2_KEPLER	01/13/2012
25	Chipset Support	D2_KEPLER	01/13/2012
26	USB HUB & MUX	D2_KEPLER	01/13/2012
27	CPU Memory S3 Support	D2_KEPLER	01/13/2012
28	DDR3 SDRAM Bank A (1 OF 2)	D2_KEPLER	01/13/2012
29	DDR3 SDRAM Bank A (2 OF 2)	D2_KEPLER	01/13/2012
30	DDR3 SDRAM Bank B (1 OF 2)	D2_KEPLER	01/13/2012
31	DDR3 SDRAM Bank B (2 OF 2)	D2_KEPLER	01/13/2012
32	DDR3 Termination	D2_KEPLER	01/13/2012
33	DDR3/FRAMEBUF VREF MARGINING	D2_KEPLER	01/13/2012
34	X29/ALS/CAMERA CONNECTOR	D2_KEPLER	01/13/2012
35	Thunderbolt Host (1 of 2)	D2_KEPLER	01/13/2012
36	Thunderbolt Host (2 of 2)	D2_KEPLER	01/13/2012
37	Thunderbolt Power Support	D2_KEPLER	01/13/2012
38	RIO CONNECTOR	D2_KEPLER	01/13/2012
39	SSD CONNECTOR	D2_KEPLER	01/13/2012
40	USB 3.0 CONNECTORS	D2_KEPLER	01/13/2012
41	SMC	D2_KEPLER	01/13/2012
42	SMC Support	D2_KEPLER	01/13/2012
43	LPC+SPI Debug Connector	D2_KEPLER	01/13/2012
44	SMBUS Connections	D2_KEPLER	01/13/2012
45	Voltage & Load Side Current Sensing	D2_SEAN	03/06/2012

Page	Contents	Sync	Date
46	High Side and CPU/AXG Current Sensing	D2_SEAN	03/05/2012
47	Thermal Sensors	D2_SEAN	03/05/2012
48	Fan Connectors	D2_KEPLER	01/13/2012
49	KEYBOARD/TRACKPAD (1 OF 2)	D2_KEPLER	01/13/2012
50	KEYBOARD/TRACKPAD (2 OF 2)	D2_KEPLER	01/13/2012
51	DIGITAL ACCELEROMETER & GYRO	D2_KEPLER	01/13/2012
52	SPI ROM	D2_KEPLER	01/13/2012
53	AUDIO: CODEC/REGULATOR	D2_CARA	03/16/2012
54	AUDIO: HEADPHONE FILTER	D2_CARA	03/16/2012
55	AUDIO: IV SENSE	D2_CARA	03/16/2012
56	AUDIO: IV SENSE FILTER	D2_CARA	03/16/2012
57	AUDIO: SPEAKER AMP	D2_CARA	03/16/2012
58	AUDIO: JACK	D2_CARA	03/16/2012
59	AUDIO: JACK TRANSLATORS	D2_CARA	03/16/2012
60	DC-In & Battery Connectors	D2_KEPLER	01/13/2012
61	PBus Supply & Battery Charger	D2_KEPLER	01/13/2012
62	System Agent Supply	D2_KEPLER	01/13/2012
63	5V / 3.3V Power Supply	D2_KEPLER	01/13/2012
64	1V5R1V35V DDR3 SUPPLY	D2_KEPLER	01/13/2012
65	CPU IMVP7 & AXG VCore Regulator	D2_SEAN	03/05/2012
66	CPU IMVP7 & AXG VCore Output	D2_SEAN	03/05/2012
67	CPU VCCIO (1V0R1V05 S0) POWER SUPPLY	D2_KEPLER	01/13/2012
68	Misc Power Supplies	D2_KEPLER	01/13/2012
69	Power FETs	D2_KEPLER	01/13/2012
70	Power Control 1/ENABLE	D2_KEPLER	01/13/2012
71	KEPLER PCI-E	D2_KEPLER	01/13/2012
72	KEPLER CORE/FB POWER	D2_SEAN	03/05/2012
73	KEPLER FRAME BUFFER I/F	D2_SEAN	03/05/2012
74	1V05 GPU / 1V35 FB POWER SUPPLY	D2_SEAN	03/05/2012
75	GDDR5 Frame Buffer A	D2_SEAN	03/05/2012
76	GDDR5 Frame Buffer B	D2_SEAN	03/05/2012
77	KEPLER EDP/DP/GPIO	D2_SEAN	03/05/2012
78	KEPLER GPIOs,CLK & STRAPS	D2_SEAN	03/05/2012
79	KEPLER PEX PWR/GNDS	D2_SEAN	03/05/2012
80	GFX IMVP VCore Regulator	D2_SEAN	03/05/2012
81	eDP Display Connector	D2_KEPLER	01/13/2012
82	eDP Mux	D2_SEAN	03/05/2012
83	eDP Muxed Graphics Support	D2_SEAN	03/05/2012
84	Thunderbolt Connector A	D2_KEPLER	01/13/2012
85	Thunderbolt Connector B	D2_KEPLER	01/13/2012
86	LCD Backlight Driver (LP8545)	D2_KEPLER	01/13/2012
87	PCH VCCIO (1.05V) POWER SUPPLY	D2_KEPLER	01/13/2012
88	Power Sequencing EG/PCH S0	D2_KEPLER	01/13/2012
89	CPU Constraints	D2_KEPLER	01/13/2012
90	Memory Constraints	D2_KEPLER	01/13/2012

Page	Contents	Sync	Date
91	PCH Constraints 1	D2_KEPLER	01/13/2012
92	PCH Constraints 2	D2_KEPLER	01/13/2012
93	Thunderbolt Constraints	D2_KEPLER	01/13/2012
94	SMC Constraints	D2_KEPLER	01/13/2012
95	GPU (Kepler) CONSTRAINTS	D2_KEPLER	01/13/2012
96	Project Specific Constraints	D2_CLEAN	03/15/2012
97	PCB Rule Definitions	D2_KEPLER	01/13/2012
98	DEBUG SENSORS AND ADC	D2_SEAN	03/05/2012
99	SMC12 SENSORS EXTENDED	D2_KEPLER	01/13/2012

CCBB

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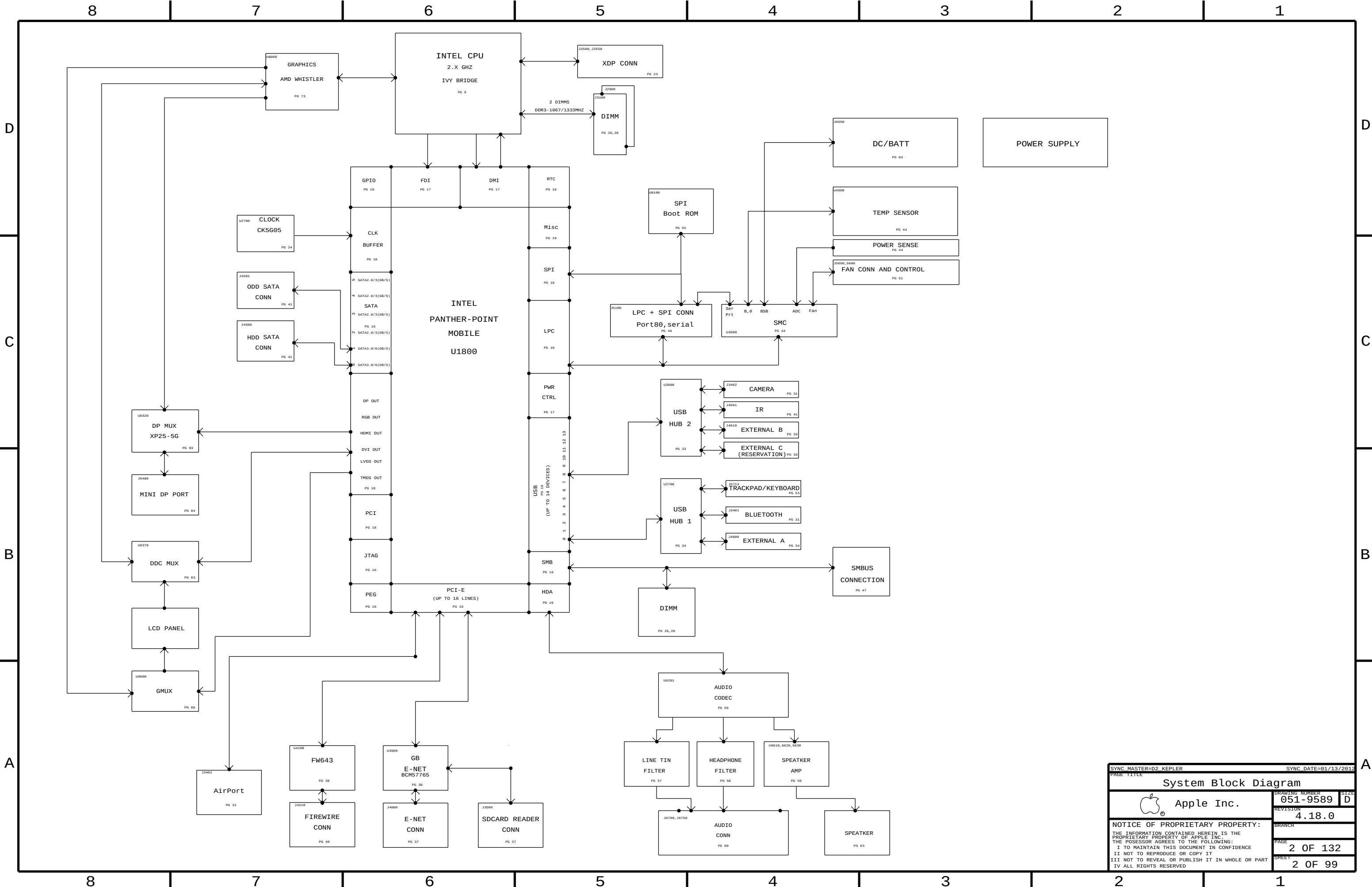
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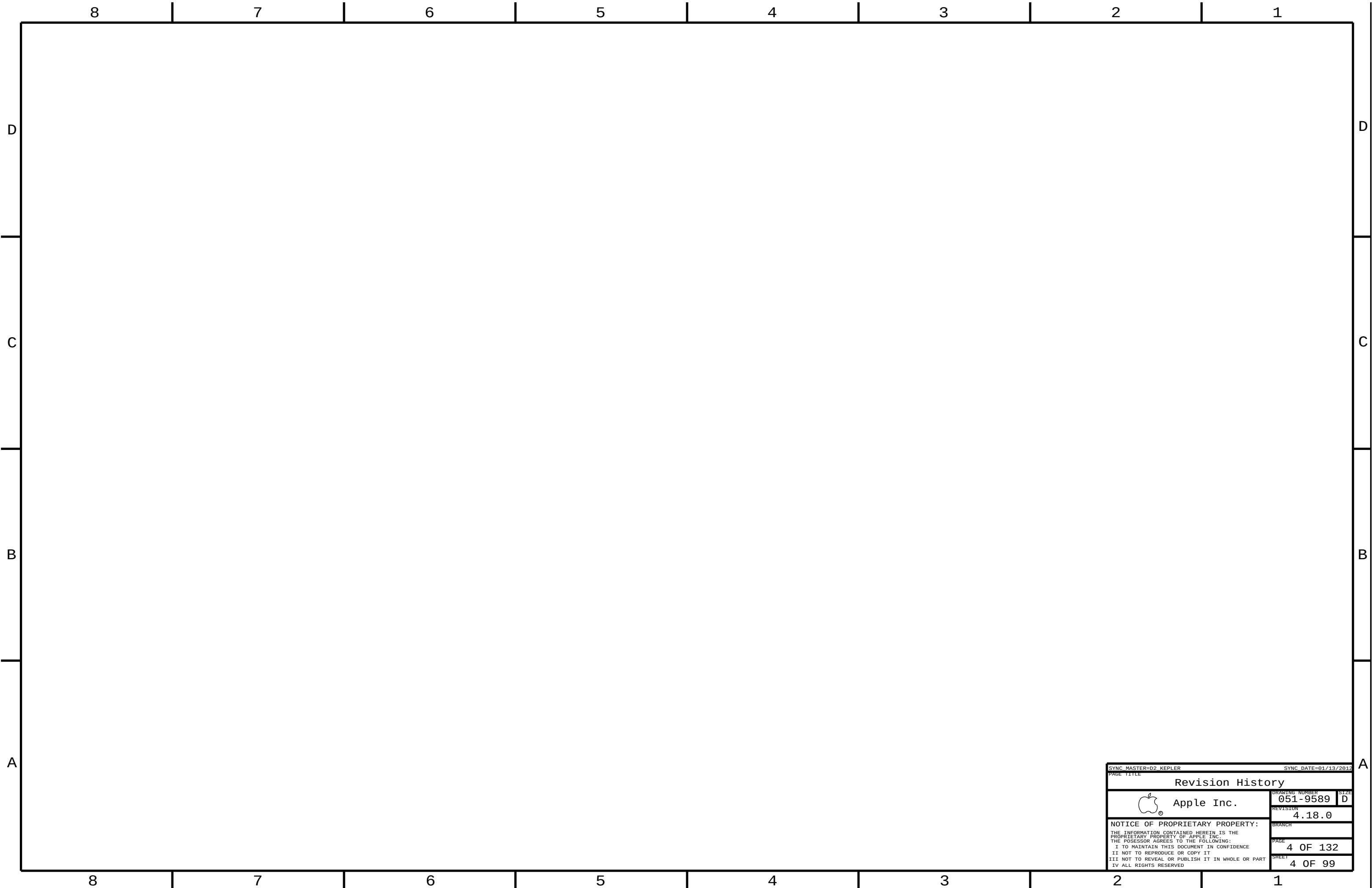
Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-9589	1	SCHEM,MLB,KEPLER_2PHASE,D2	SCH	CRITICAL	
820-3332	1	PCBF,MLB,KEPLER_2PHASE,D2	PCB	CRITICAL	

DRAWING
TITLE=MLB
ABBREV=ABBREV
LAST_MODIFIED=Wed May 9 13:50:52 2012

DRAWING TITLE	
SCHEM, MLB, KEPLER, 2PHASE, D2	
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	051-9589
	SIZE
	D
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	4.18.0
	BRANCH
	PAGE
	1 OF 132
	SHEET
	1 OF 99

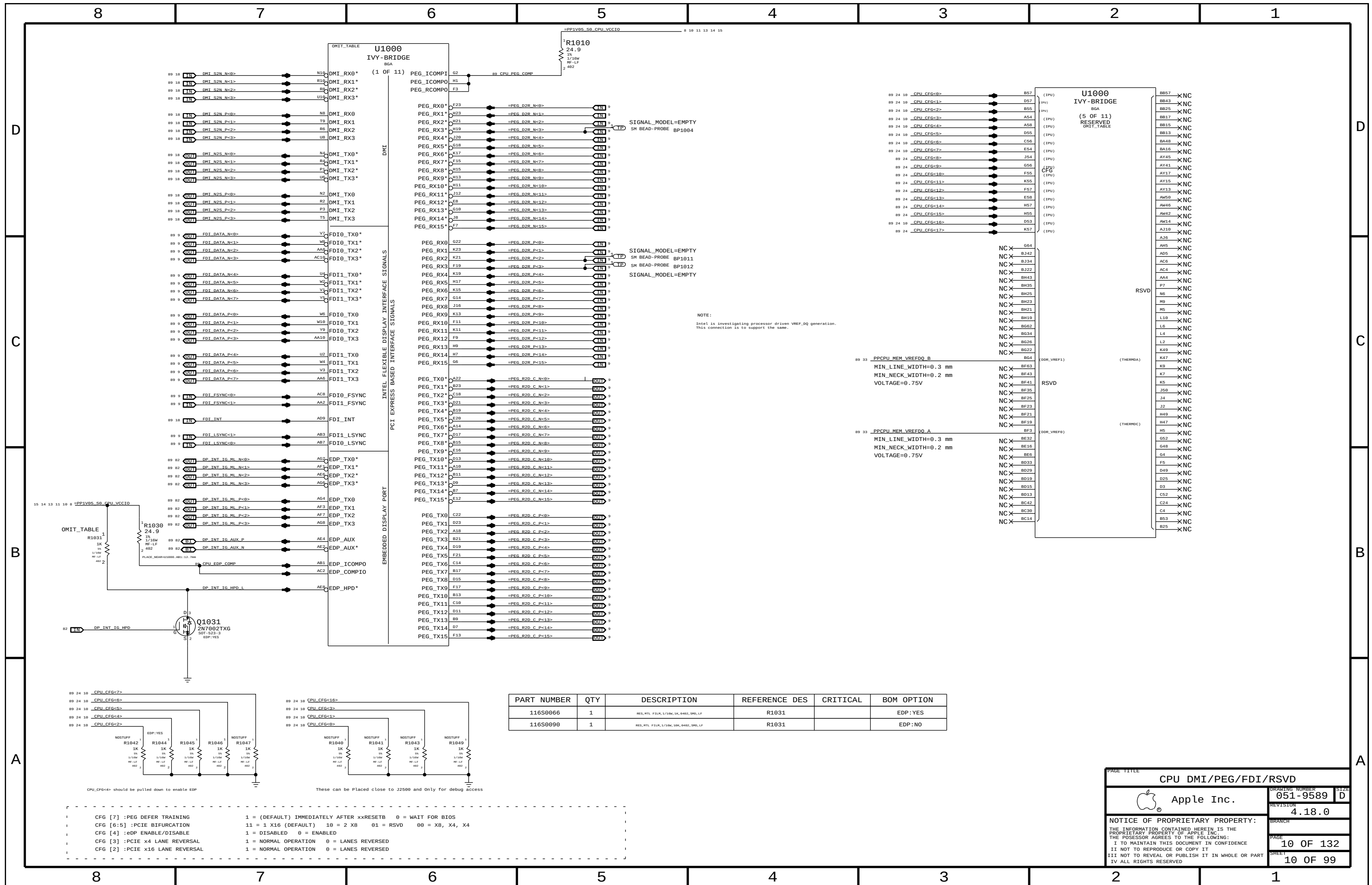


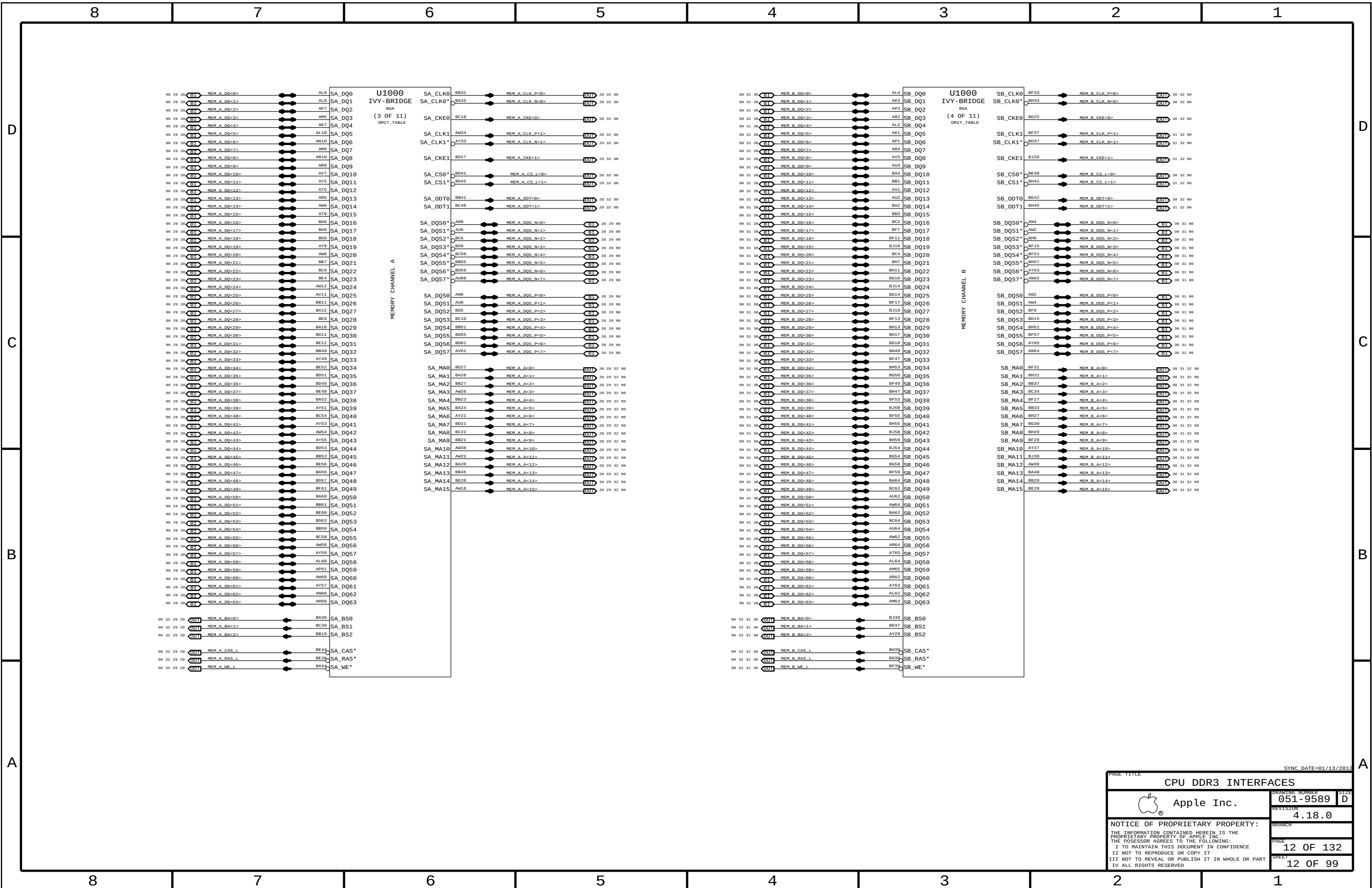


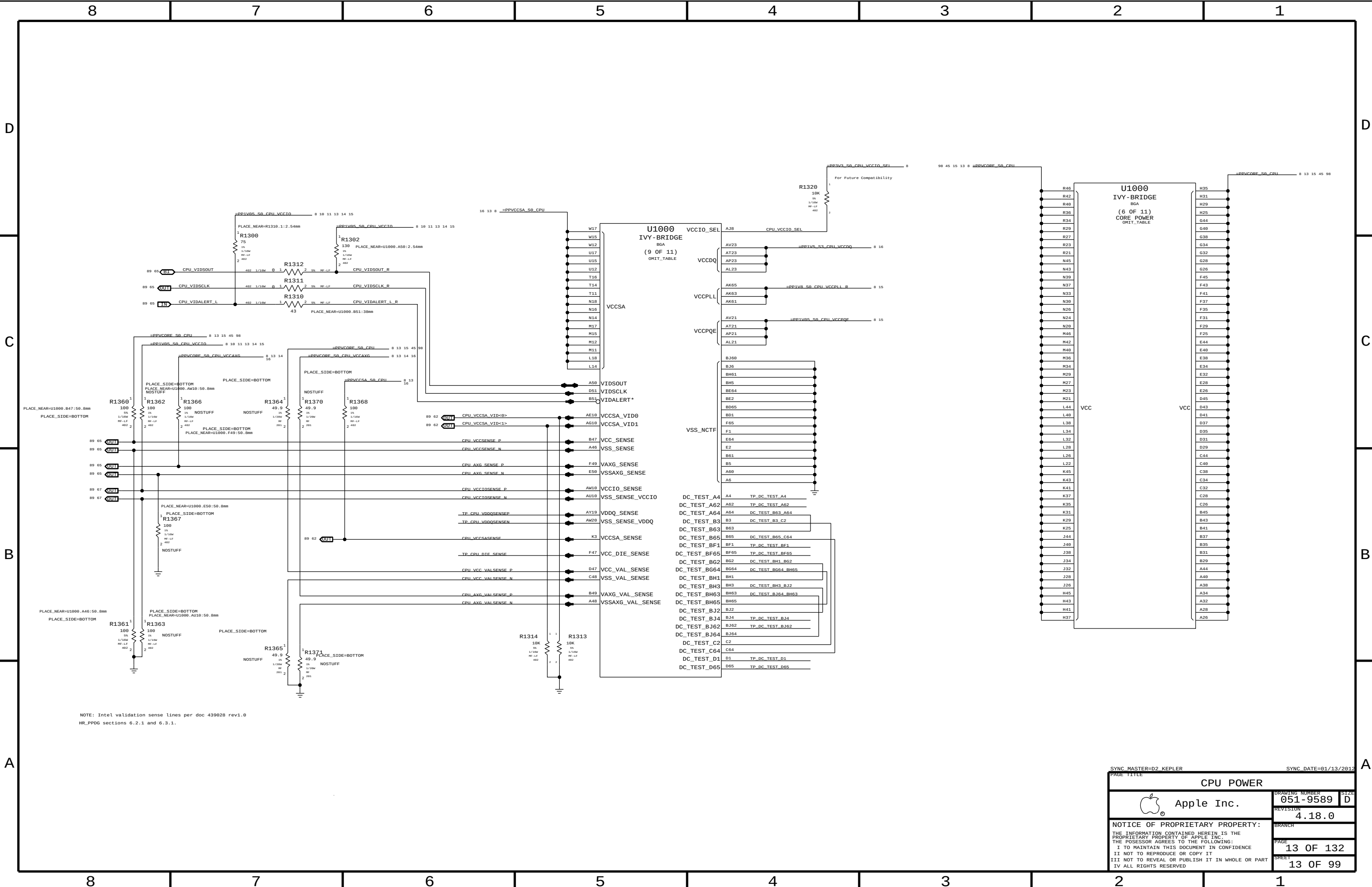
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SYNC DATE=01/13/2012

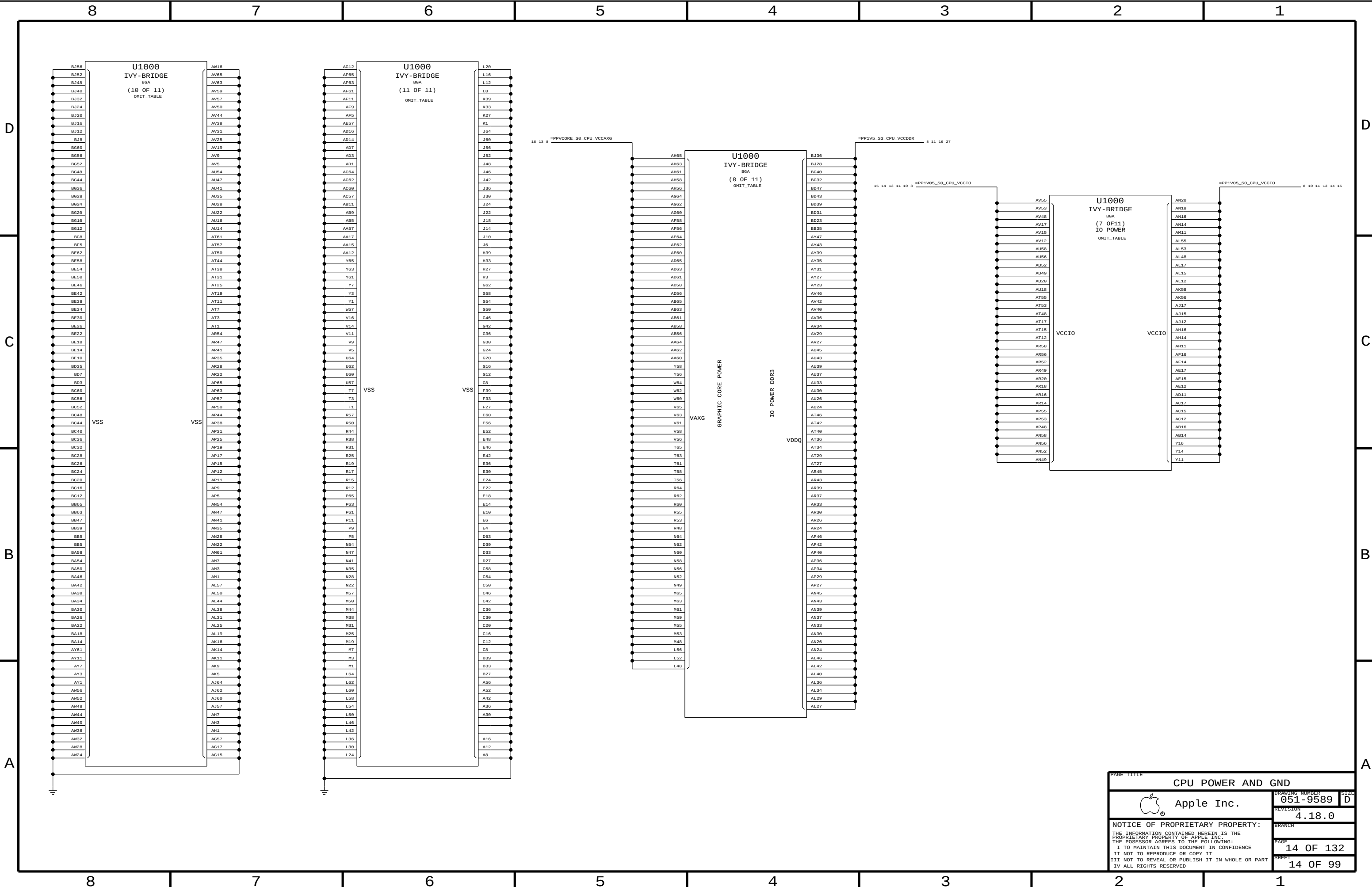
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	PAGE 4 OF 132
SHEET 4 OF 99	





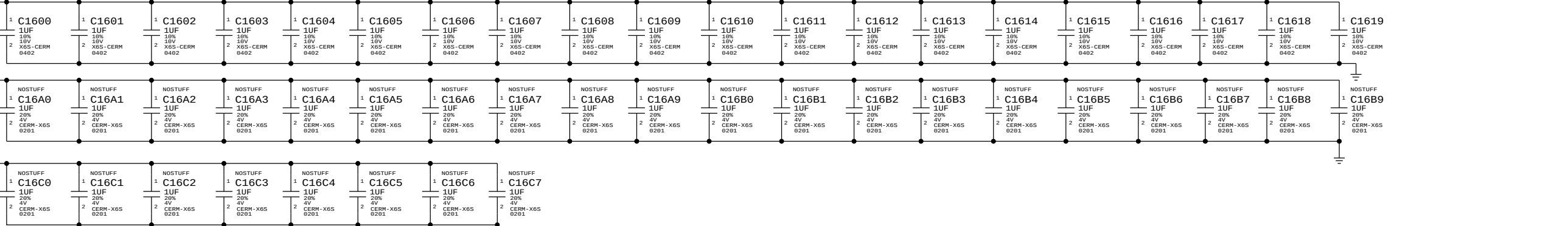


NOTE: Intel validation sense lines per doc 439928 rev1.0
HR_PPDG sections 6.2.1 and 6.3.1.

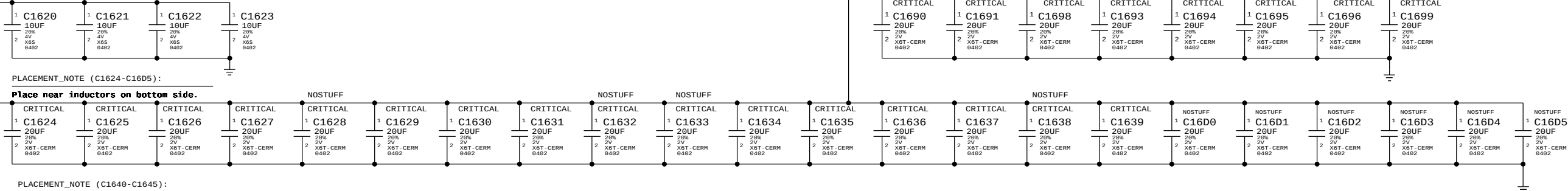


Apple Implementation: 8x 270uF 6m0hm, 0x 470uF 4m0hm, 16x 22uF 0402, 4x 10uF 0402, 20x 1uF 0402, 28x 1uF 0201 (NOSTUFF), 4x 22uF 0402 (NOSTUFF)

Place on bottom side of U1000



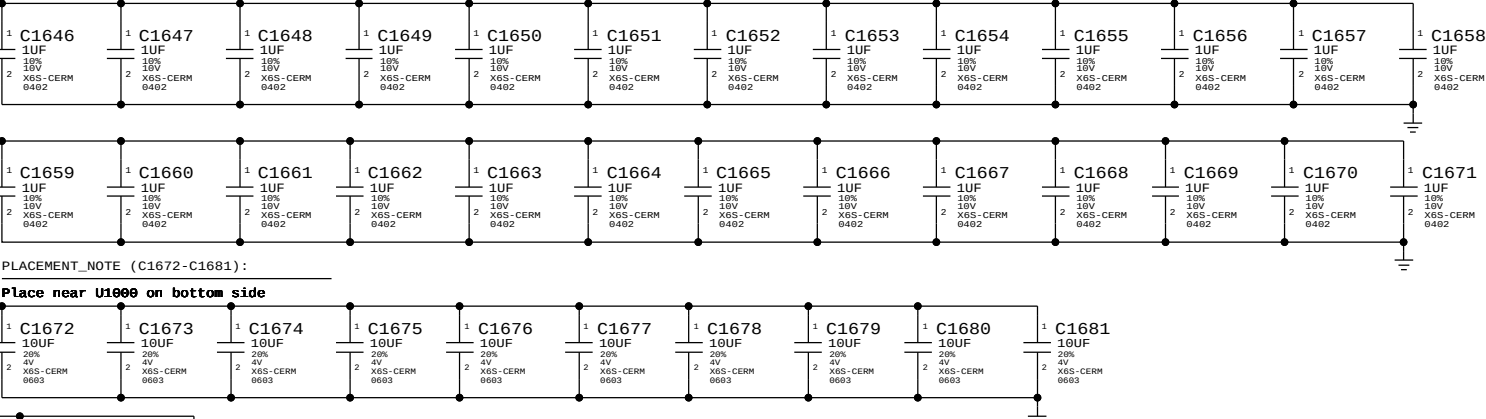
Place near U1000 on bottom side



1 CRITICAL C1640 270UF 20% 2V TANT CASE-B2-SM 1 CRITICAL C1641 270UF 20% 2V TANT CASE-B2-SM 1 CRITICAL C1642 270UF 20% 2V TANT CASE-B2-SM 1 CRITICAL C1643 270UF 20% 2V TANT CASE-B2-SM 1 CRITICAL C1644 270UF 20% 2V TANT CASE-B2-SM 1 CRITICAL C1645 270UF 20% 2V TANT CASE-B2-SM 1 CRITICAL C1688 270UF 20% 2V TANT CASE-B2-SM 1 CRITICAL C1689 270UF 20% 2V TANT CASE-B2-SM

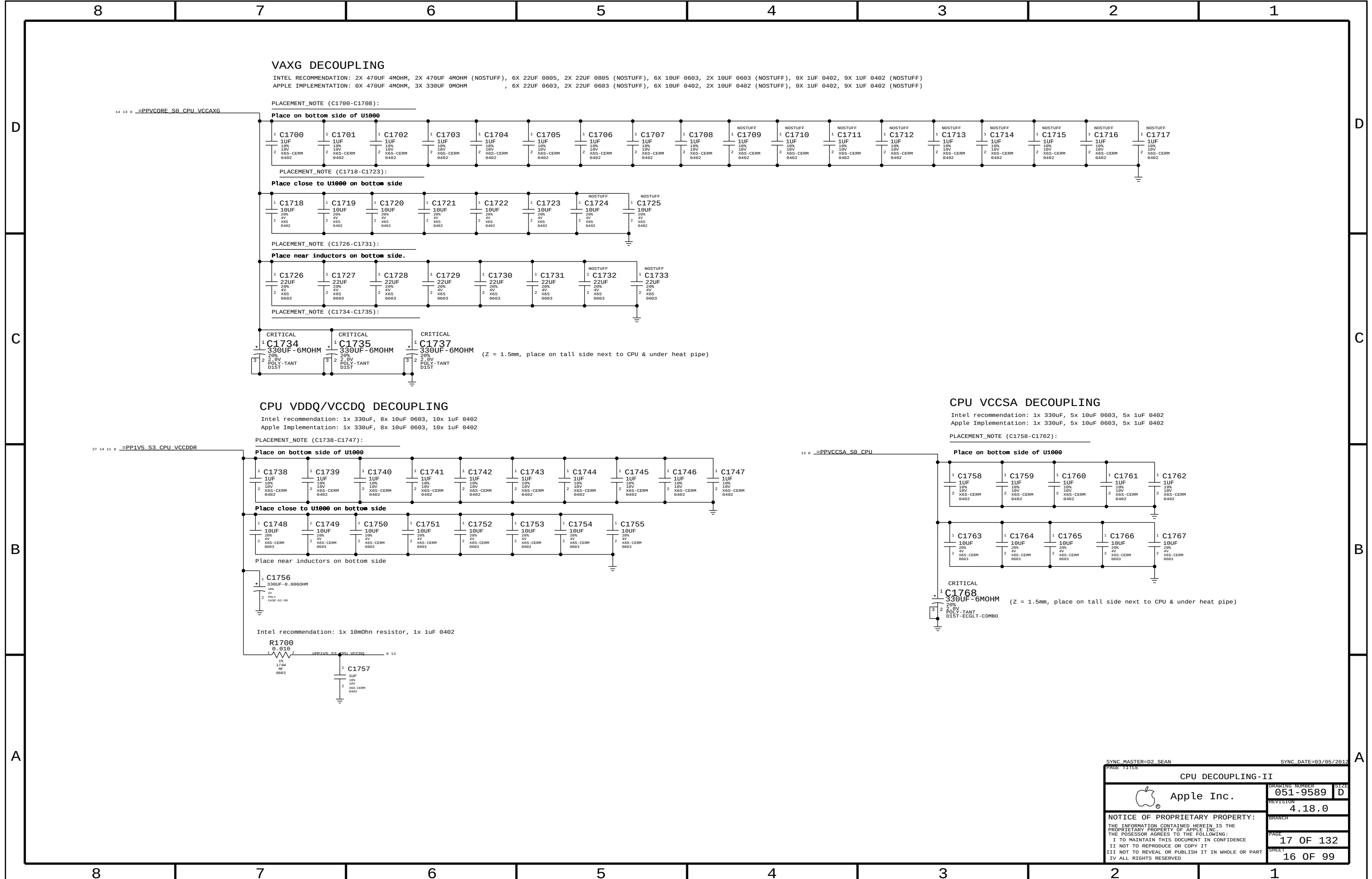
Apple Implementation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402

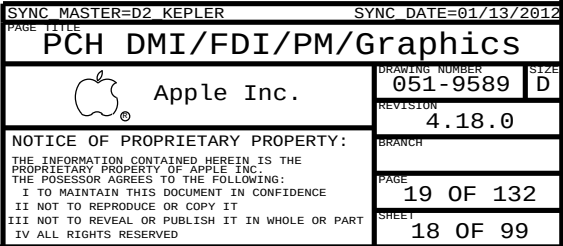
Place on bottom side of U1000

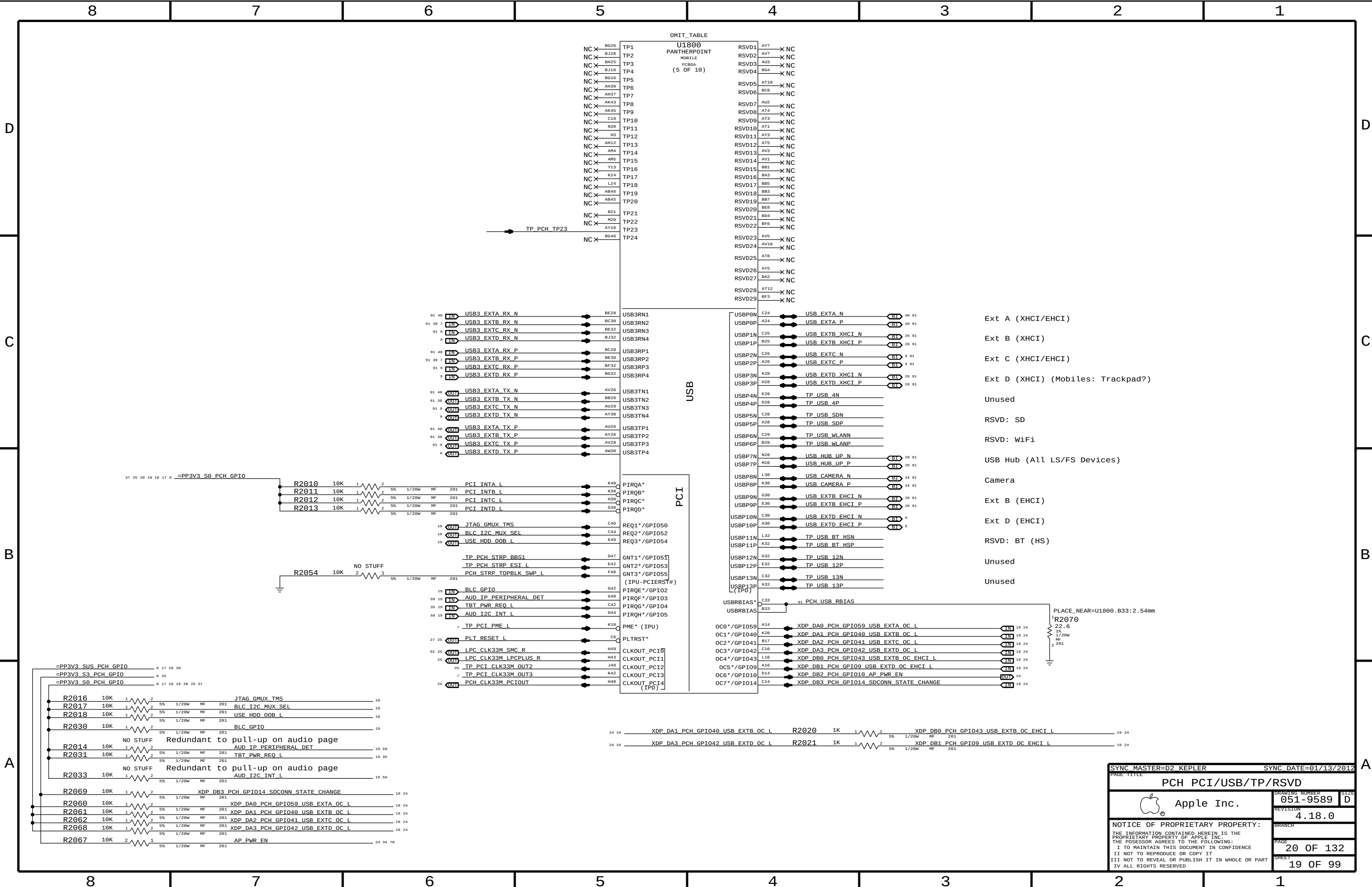


3 2 2.0V
POLY-TANT
D15T-ECGLT-COMBO

[illegible]







SYNC MASTER=D2 KEPLER

SYNC DATE=01/13/2012

PAGE TITLE

PCH PCI/USB/TP/RSVD

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DRAWING NUMBER
051-9589

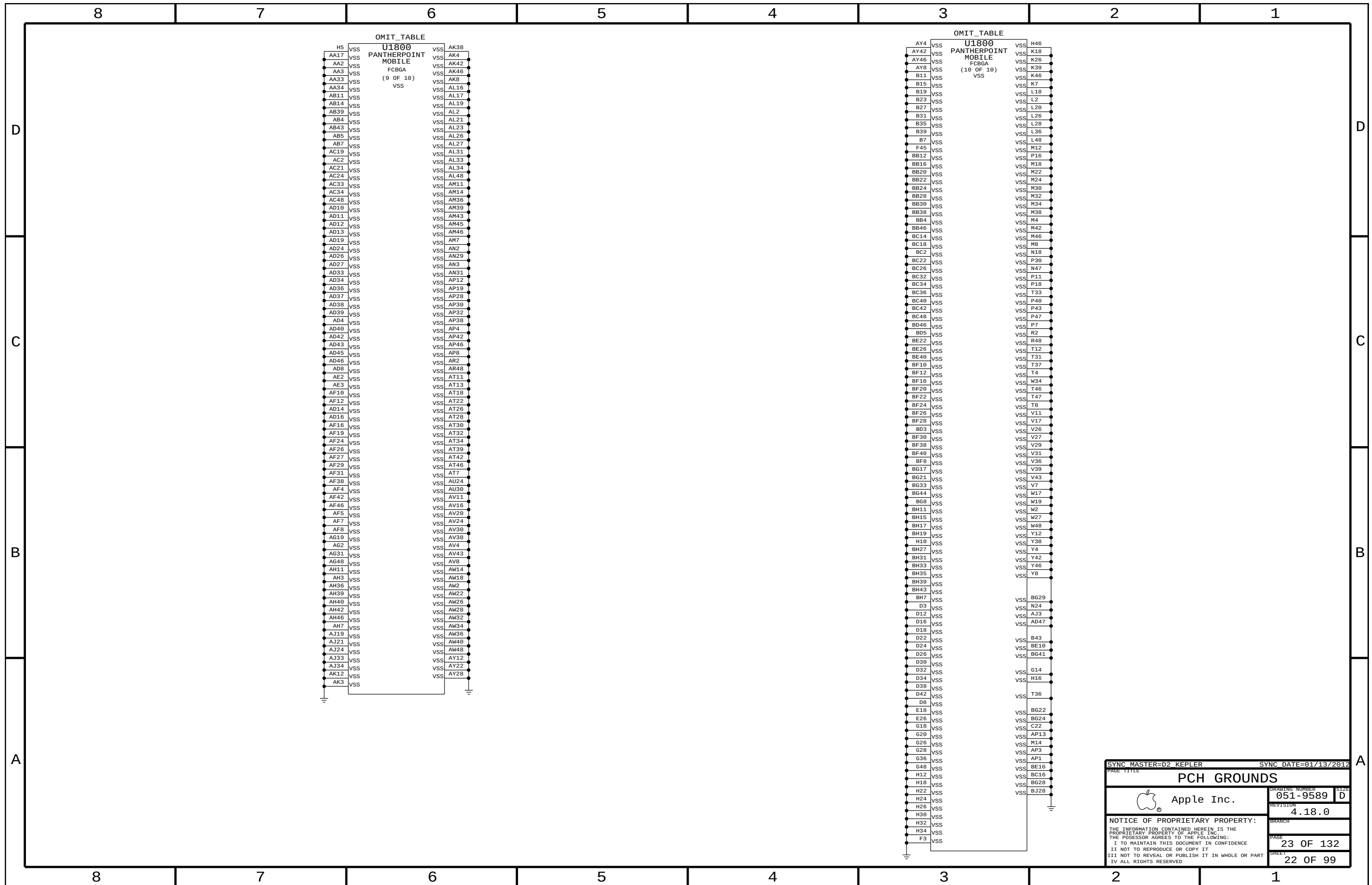
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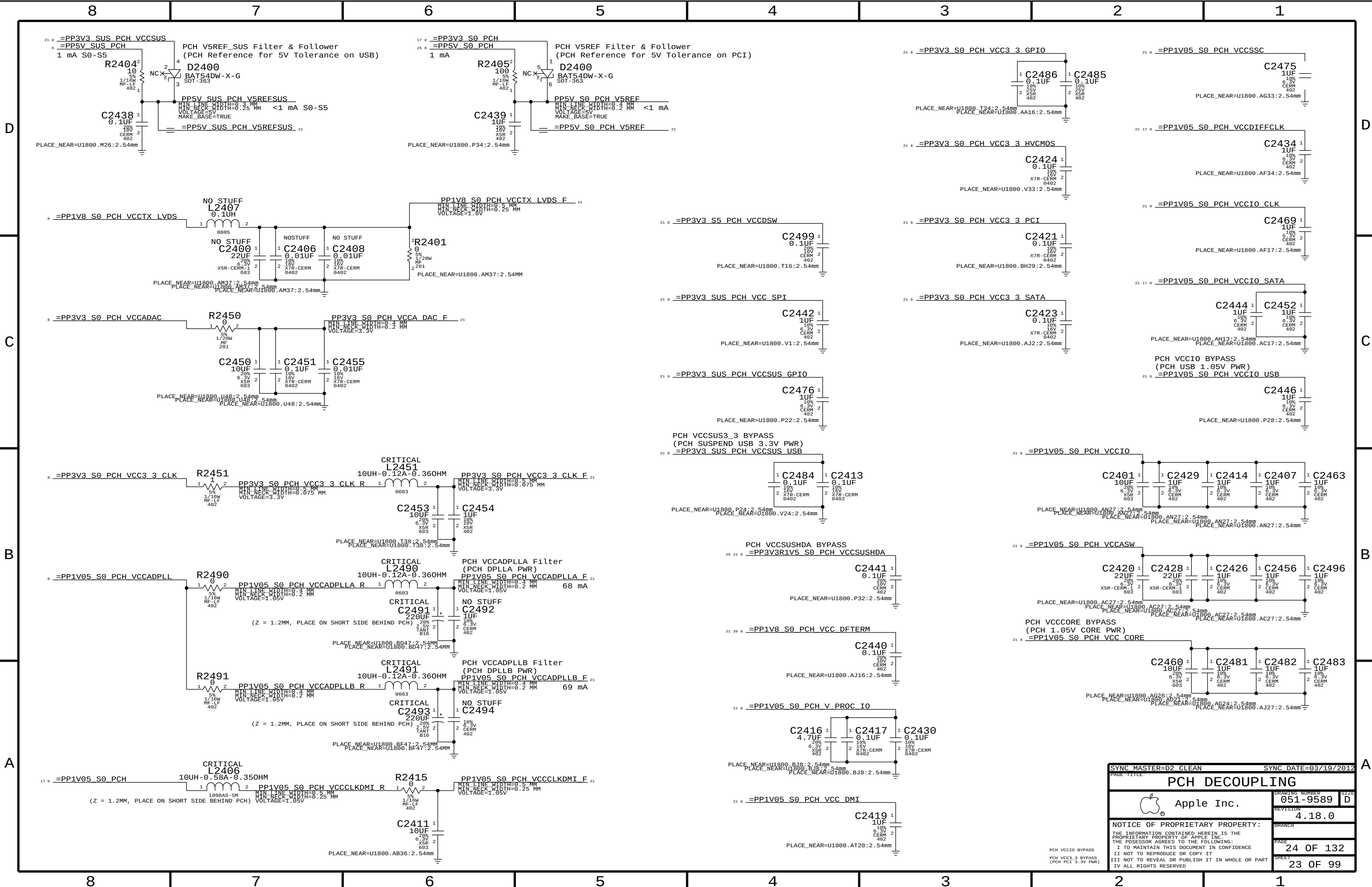
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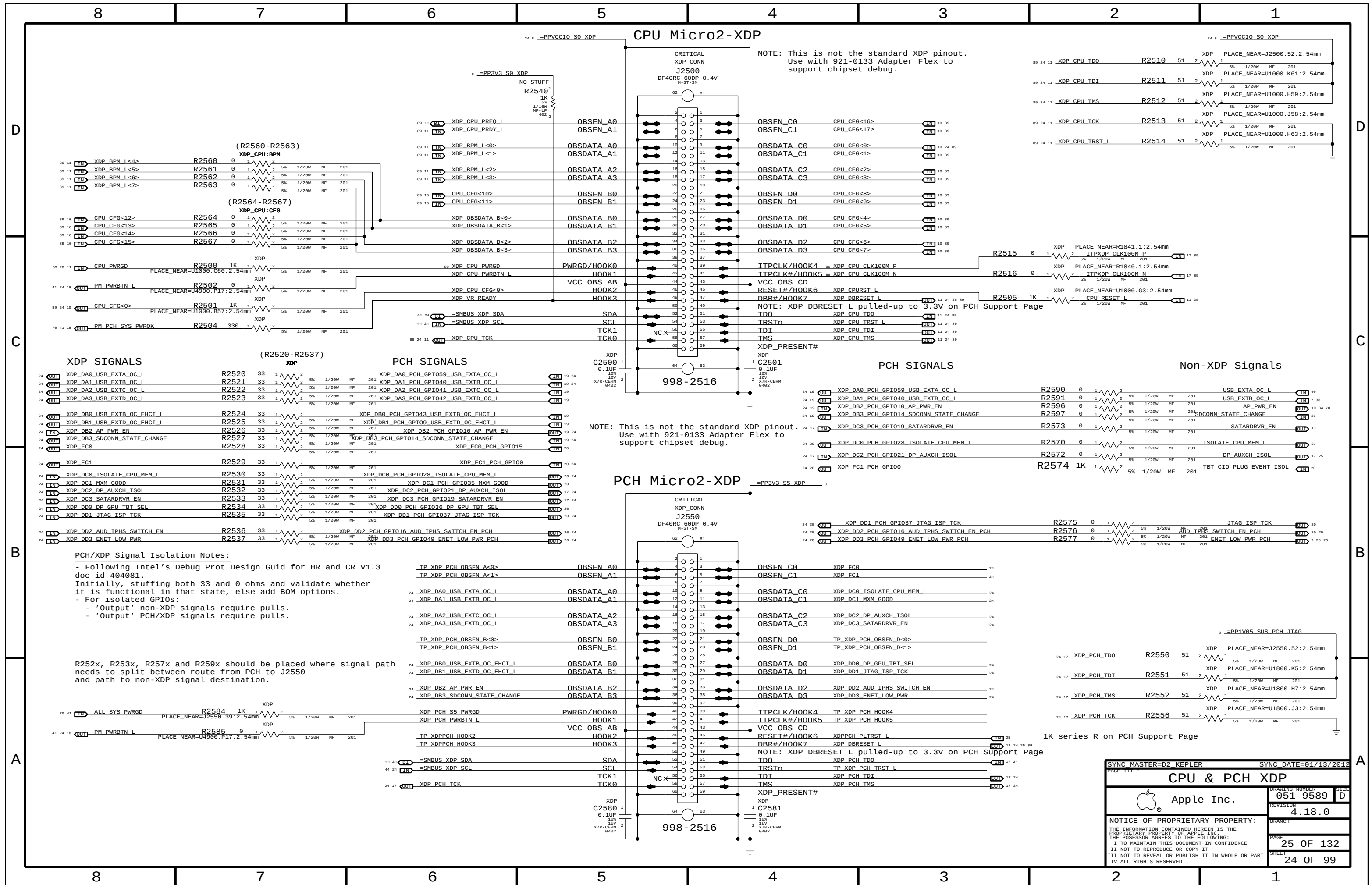
PAGE
20 OF 132

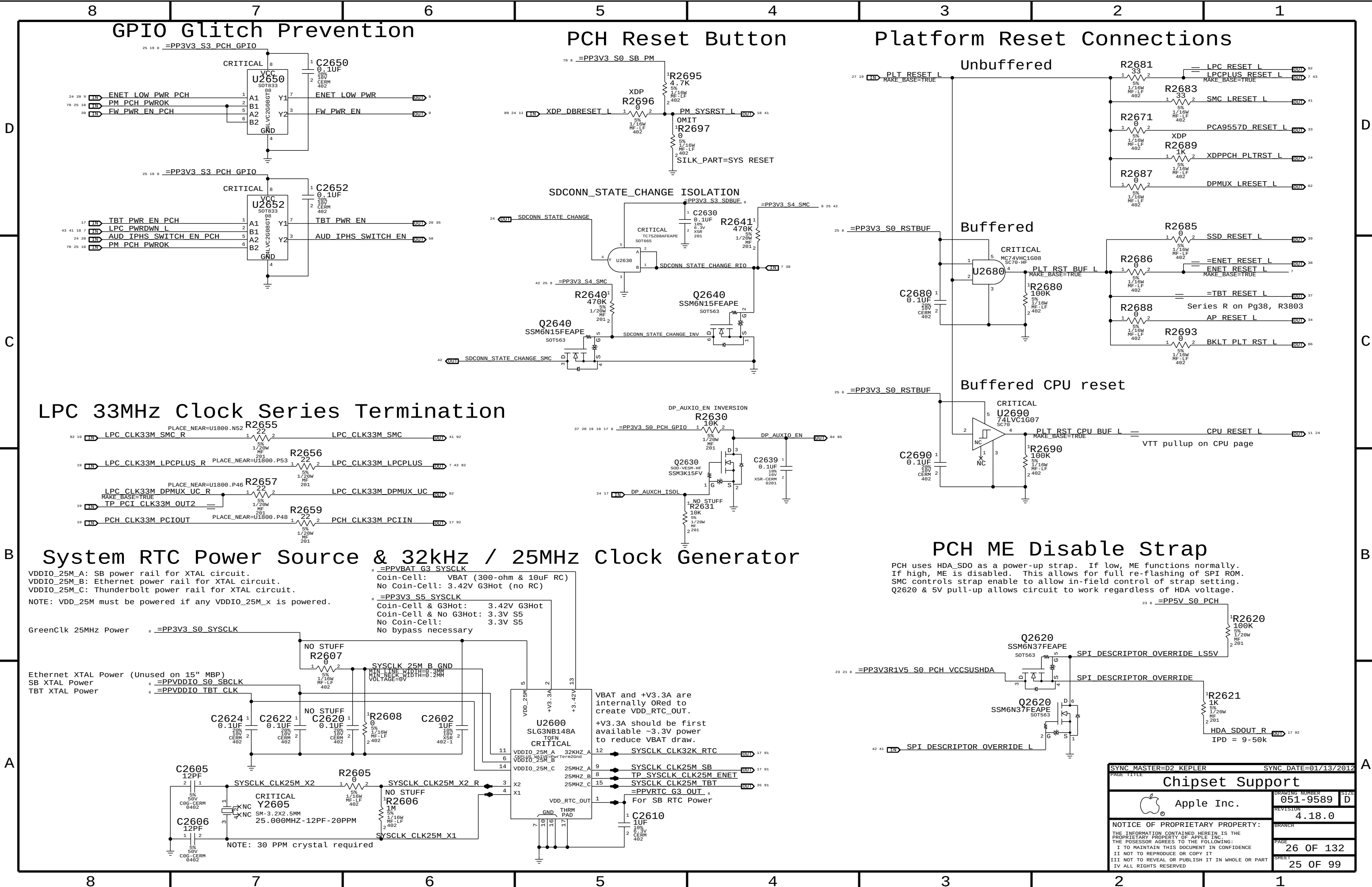
SHEET
19 OF 99





SYNC MASTER=D2_CLEAN		SYNC DATE=03/19/2012	
PAGE TITLE		PCH DECOUPLING	
 Apple Inc.		DRAWING NUMBER	051-9589
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		PAGE	24 OF 132
		SHEET	23 OF 99





GPIO Glitch Prevention

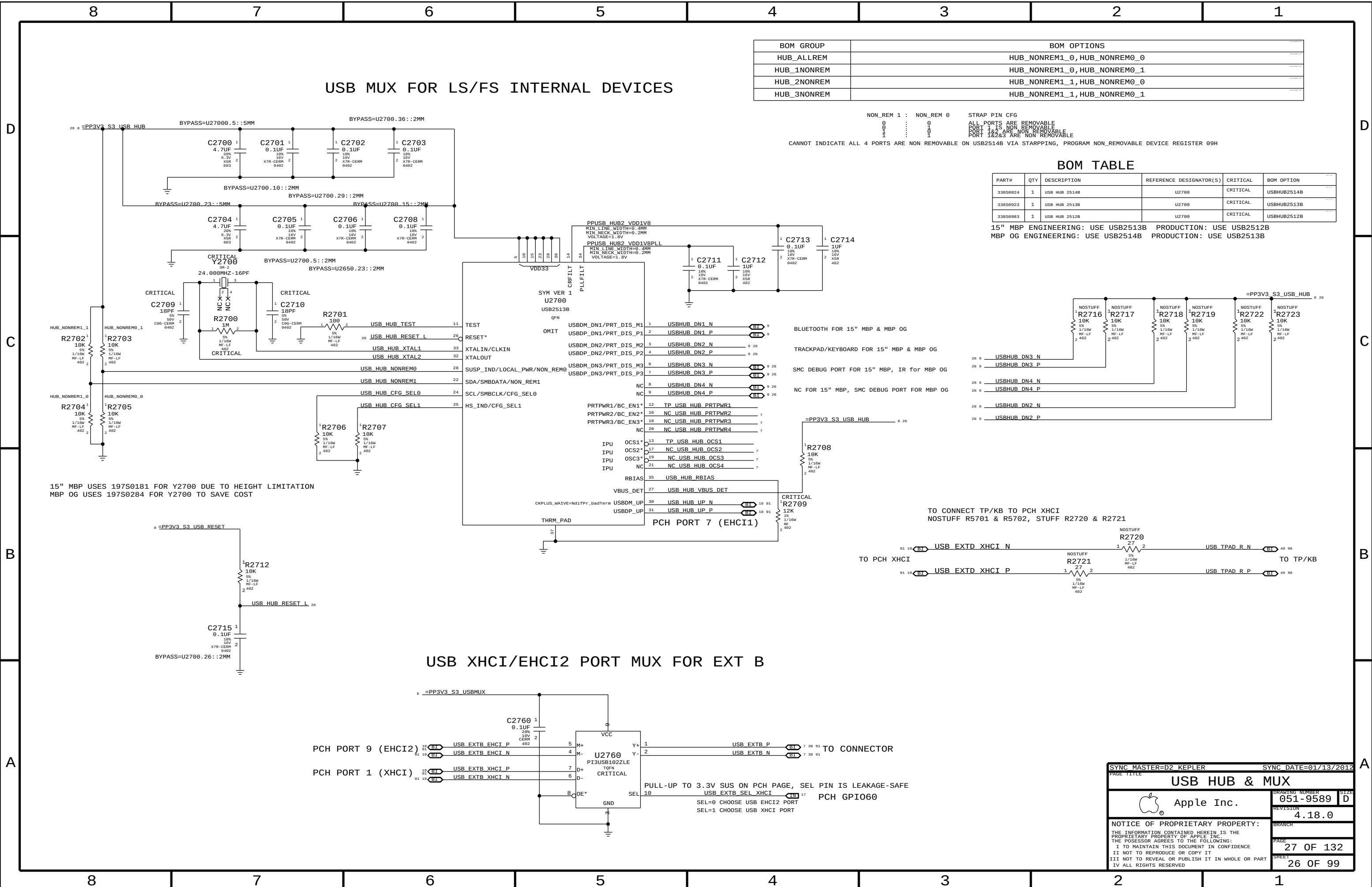
PCH Reset Button

Platform Reset Connections

System RTC Power Source & 32kHz / 25MHz Clock Generator

PCH ME Disable Strap

SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE		Chipset Support	
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BOM GROUP	BOM OPTIONS
HUB_ALLREM	HUB_NONREM1_0, HUB_NONREM0_0
HUB_1NONREM	HUB_NONREM1_0, HUB_NONREM0_1
HUB_2NONREM	HUB_NONREM1_1, HUB_NONREM0_0
HUB_3NONREM	HUB_NONREM1_1, HUB_NONREM0_1

NON_REM 1 : NON_REM 0
0 0
1 1

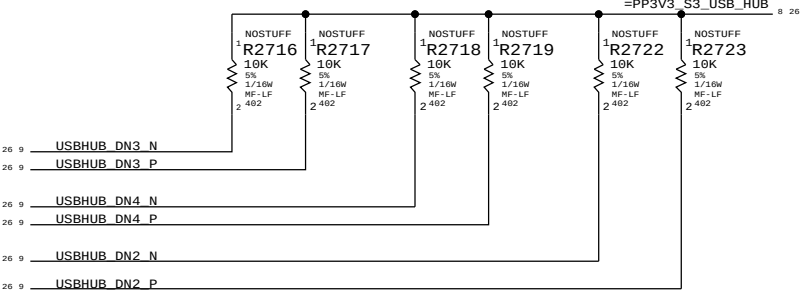
STRAP PIN CFG
ALL PORTS ARE REMOVABLE
PORT 1&2 ARE NON REMOVABLE
PORT 1&2&3 ARE NON REMOVABLE

CANNOT INDICATE ALL 4 PORTS ARE NON REMOVABLE ON USB2514B VIA STARPPING, PROGRAM NON_REMOVABLE DEVICE REGISTER 09H

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S9824	1	USB HUB 2514B	U2700	CRITICAL	USBHUB2514B
338S9923	1	USB HUB 2513B	U2700	CRITICAL	USBHUB2513B
338S9983	1	USB HUB 2512B	U2700	CRITICAL	USBHUB2512B

15" MBP ENGINEERING: USE USB2513B PRODUCTION: USE USB2512B
MBP OG ENGINEERING: USE USB2514B PRODUCTION: USE USB2513B

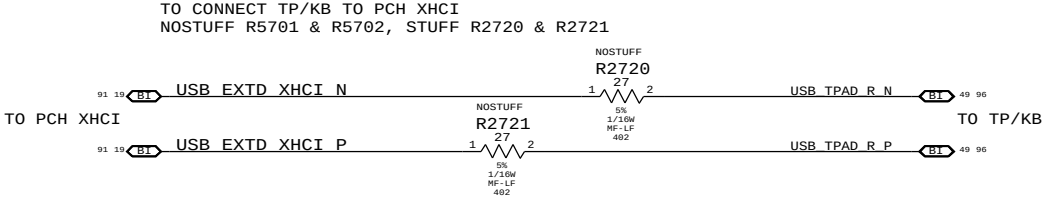


BLUETOOTH FOR 15" MBP & MBP OG

TRACKPAD/KEYBOARD FOR 15" MBP & MBP OG

SMC DEBUG PORT FOR 15" MBP, IR for MBP OG

NC FOR 15" MBP, SMC DEBUG PORT FOR MBP OG



SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE		USB HUB & MUX	
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		PAGE	27 OF 132
		SHEET	26 OF 99

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0365	1	RES,MTL FILM,1/16W,33.2K,1,9402,SMD,LF	R2821		PPDDR:1V5
114S0376	1	RES,MTL FILM,1/16W,43.2K,1,9402,SMD,LF	R2821		PPDDR:1V35

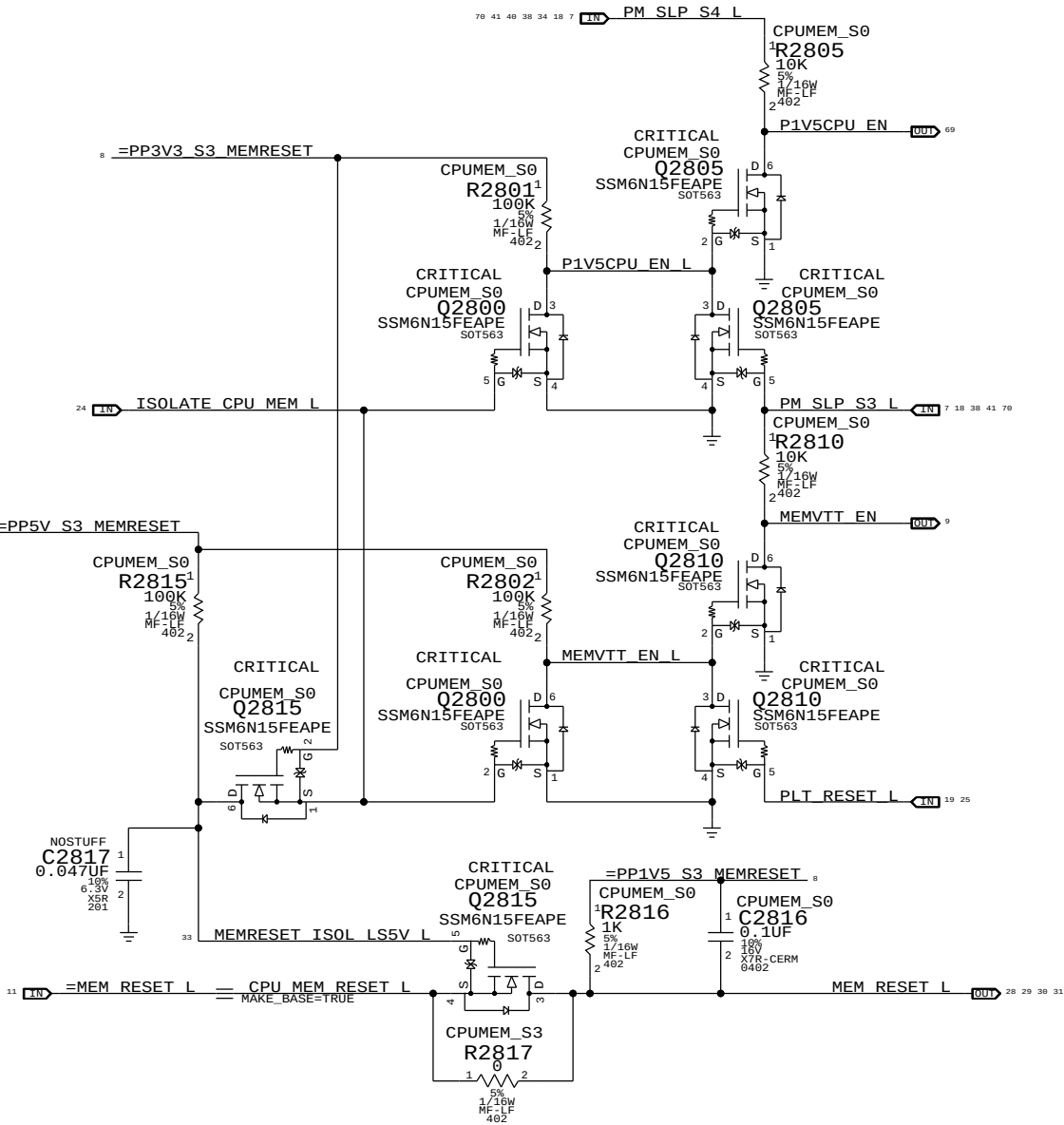
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.

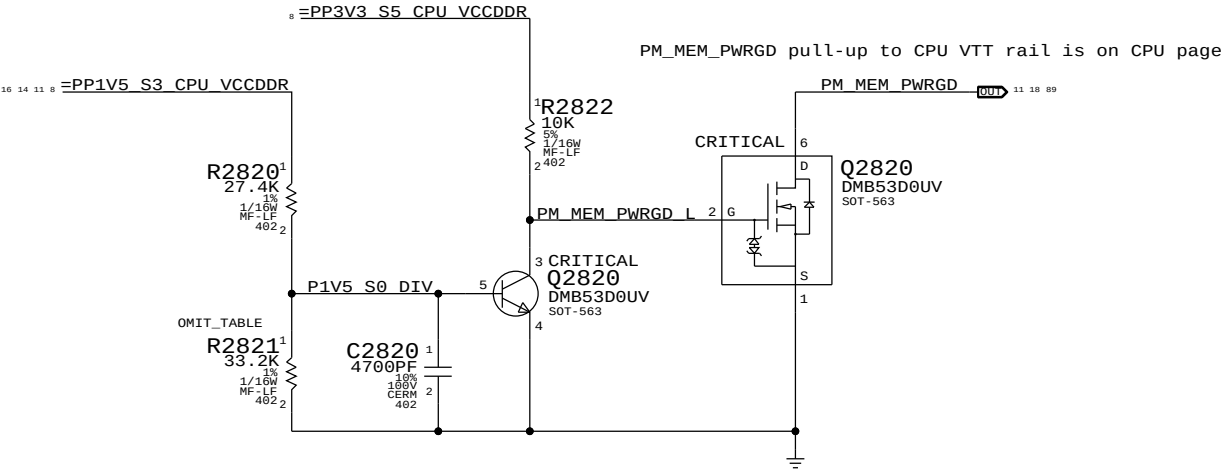
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

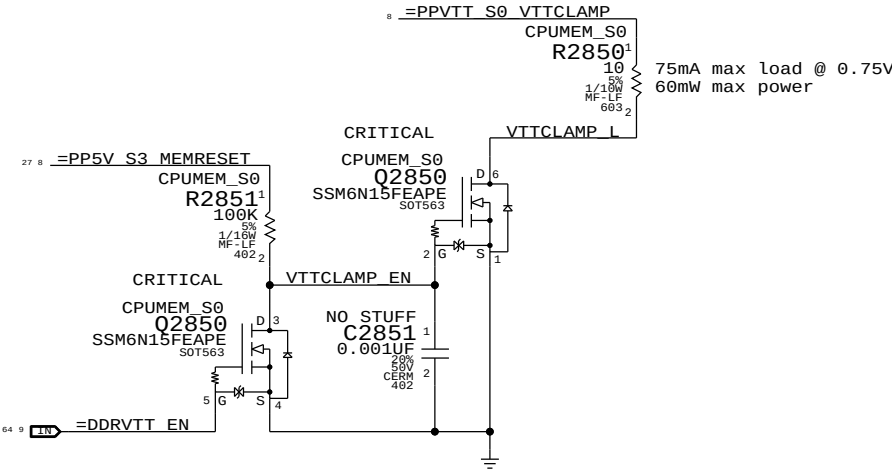


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

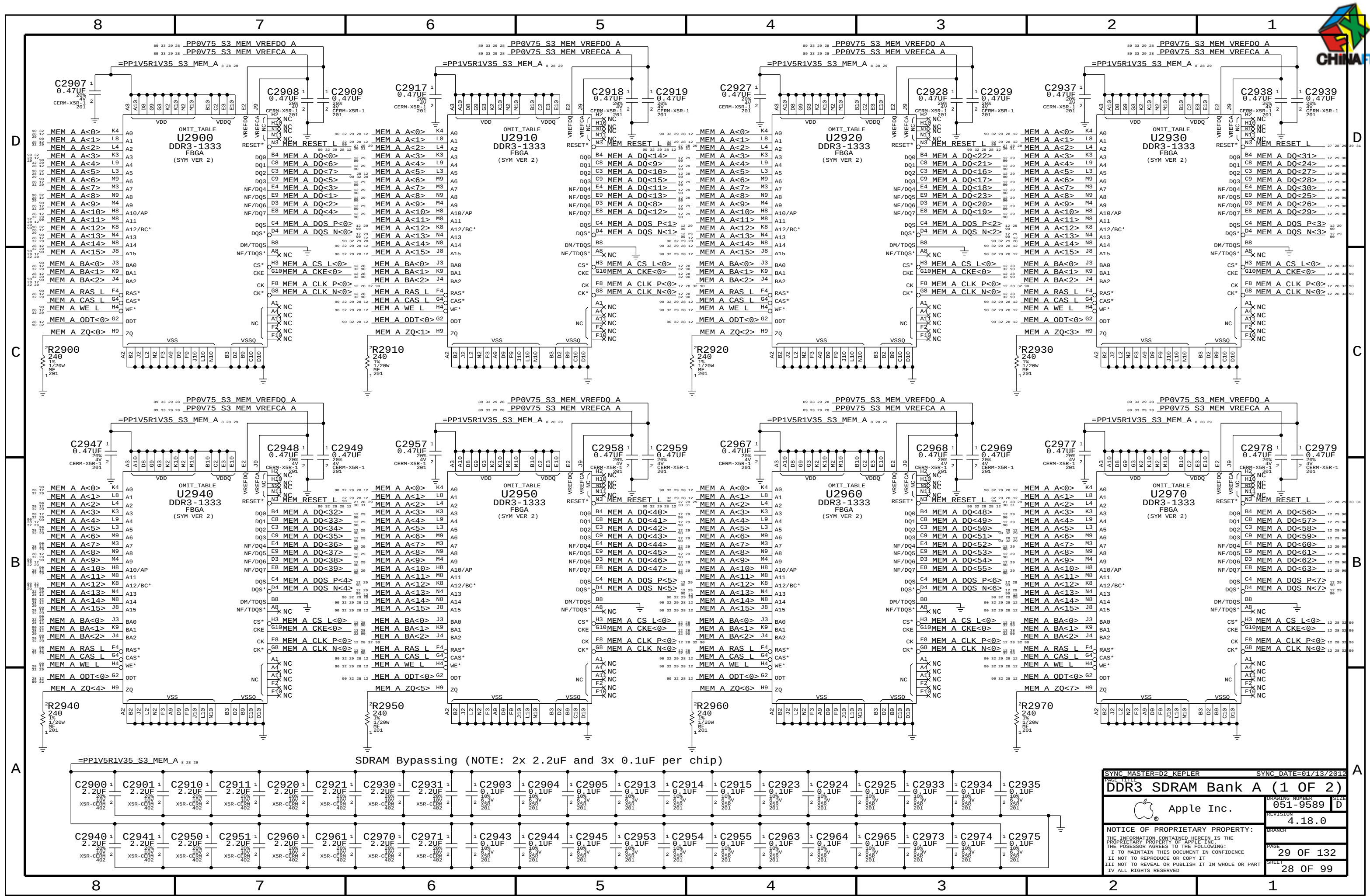


Step	ISOLATE_CPU_MEM_L	PLT_RST_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

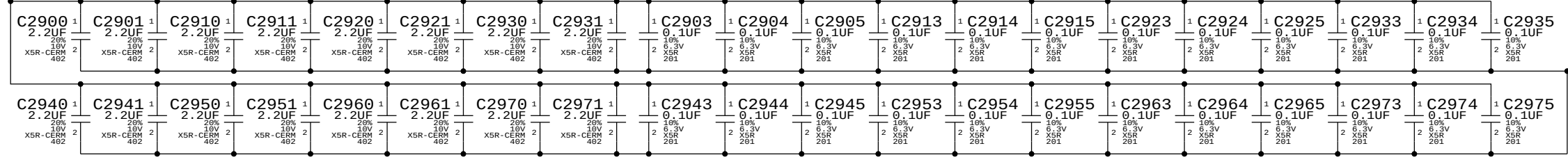
(*) CPU_MEM_RESET_L asserts due to loss of PMEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

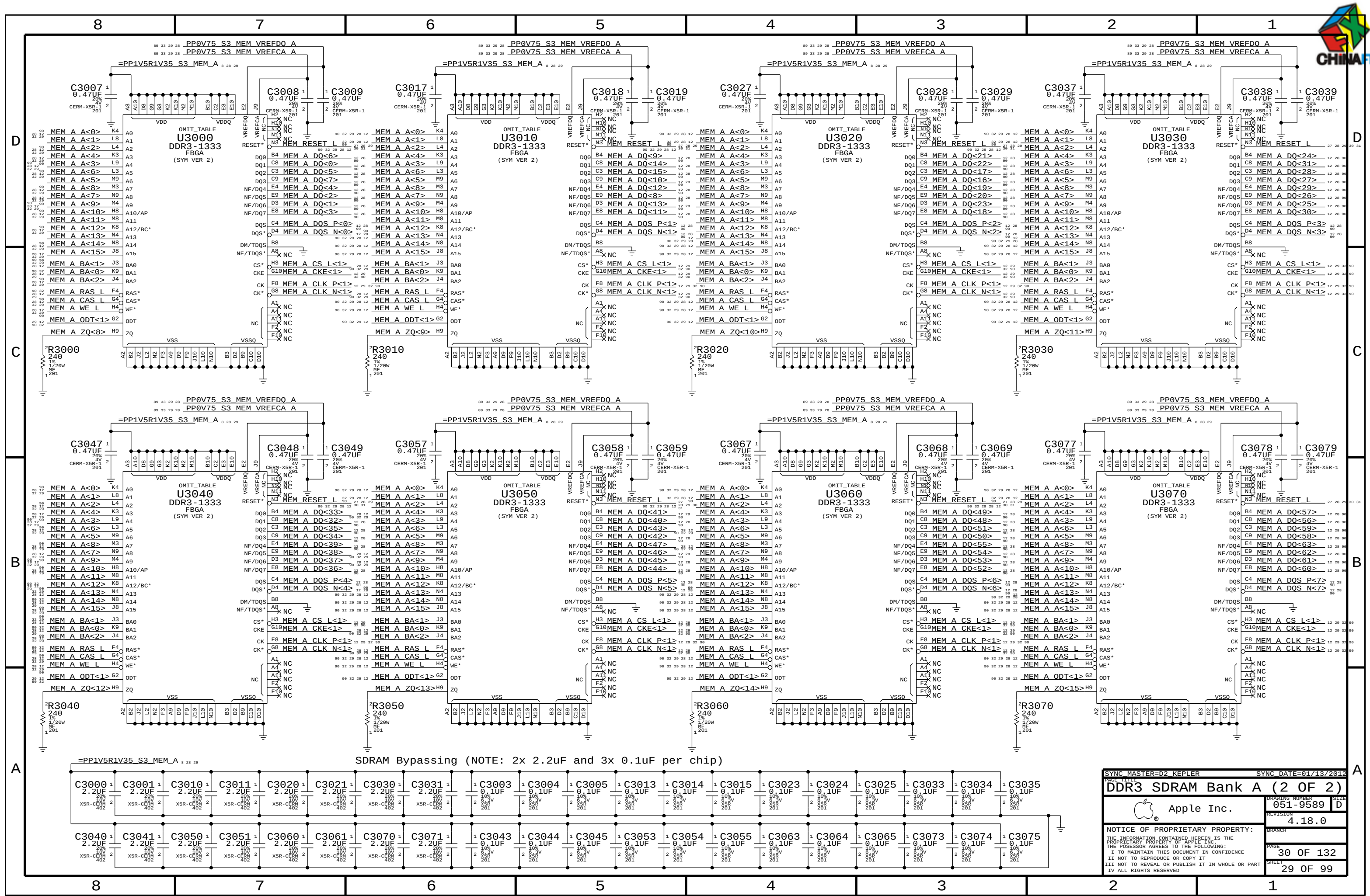
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SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)

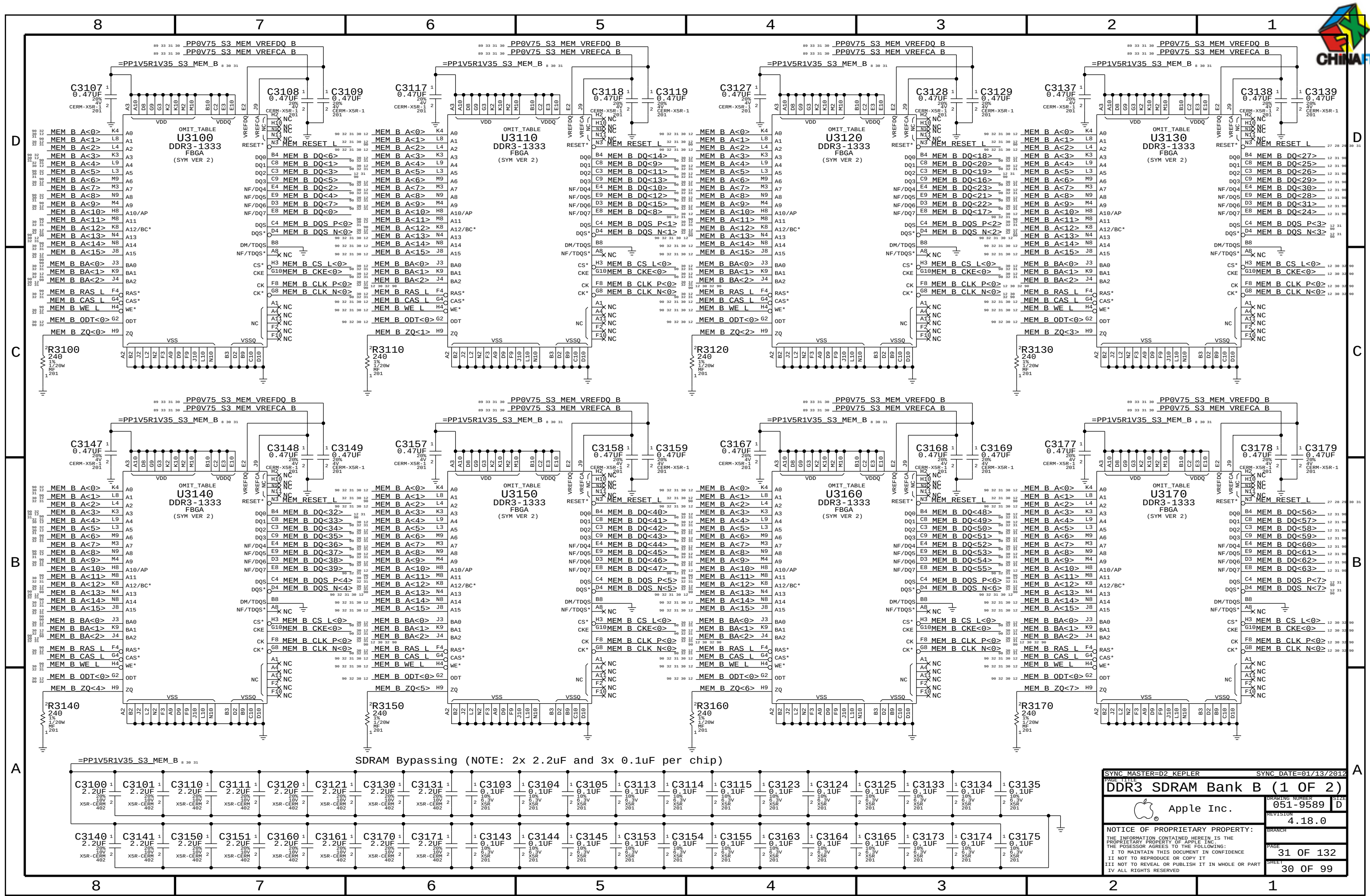


SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
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		REVISION	4.18.0
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SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)

SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE			
DDR3 SDRAM Bank A (2 OF 2)		DRAWING NUMBER	051-9589
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SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)

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SYNC DATE=01/13/2012

PAGE TITLE		
DDR3 SDRAM Bank B (1 OF 2)		
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PAGE		31 OF 132
SHEET		30 OF 99

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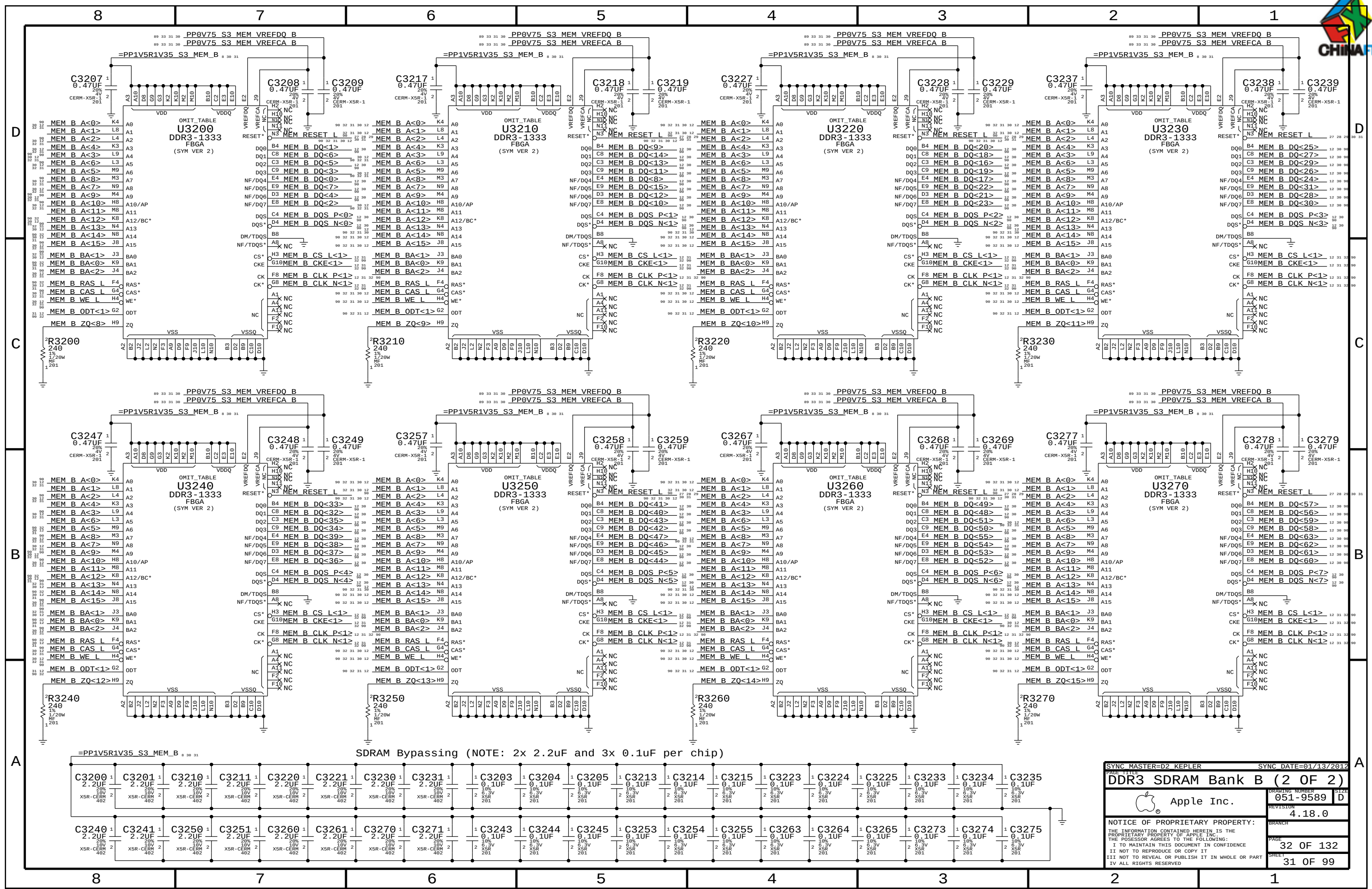
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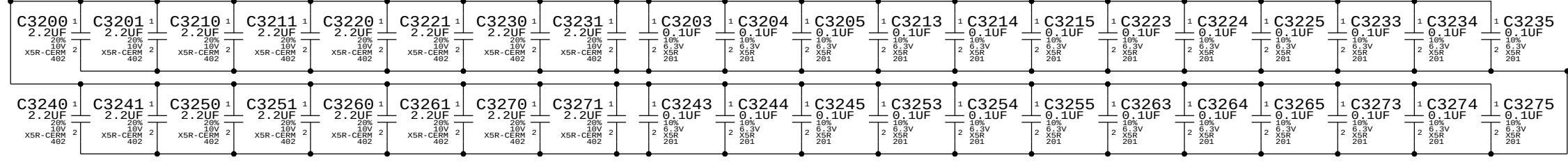
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SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)



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JEDEC 4.20.18 Unbuffered SODIMM Raw Card F spec recommends 36 Ohm term to VTT for CS,CKE,ODT and 36 Ohm for BA,A,RAS,CAS,WE

D

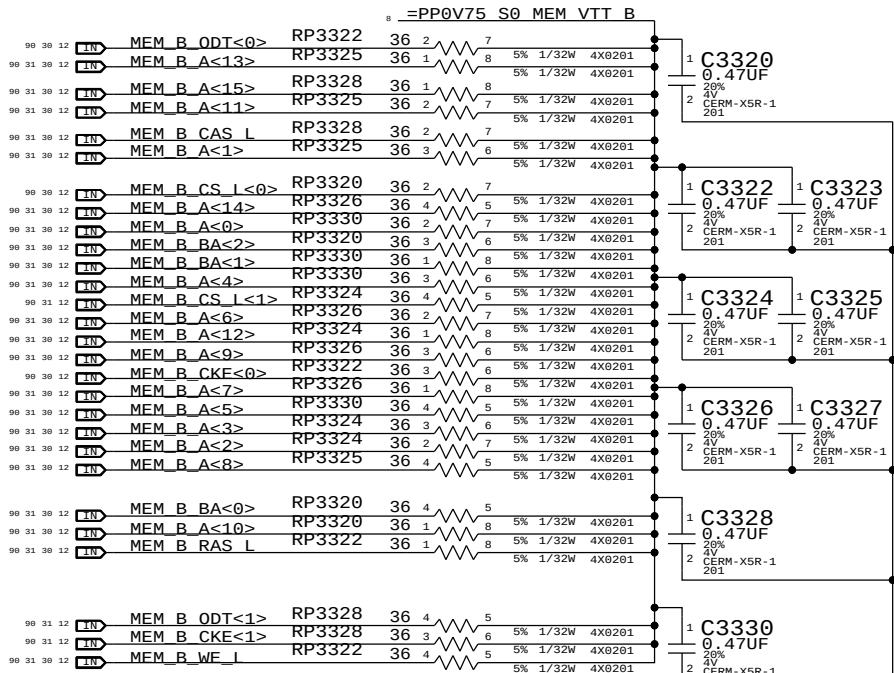
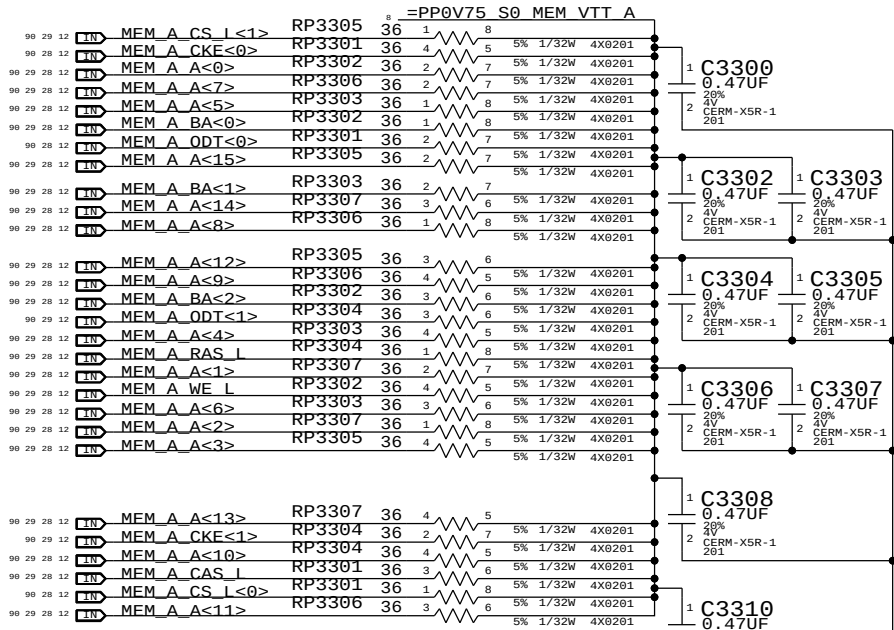
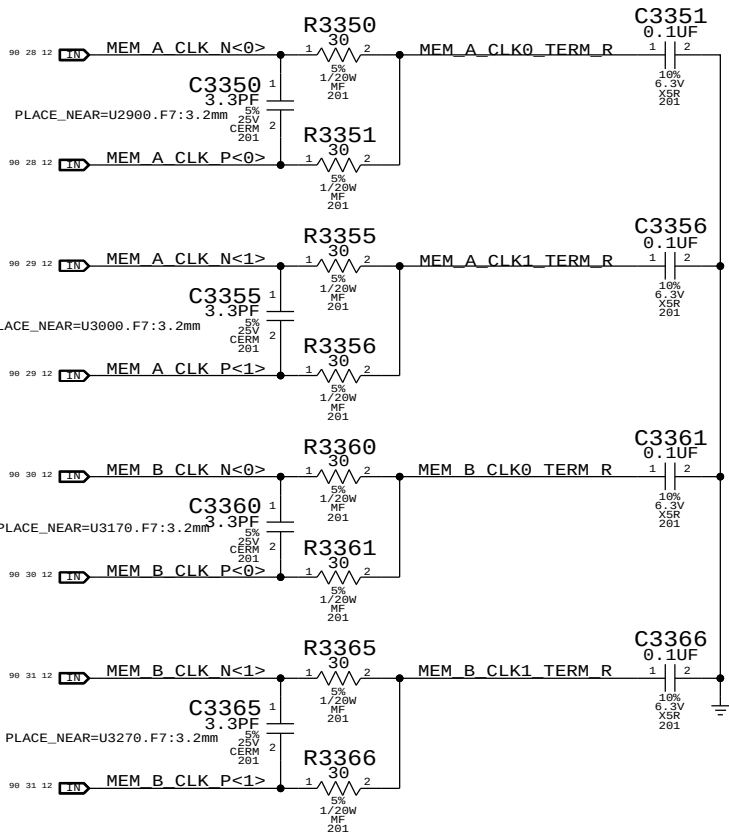
C

B

A

MEM Clock Termination

Place RC end termination after last DRAM
Place Source Cterm at neckdown at first DRAM



D

C

B

A

SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
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NOTE: Must not enable more than two S0-DIMM margining buffers at once or VRef source may be overloaded.

Page Notes

Power aliases required by this page:
 - =PP3V3_S3_VREFMRGN
 - =PPVTT_S3_DDR_BUF
 - =PPDDR_S3_MEMVREF

Signal aliases required by this page:
 - =I2C_VREFDACS_SCL
 - =I2C_VREFDACS_SDA
 - =I2C_PCA9557D_SCL
 - =I2C_PCA9557D_SDA

BOM options provided by this page:
 DDRVREF_DAC - Stuffs Apple margining circuit.
 VREFDQ:LDO_DAC - LDO outputs sent to DQ inputs.
 VREFDQ:M1_M3 - CPU margined DDR voltage divider sent to DQ inputs.
 VREFDQ:M1_DAC - DAC margined DDR voltage divider sent to DQ inputs.
 VREFCA:LDO - LDO outputs sent to CA inputs.
 VREFCA:LDO_DAC - DAC margined LDO outputs sent to CA inputs.

RST* on 'platform reset' so that system watchdog will disable margining.
 NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3403, R3405		VREFDQ:LDO
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3409, R3411		VREFCA:LDO

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0218	4	RES, MTL FILM, 1K, 1%, 0402, SM, LF	R3421, R3422, R3441, R3442		VREFDQ:M1_DAC
114S0171	2	RES, MTL FILM, 332, 1%, 0402, SM, LF	R3404, R3406		VREFDQ:M1_DAC

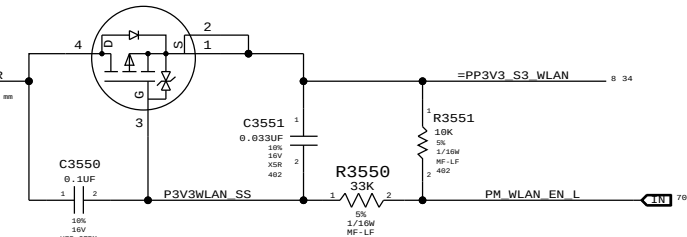
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE			
DDR3/FRAMEBUF VREF MARGINING			
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
11750002	4	RES, 000M, 0301	L3570, L3571, L3573, L3574		

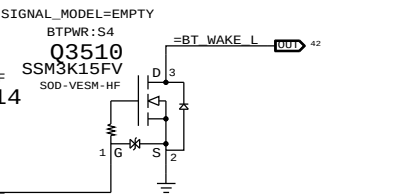
3V S3 WLAN FET	
MOSFET	DMP2018LFK
CHANNEL	P-TYPE
RDS(ON) @ 2.5V	14 mOhm Typ 20 mOhm MAX
LOADING	1 A (EDP)

CRITICAL
DMP2018LFK
DFN2563-6



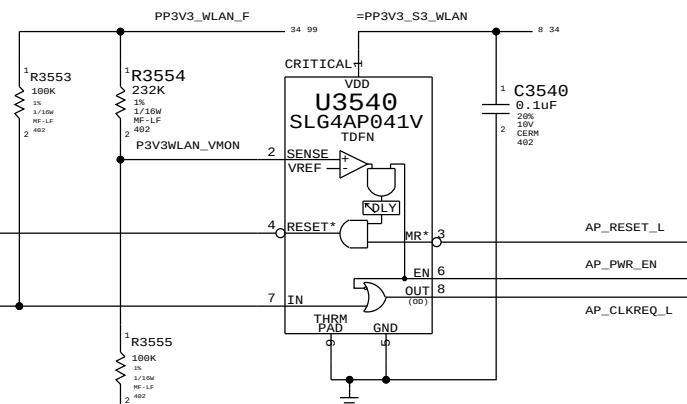
AIRPORT

BLUETOOTH



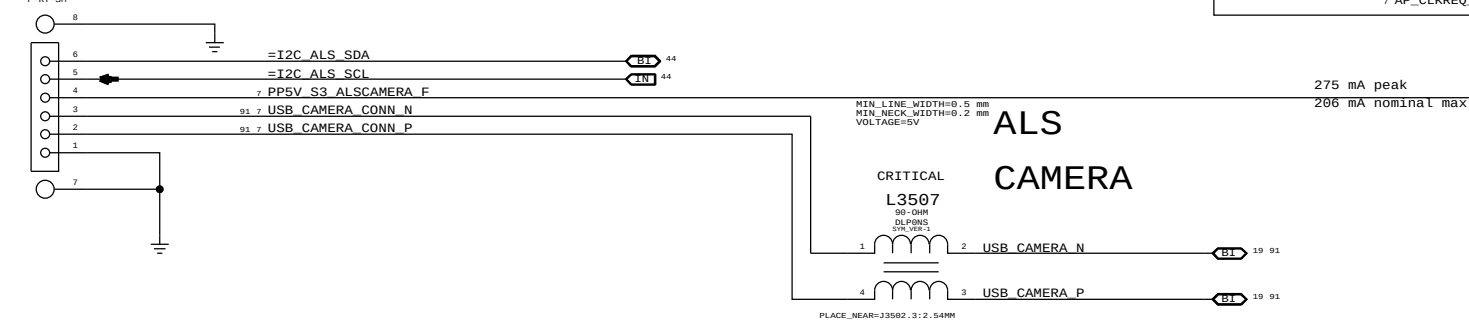
Supervisor & CLKFREQ # Isolation

Delay = 130 ms +/- 20%



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
51859767	1	CONN, HDR, TACH-ALS, P=0.40MM, 6P, 1E	33502	CRITICAL	

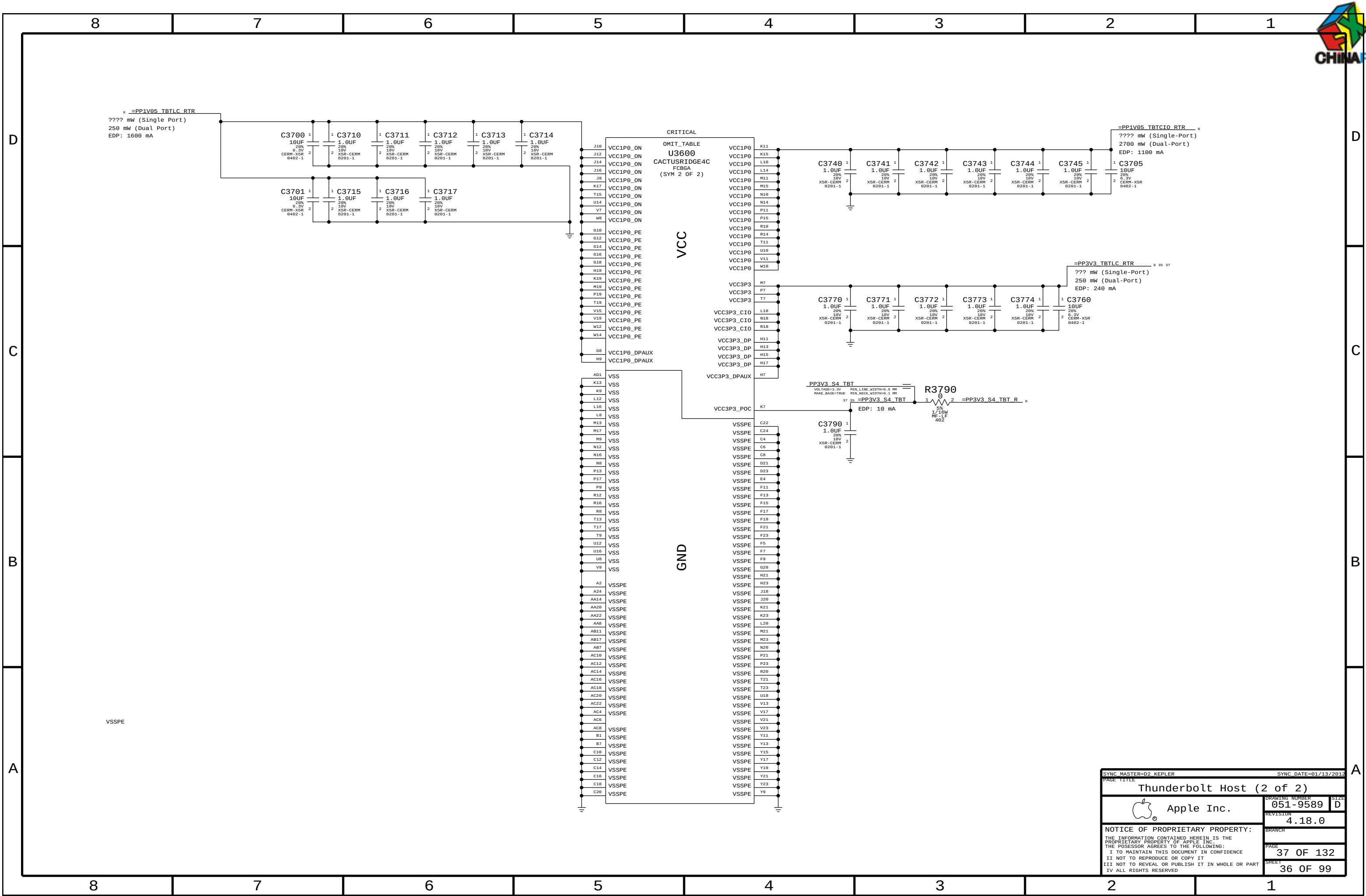
CRITICAL
J3502
CCR20-6K710S
F-RT-SM1



ALS
CAMERA

SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
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X29/ALS/CAMERA CONNECTOR			
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Thunderbolt Host (2 of 2)			
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Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFTET (3.3V FET Input)
- =PP3V3_TBTLC_FET (3.3V FET Output)
- =PP3V3_S0_TBTMWRCTL
- =PP1V05_TBT_P1V05TBTFTET (1.05V FET Input)
- =PP1V05_TBTLC_FET (1.05V FET Output)

Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

Thunderbolt 15V Boost Regulator

SI8409DB:
Vds(max): -30V
Vgs(max): +/-12V
Vgs(th): -1.4V
Rds(on): 46mOhm @ 4.5V Vgs
Id(max): 3.7A @ 70C

CRITICAL
TBTBST:Y
L3895
3.3uH-6.5A

TBTBST BOOST
MIN LINE WIDTH=0.5 mm
MIN NECK WIDTH=0.25 mm
NOISE=TRUE
DIOT=TRUE

CRITICAL
TBTBST:Y
U3890
LT3957
QFN

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
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281-2

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281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

CRITICAL
TBTBST:Y
R3889
1/20W
281-2

Supervisor & CLKREQ# Isolation

TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.

1.05V TBT "CIO" Switch

1.05V TBT "LC" Switch

Max Current = 2A (85C)

U3815

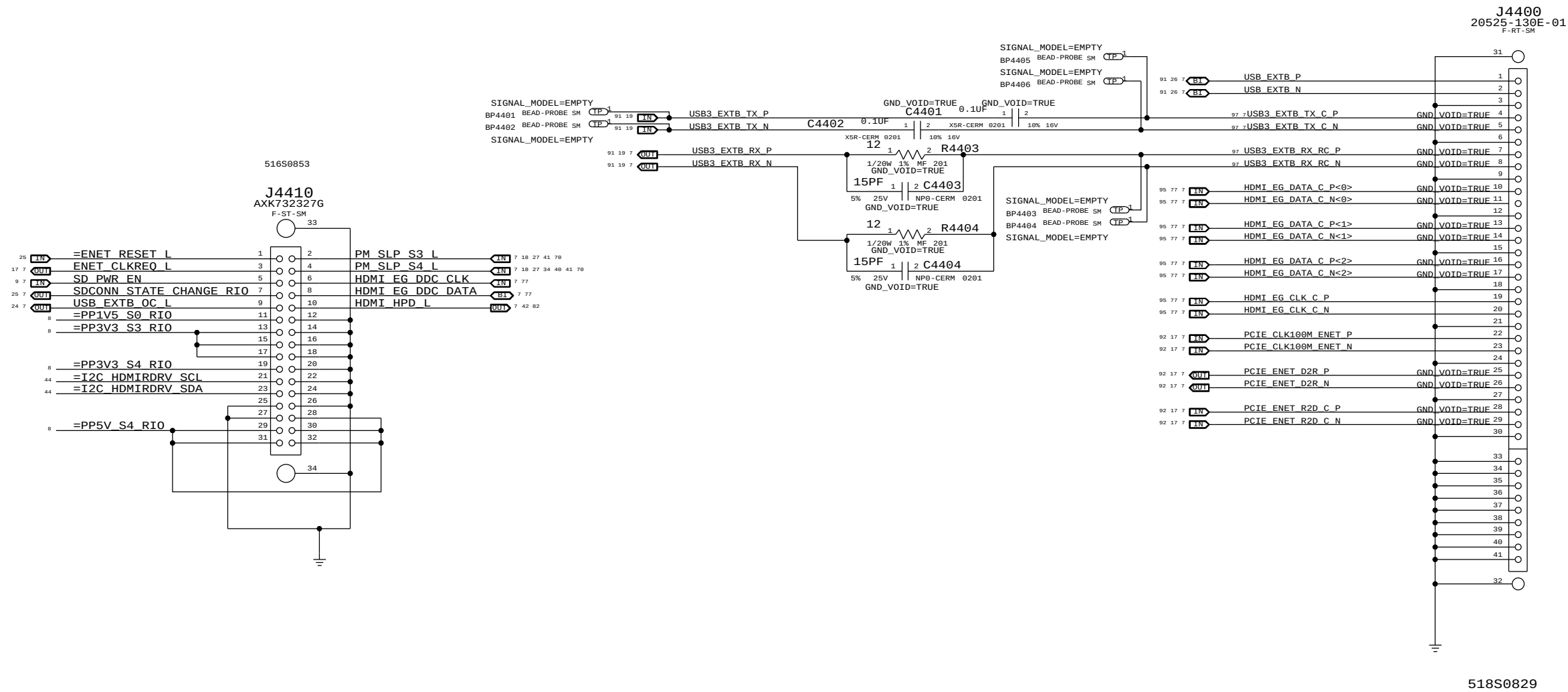
Part	TPS22924C
Type	Load Switch
R(on)	20.3 mOhm Typ
@ 1.0V	28.6 mOhm Max

C3816 must be 10%
RC guarantees minimum 5ms to reach 0.5V

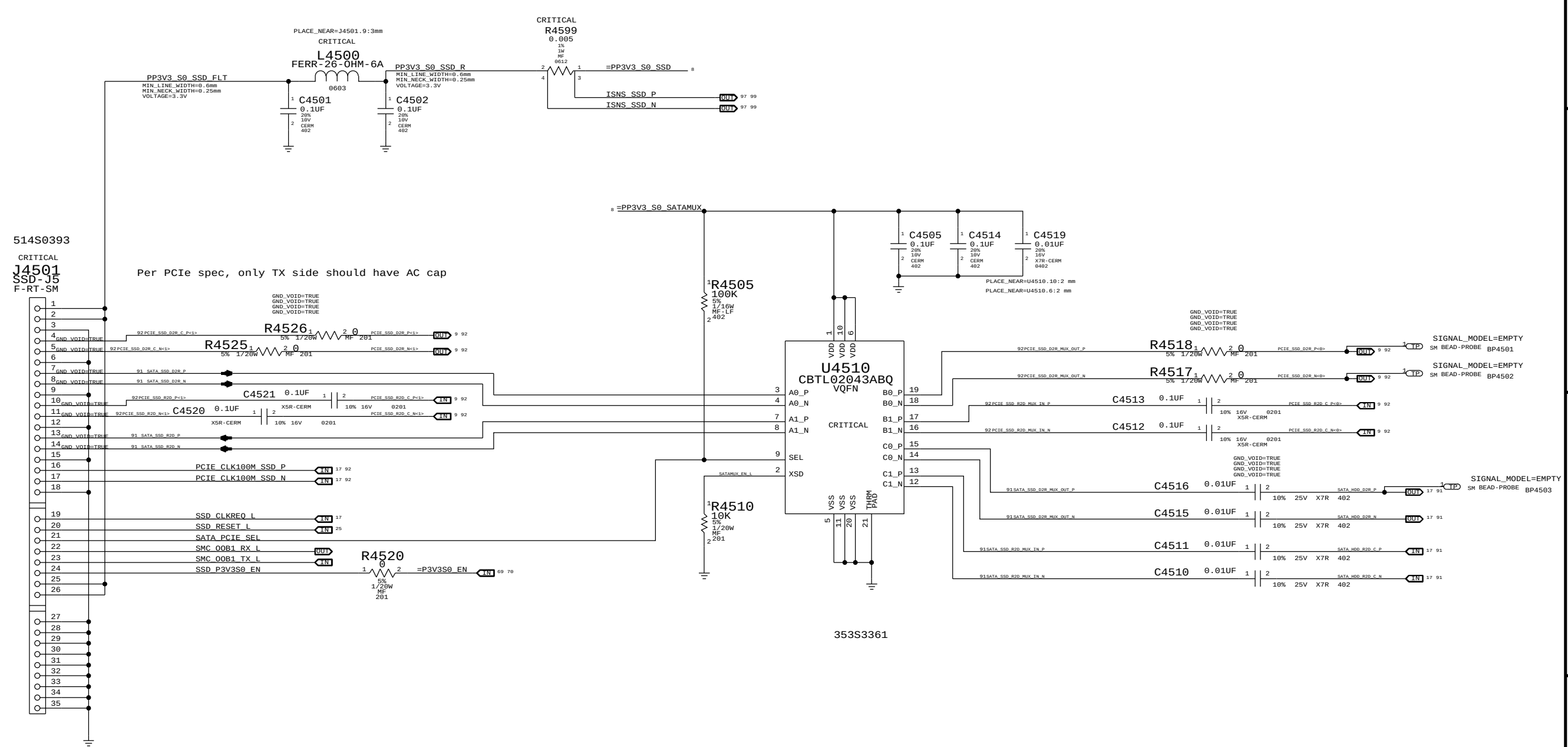
U3820

Part	TPS22920
Type	Load Switch
R(on)	8 mOhm Typ
@ 1.05V	11.5 mOhm Max

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		SHEET	38 OF 99



PCIE/SATA GUMSTICK2 CONNECTOR

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		PAGE	45 OF 132
		SHEET	39 OF 99



D

C

B

A

D

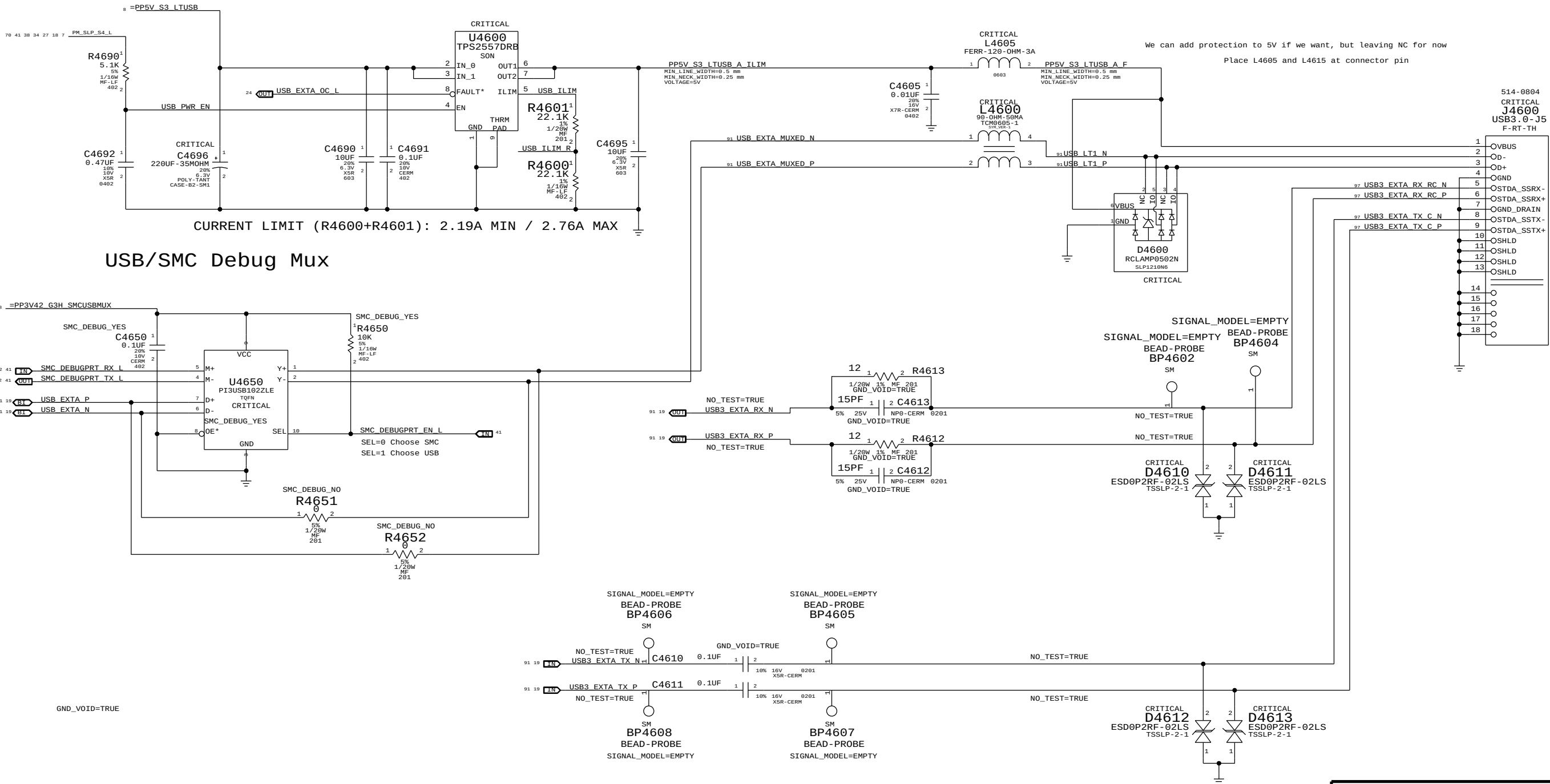
C

B

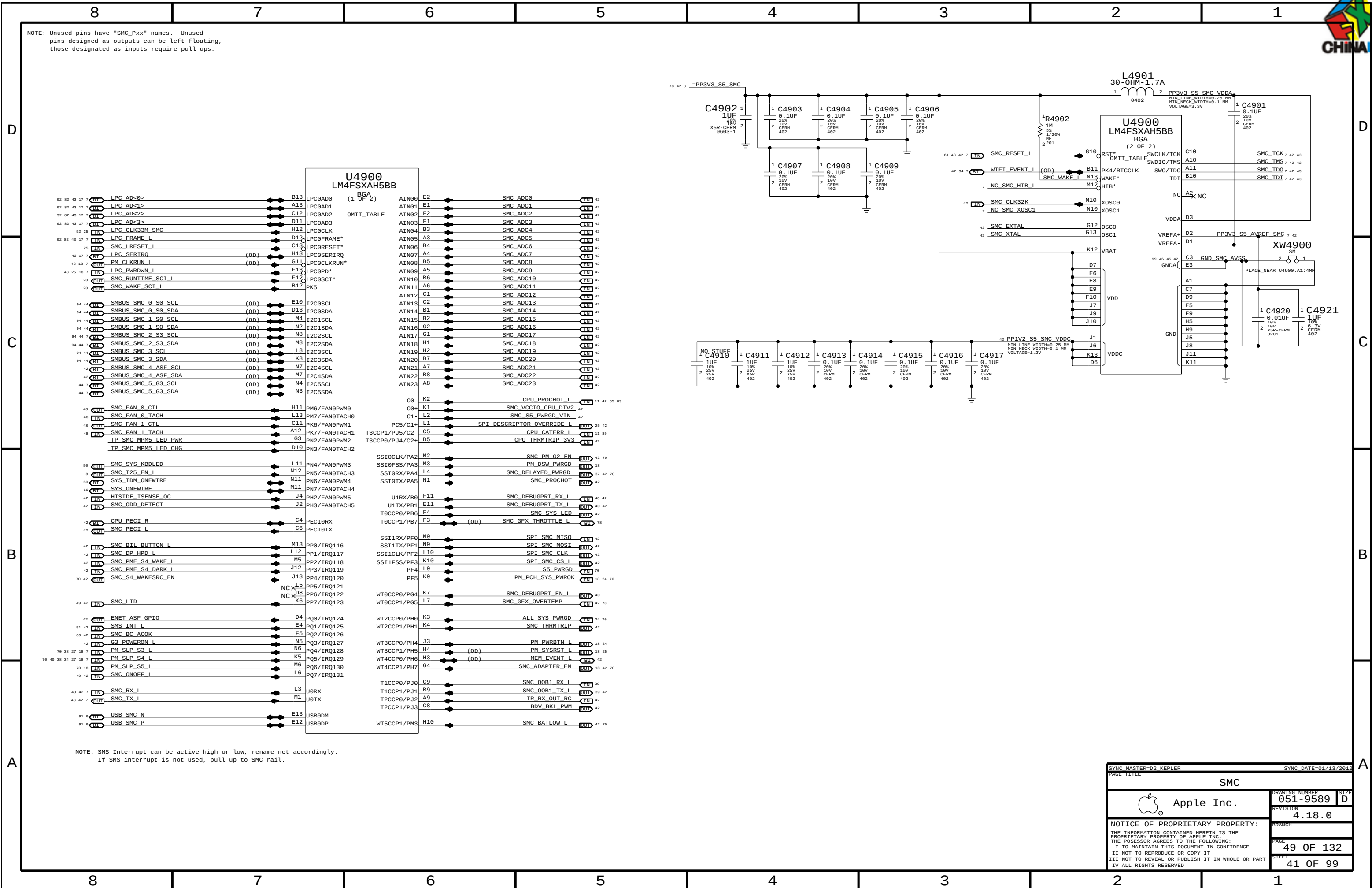
A

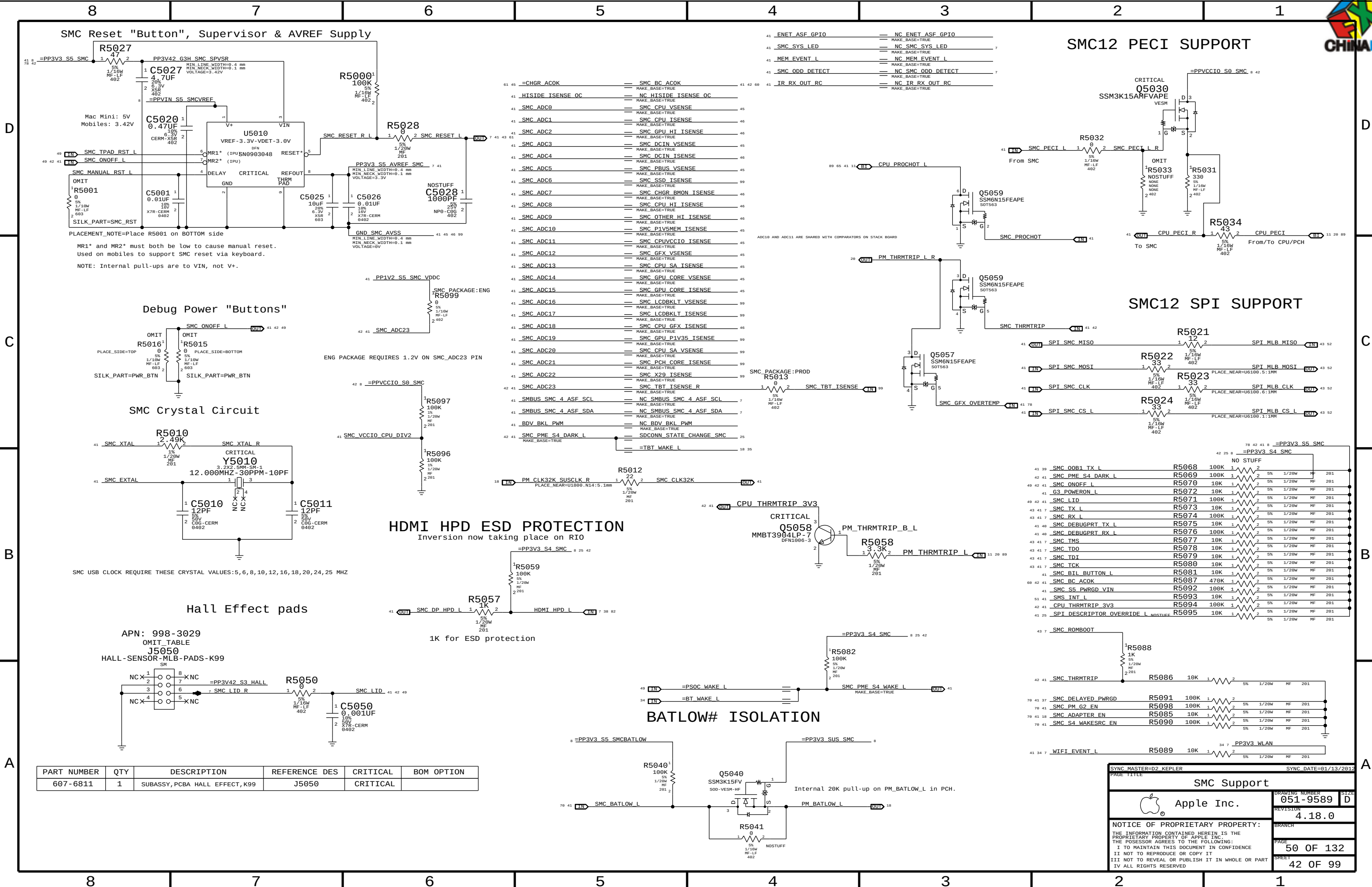
USB Port Power Switch

Left USB Port A

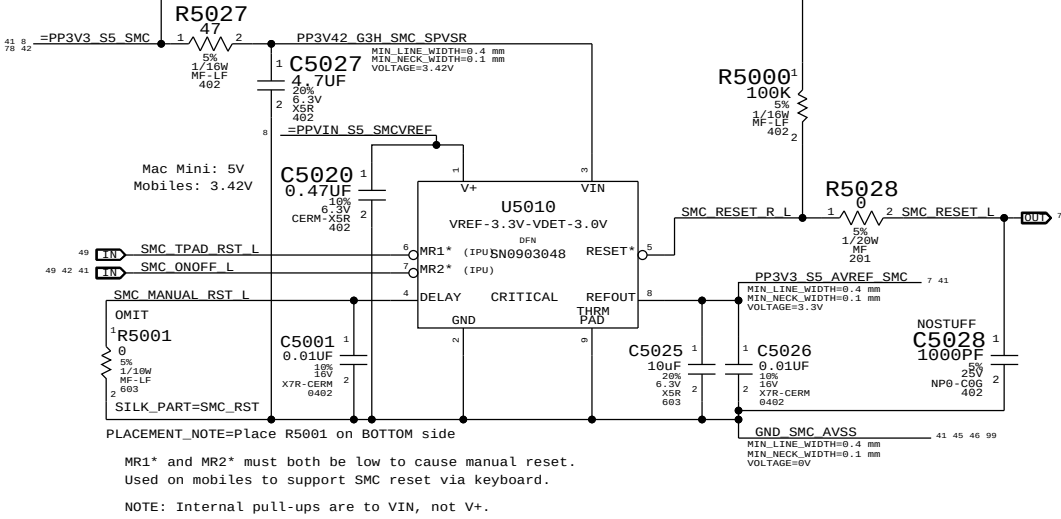


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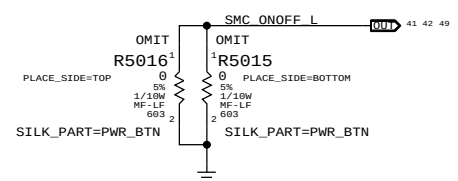




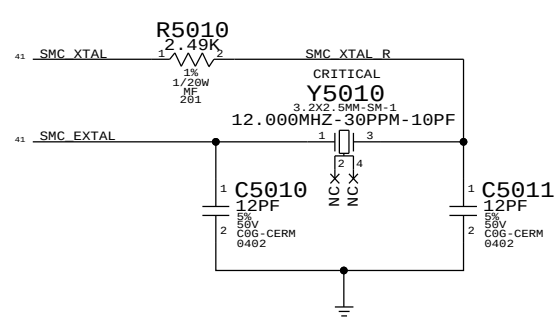
SMC Reset "Button", Supervisor & AVREF Supply



Debug Power "Buttons"

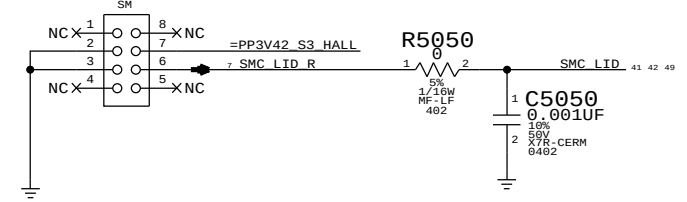


SMC Crystal Circuit



Hall Effect pads

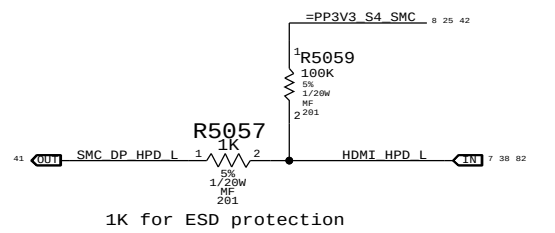
APN: 998-3029
OMIT_TABLE
J5050
HALL-SENSOR-MLB-PADS-K99



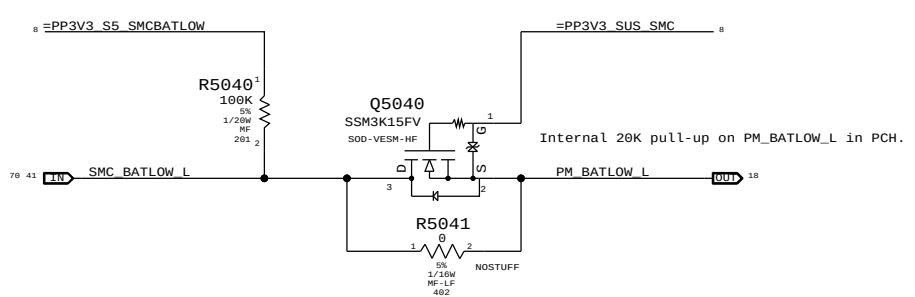
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
607-6811	1	SUBASSY,PCBA HALL EFFECT,K99	J5050	CRITICAL	

HDMI HPD ESD PROTECTION

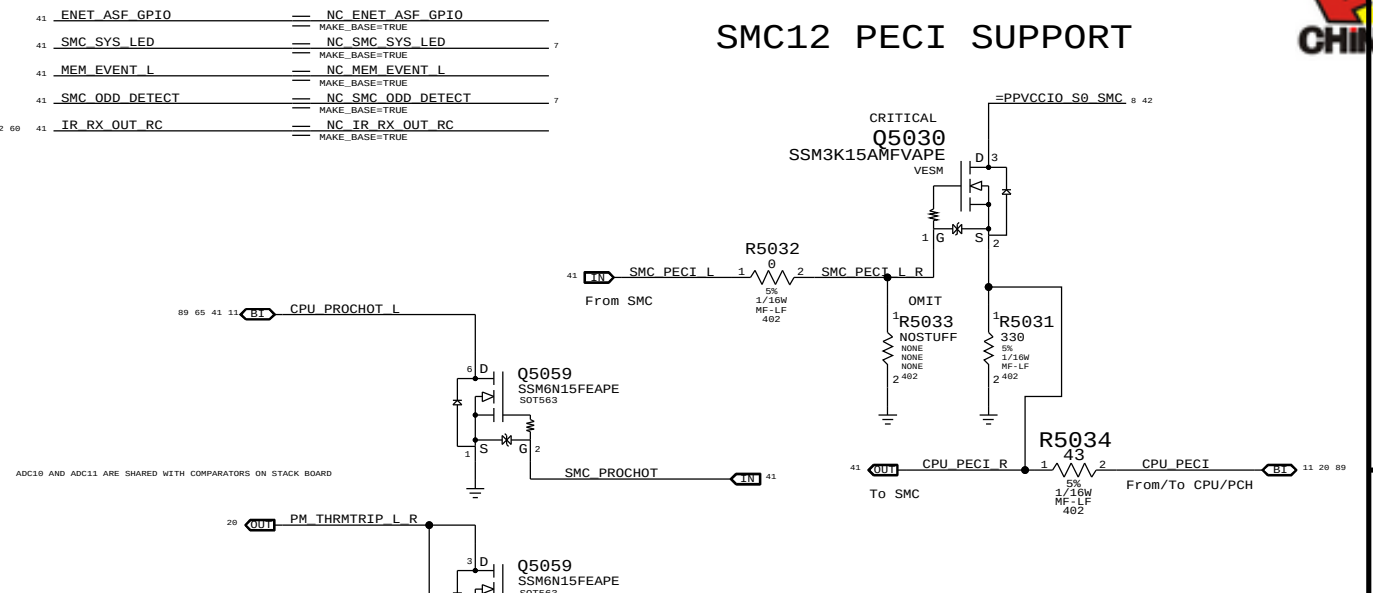
Inversion now taking place on R10



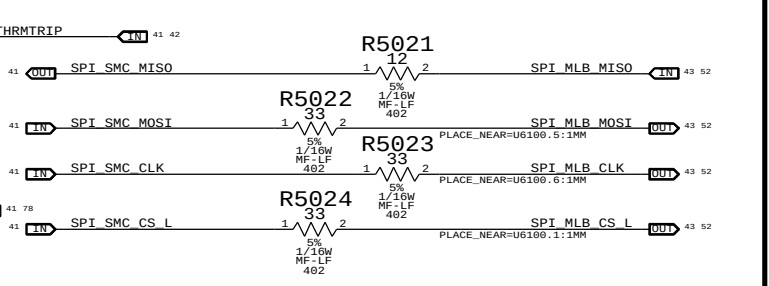
BATLOW# ISOLATION



SMC12 PEI SUPPORT



SMC12 SPI SUPPORT



SMC	Value	Pin	Value	Pin	Value	Pin	Value	Pin
SMC_00B1 TX_L	100K	1	100K	2	100K	3	100K	4
SMC_PME S4 DARK_L	100K	1	100K	2	100K	3	100K	4
SMC_ONOFF_L	10K	1	10K	2	10K	3	10K	4
G3 POWERON_L	10K	1	10K	2	10K	3	10K	4
SMC_LID	100K	1	100K	2	100K	3	100K	4
SMC_TX_L	10K	1	10K	2	10K	3	10K	4
SMC_RX_L	10K	1	10K	2	10K	3	10K	4
SMC_DEBUGPRT_TX_L	10K	1	10K	2	10K	3	10K	4
SMC_DEBUGPRT_RX_L	100K	1	100K	2	100K	3	100K	4
SMC_TMS	10K	1	10K	2	10K	3	10K	4
SMC_TDO	10K	1	10K	2	10K	3	10K	4
SMC_TDI	10K	1	10K	2	10K	3	10K	4
SMC_TCK	10K	1	10K	2	10K	3	10K	4
SMC_BIL_BUTTON_L	10K	1	10K	2	10K	3	10K	4
SMC_BC ACOK	470K	1	470K	2	470K	3	470K	4
SMC_S5_PWRGD_VIN	100K	1	100K	2	100K	3	100K	4
SMC_INT_L	10K	1	10K	2	10K	3	10K	4
CPU_THRMTRIP_3V3	100K	1	100K	2	100K	3	100K	4
SPI_DESCRIPTOR_OVERRIDE_L	10K	1	10K	2	10K	3	10K	4

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SMC Support

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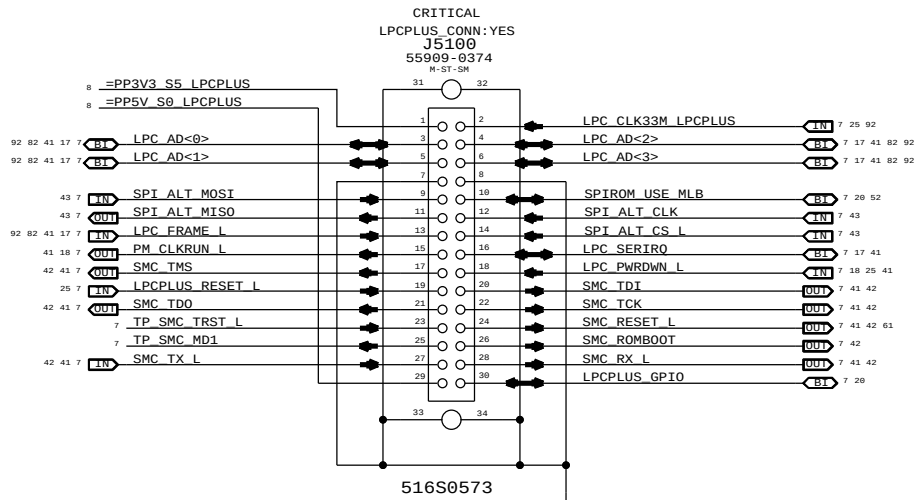
PAGE

50 OF 132

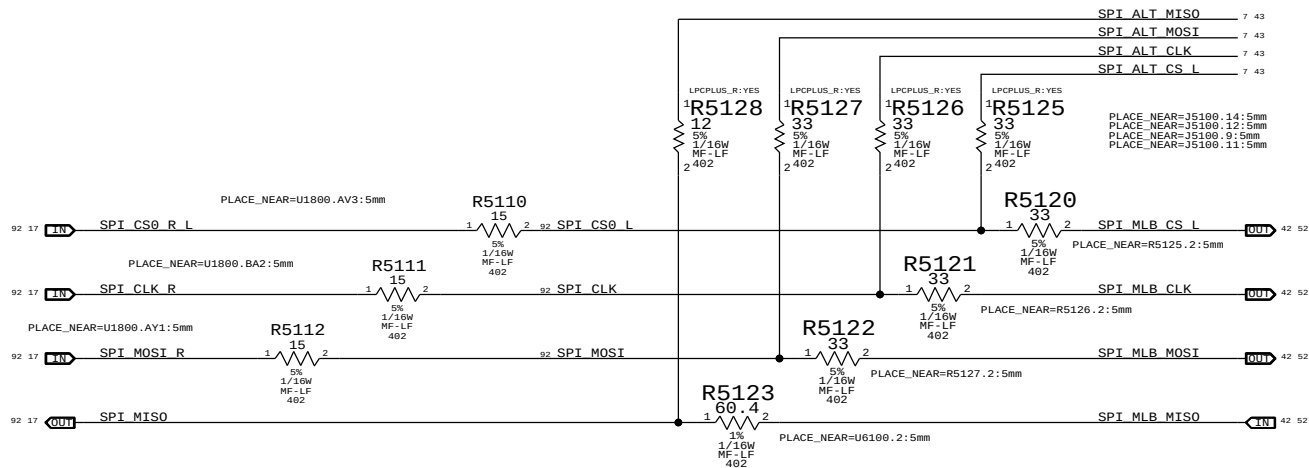
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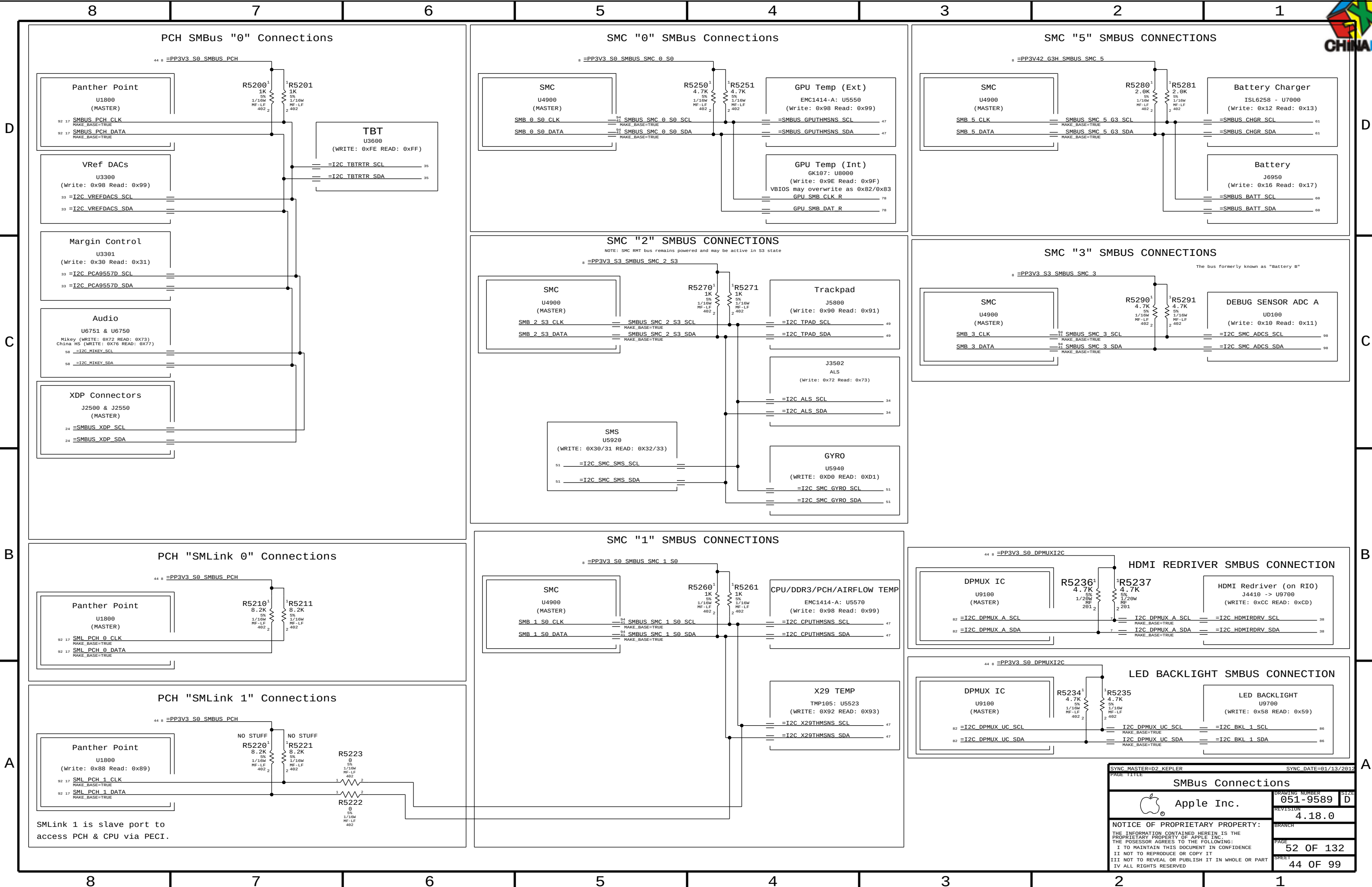
42 OF 99

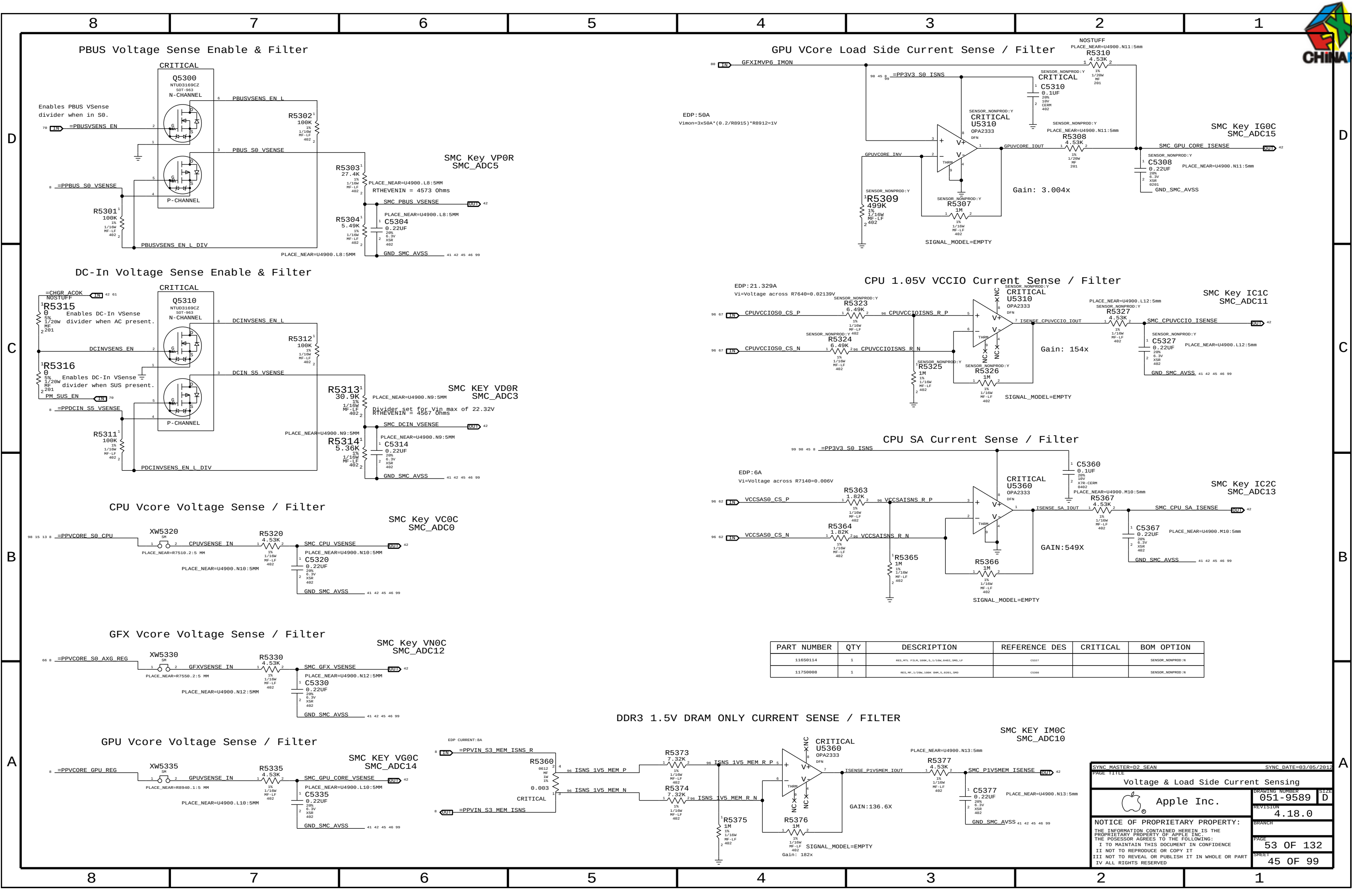
LPC+SPI Connector



SPI Bus Series Termination

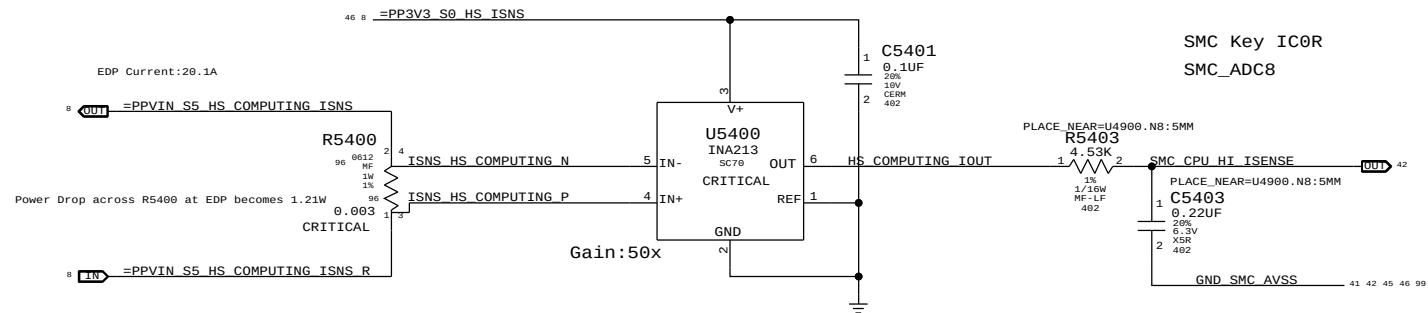




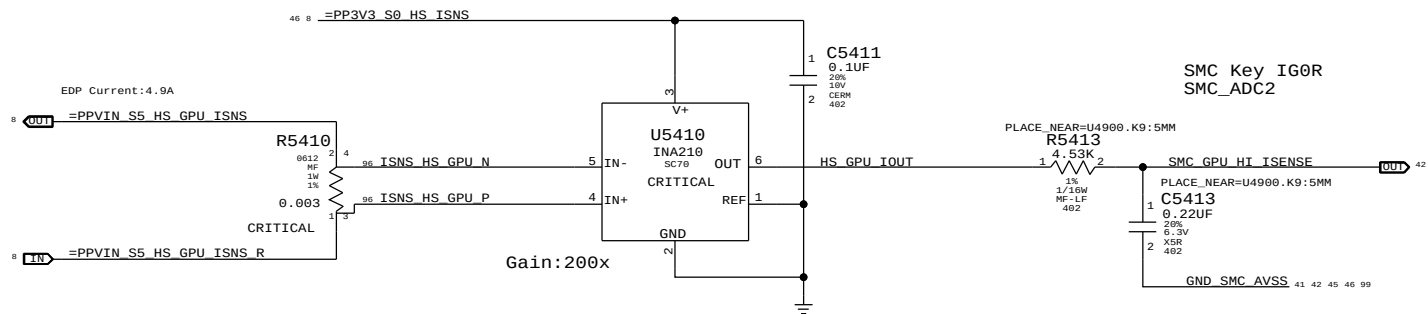




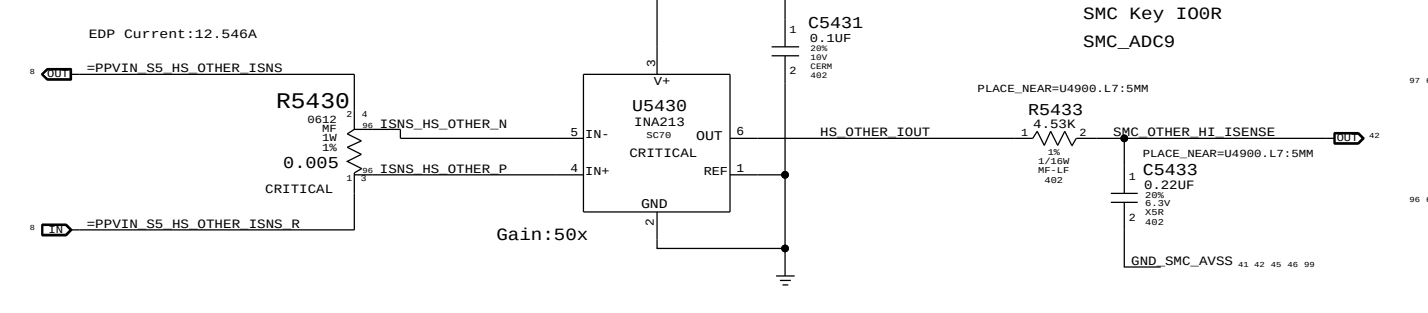
COMPUTING High Side Current Sense / Filter



GRAPHICS High Side Current Sense / Filter



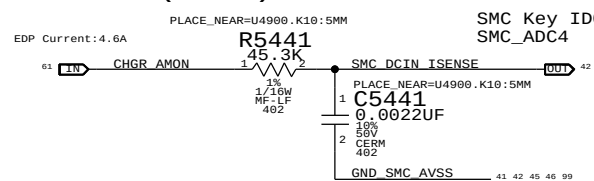
OTHER High Side Current Sense / Filter



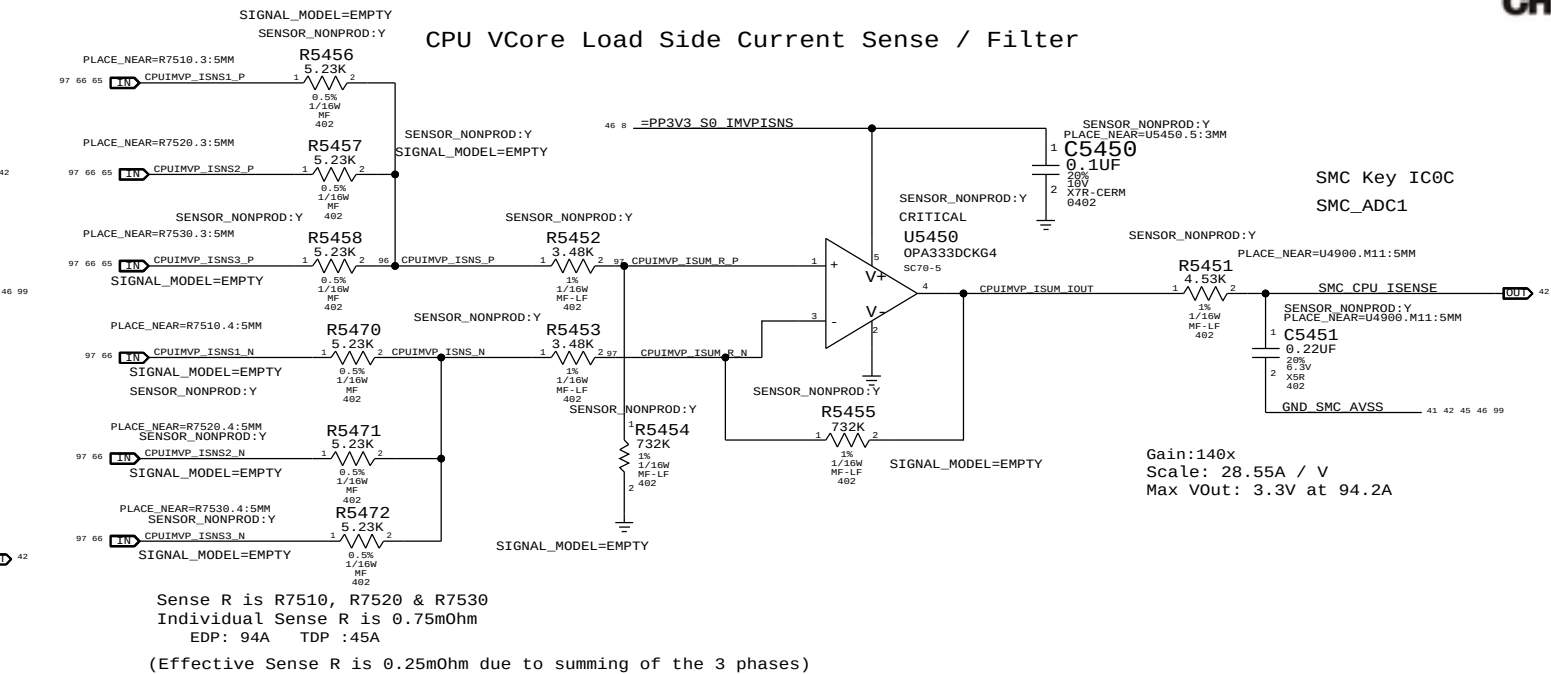
CHARGER BMON HIGH SIDE (BATTERY DISCHARGE) CURRENT SENSE & FILTER



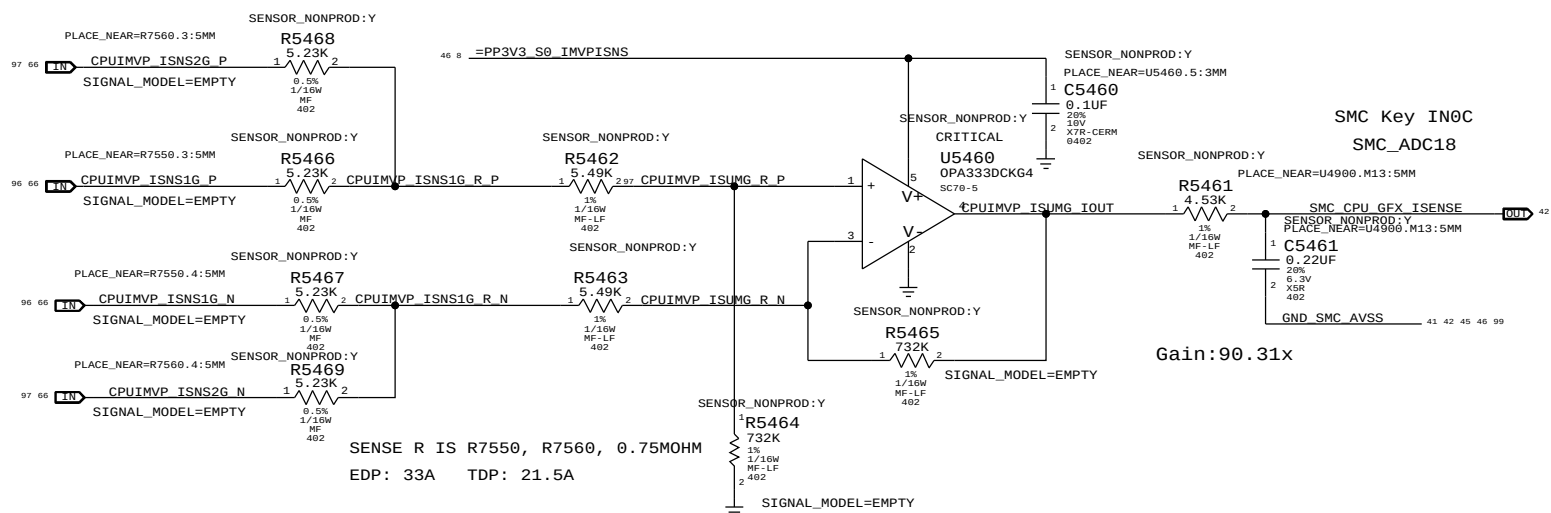
DC-IN (AMON) Current Sense Filter



CPU VCore Load Side Current Sense / Filter

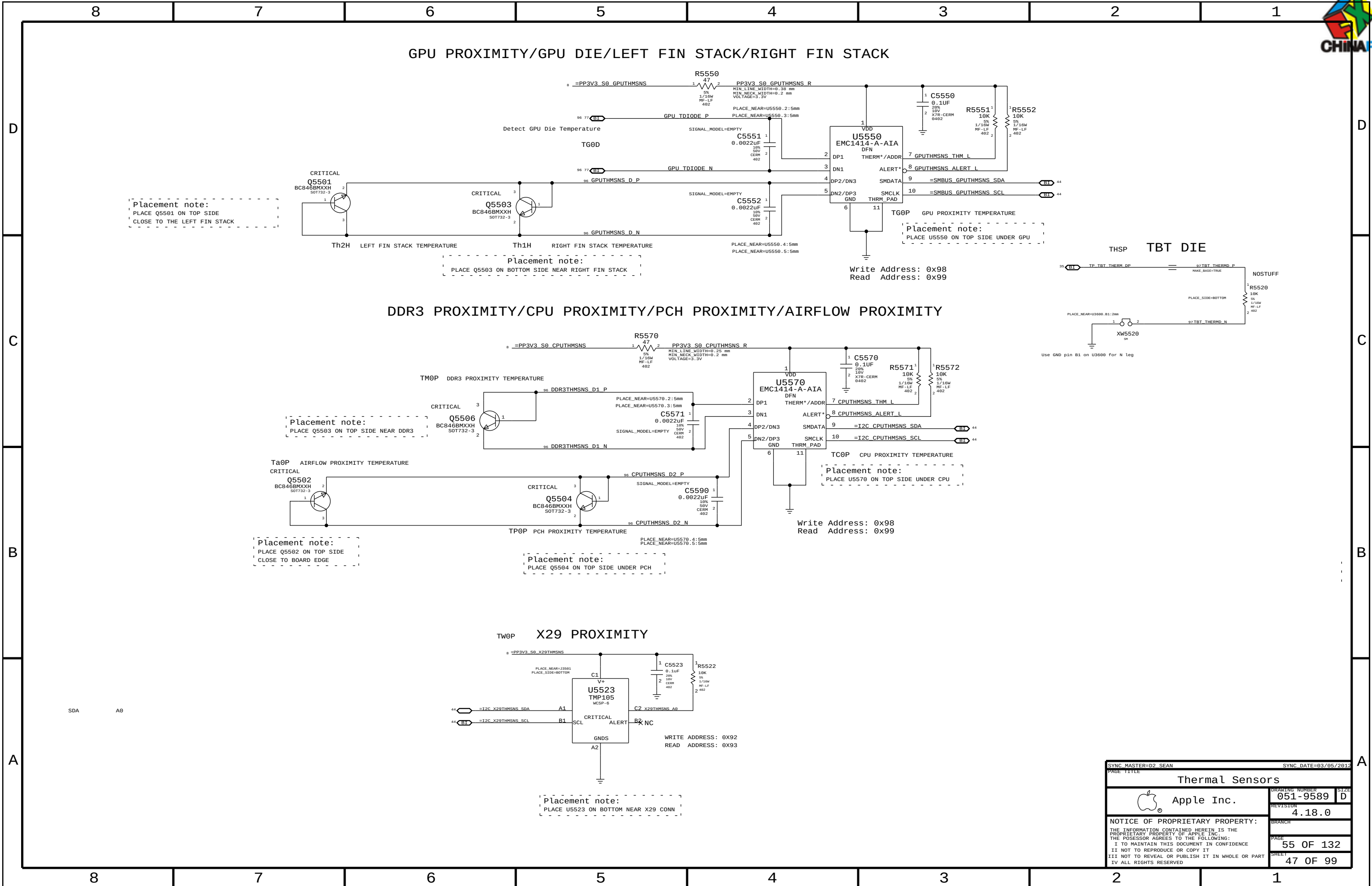


GFX/IG VCore Load Side Current Sense / Filter



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
11659114	2	RES,MTL, F,10K, 5%, 1/16W, 0402, SMD, LF	C5451, C5461		SENSOR_NONPROD:N

SYNC MASTER=D2 SEAN		SYNC DATE=03/05/2012	
PAGE TITLE		High Side and CPU/AXG Current Sensing	
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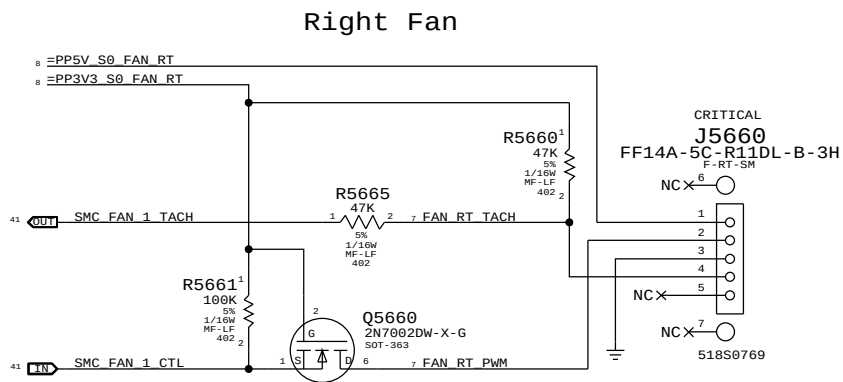
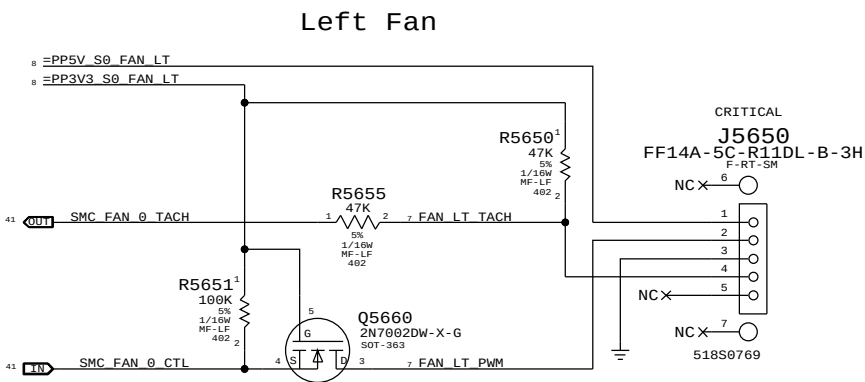
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SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE		Fan Connectors	
		DRAWING NUMBER	051-9589
		REVISION	4.18.0
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		PAGE	56 OF 132
		SHEET	48 OF 99



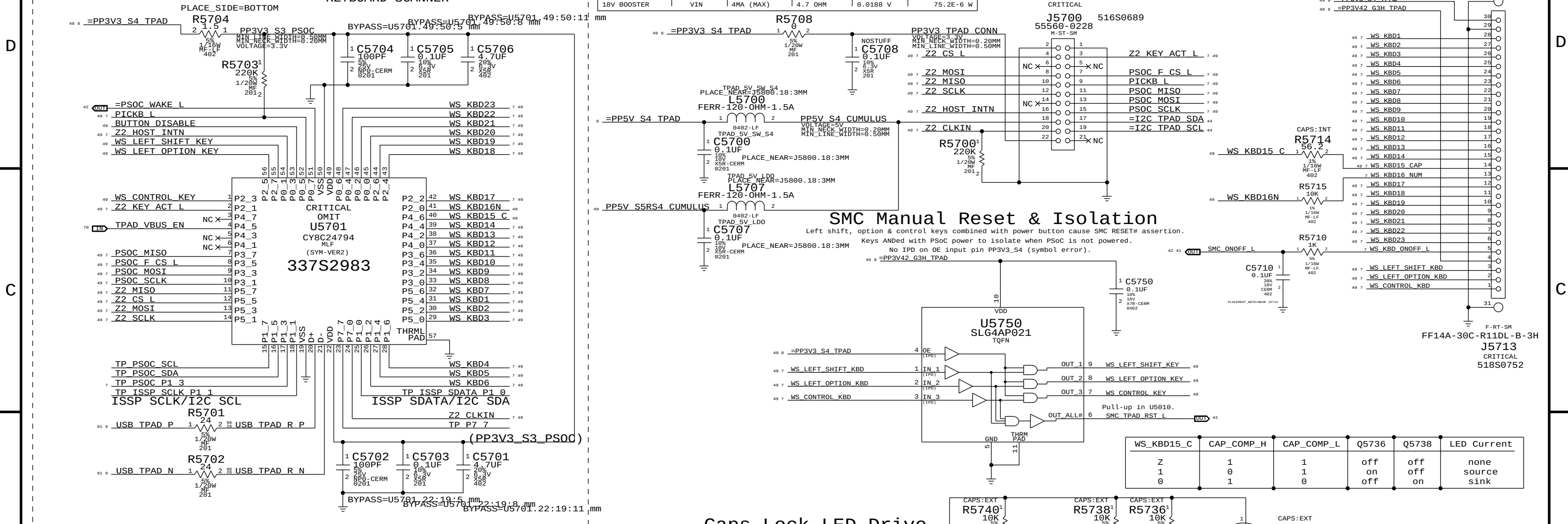
PSOC USB CONTROLLER

- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER

IC	PIN	NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA		2.55 KOHM	0.0255 V	0.255E-6 W
		80UA			0.204 V	16.32E-6 W
3V3 LDO	VDD	60MA (MAX)	10 OHM		0.6 V	36E-3 W
	VOUT	60MA (MAX)	0.2 OHM		0.012 V	0.72E-3 W
PSOC	VDD	8MA (TYP)	1.5 OHM		0.012 V	96E-6 W
		14MA (MAX)			0.021 V	294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM		0.0188 V	75.2E-6 W

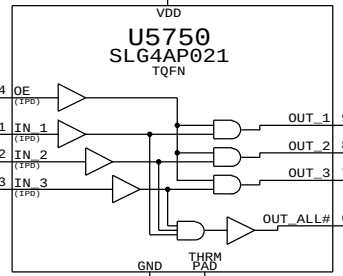
IPD Flex Connector

Keyboard Connector



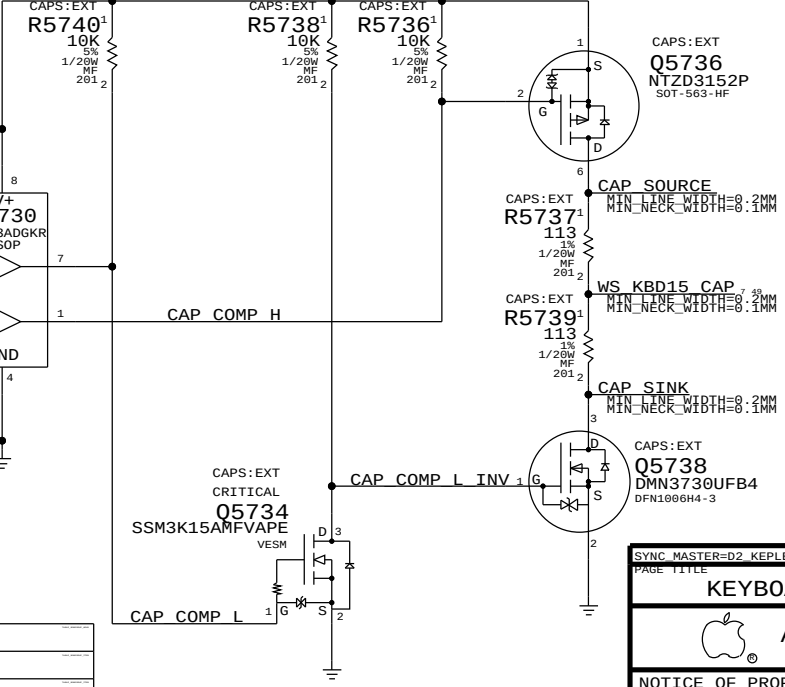
SMC Manual Reset & Isolation

Left shift, option & control keys combined with power button cause SMC RESET# assertion.
Keys ANDed with PSoC power to isolate when PSoC is not powered.
No IPD on OE input pin PP3V3_S4 (symbol error).



WS_KBD15_C	CAP_COMP_H	CAP_COMP_L	Q5736	Q5738	LED Current
Z	1	1	off	off	none
1	0	1	on	off	source
0	1	0	off	on	sink

Caps Lock LED Drive



All RC values are TBD

5V TRACKPAD S4 FET

MOSFET	SIA413
CHANNEL	P-TYPE 12V
RDS(ON)	29 mOhm @4.5V
LOADING	16 mA (EDP)

TPAD Buttons Disable

PLACE THESE COMPONENTS CLOSE TO J5800
THIS ASSUMES THERE'S A PP3V42_G3H PULL UP ON MLB

THE TPAD BUTTONS WILL BE DISABLE
WHEN THE LID IS CLOSED
LID OPEN => SMC_LID_LC ~ 3.42V
LID CLOSE => SMC_LID_LC < 0.50V

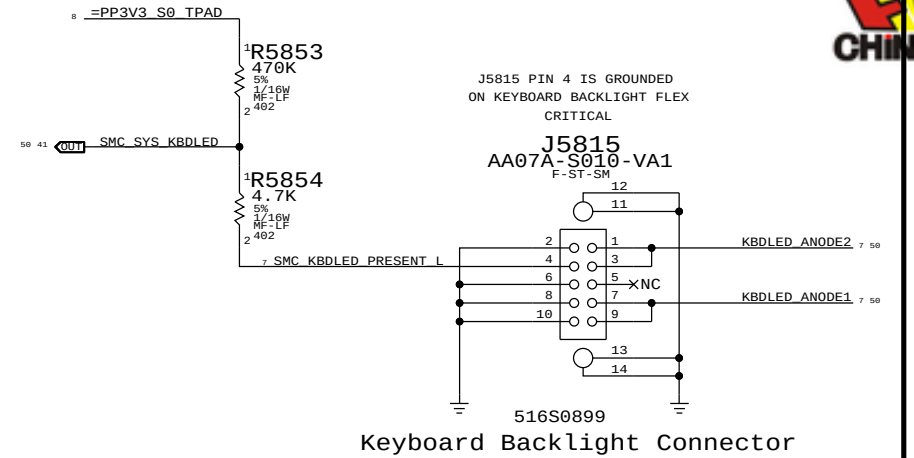
BOM Options available to CSA 5

TPAD_5V:SW_S4 Original implementation off PP5V_S4
TPAD_5V:LD0_S4 PP5V_S5 LDO power in S4 only
TPAD_5V:LD0_S5 PP5V_S5 LDO power

BOM GROUP	BOM OPTIONS
TPAD_5V:SW_S4	TPAD_5V_SW_S4
TPAD_5V:LD0_S4	TPAD_5V_FET, TPAD_5V_LDO
TPAD_5V:LD0_S5	TPAD_5V_NO_FET, TPAD_5V_LDO

SYNC MASTER=D2 KEPLER
PAGE TITLE
KEYBOARD/TRACKPAD (1 OF 2)
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REVISION
4.18.0
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PAGE
57 OF 132
SHEET
49 OF 99

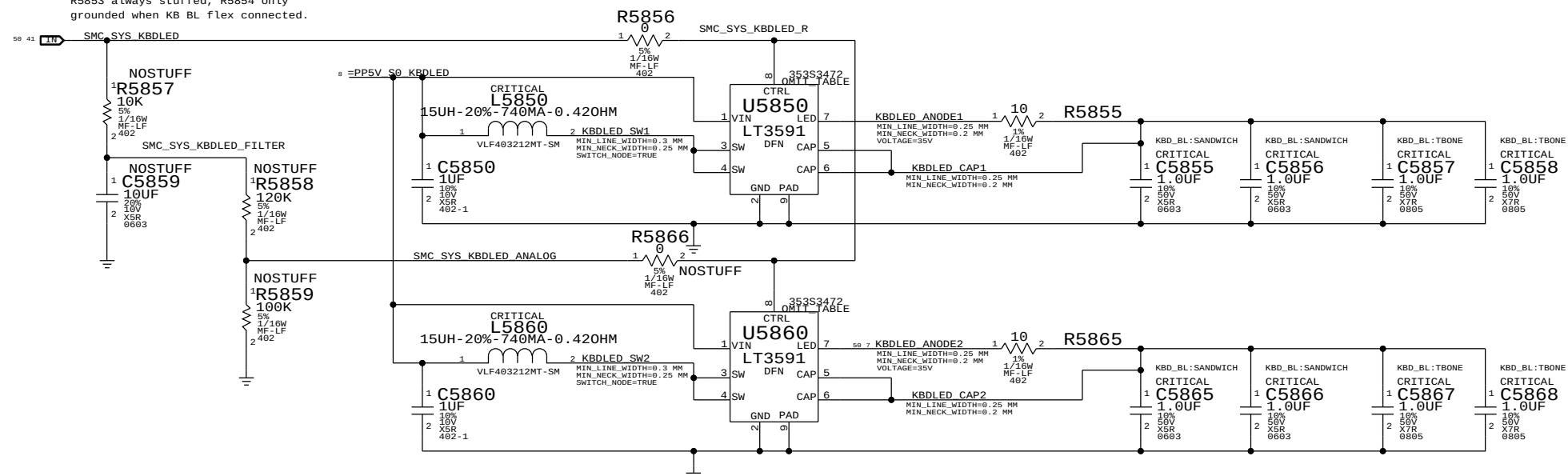


Keyboard Backlight Driver & Detection

To detect Keyboard backlight, SMC will tristate and read SMC_SYS_KBDLED:

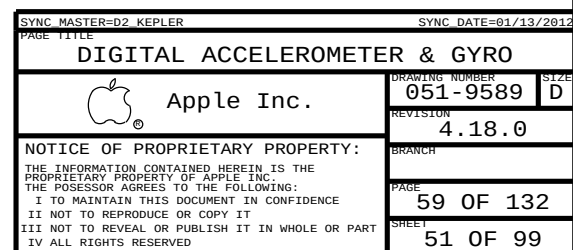
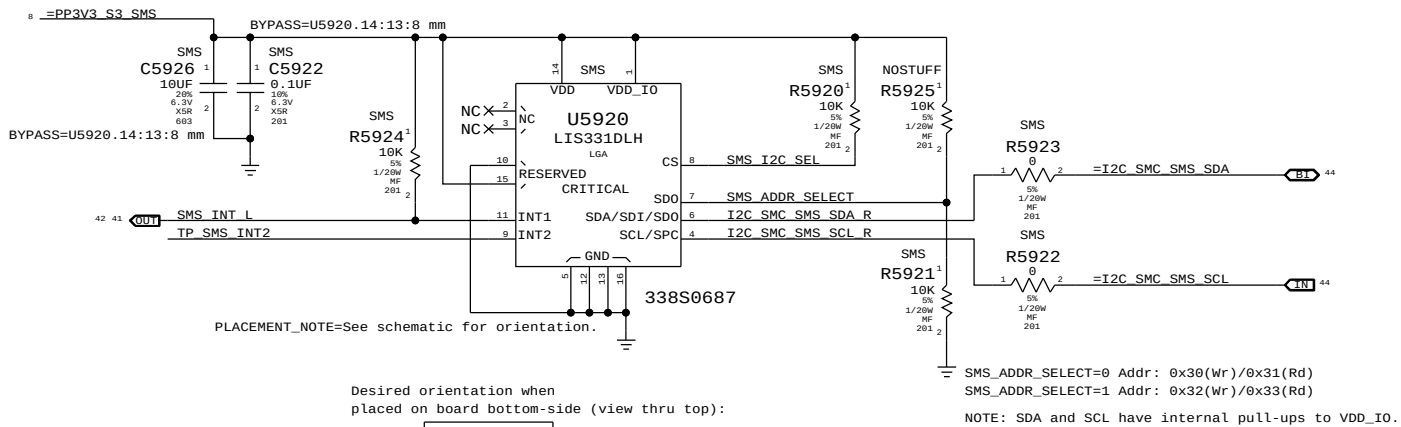
If LOW, keyboard backlight present
If HIGH, keyboard backlight not present

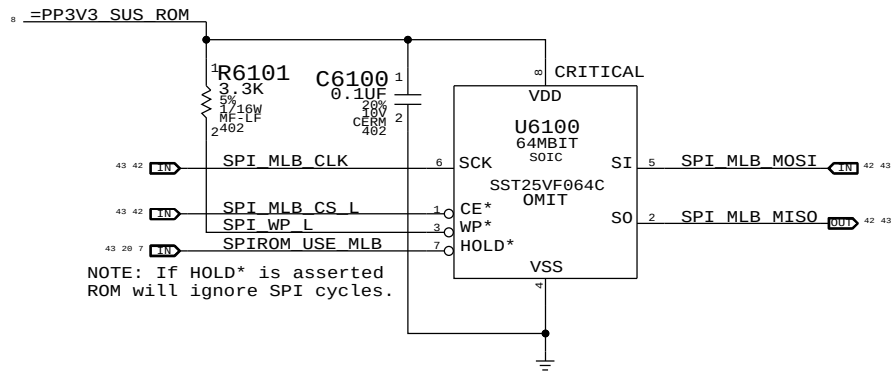
R5853 always stuffed, R5854 only grounded when KB BL flex connected.



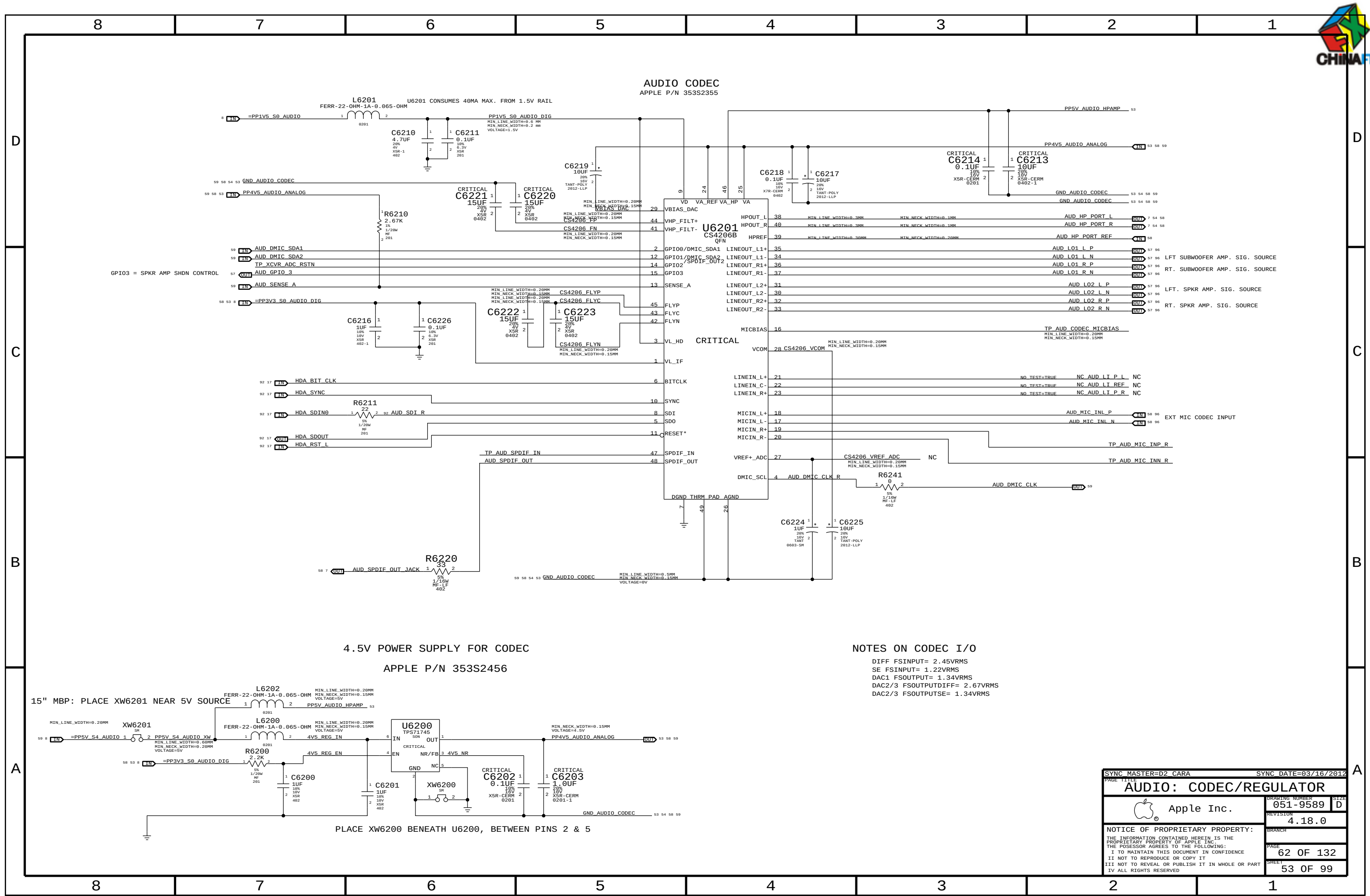
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S1612	2	IC, DC/DC CVTR, 800ST, WHITE LED, 3MHZ, 50NB	US850, US860	CRITICAL	

SYNC MASTER-D2 KEPLER	SYNC DATE=01/13/2012
PAGE TITLE	
KEYBOARD/TRACKPAD (2 OF 2)	
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	REVISION 4.18.0
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BRANCH	
PAGE	
58 OF 132	
SHEET	
50 OF 99	





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SPI ROM			
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		PAGE	61 OF 132
		SHEET	52 OF 99



4.5V POWER SUPPLY FOR CODEC
APPLE P/N 353S2456

NOTES ON CODEC I/O
DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

PAGE TITLE		SYNC DATE=03/16/2012	
AUDIO: CODEC/REGULATOR		DRAWING NUMBER	051-9589
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		PAGE	62 OF 132
		SHEET	53 OF 99



D

C

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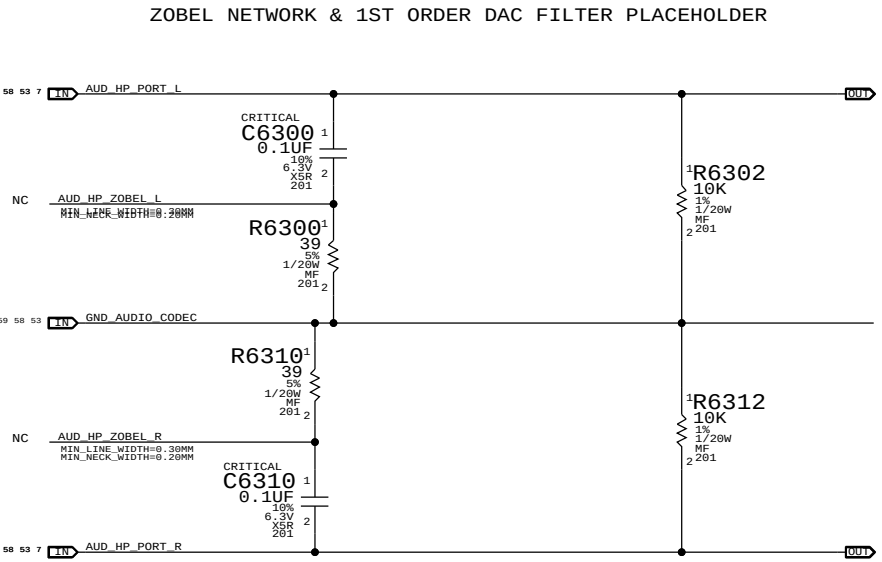
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SYNC MASTER=D2_CARA		SYNC DATE=03/16/2012	
PAGE TITLE		AUDIO: HEADPHONE FILTER	
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		PAGE	63 OF 132
		SHEET	54 OF 99



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

SYNC MASTER=D2_CARA

SYNC DATE=03/16/2012

PAGE TITLE		AUDIO: IV SENSE	
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		PAGE	64 OF 132
		SHEET	55 OF 99

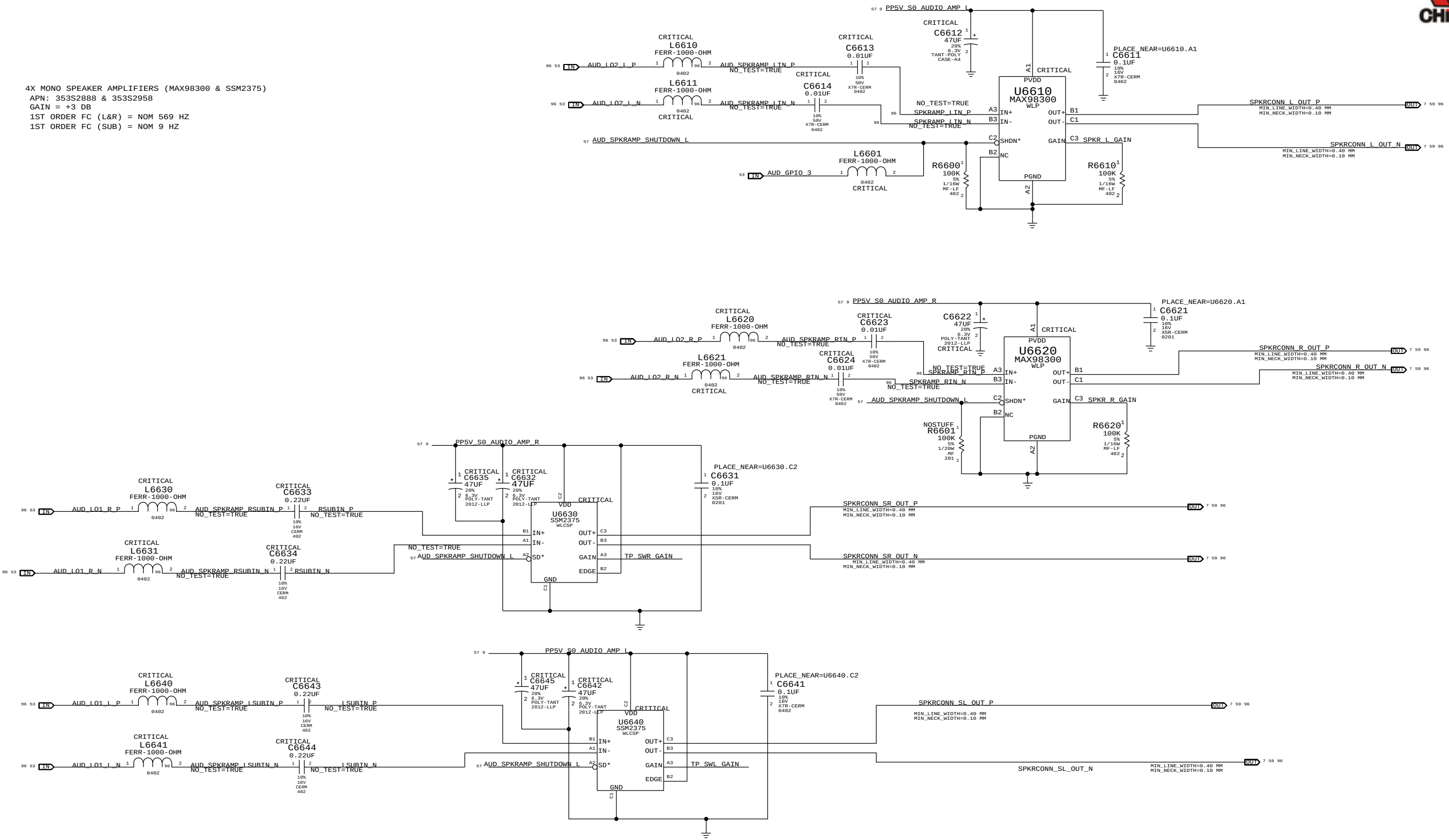


8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

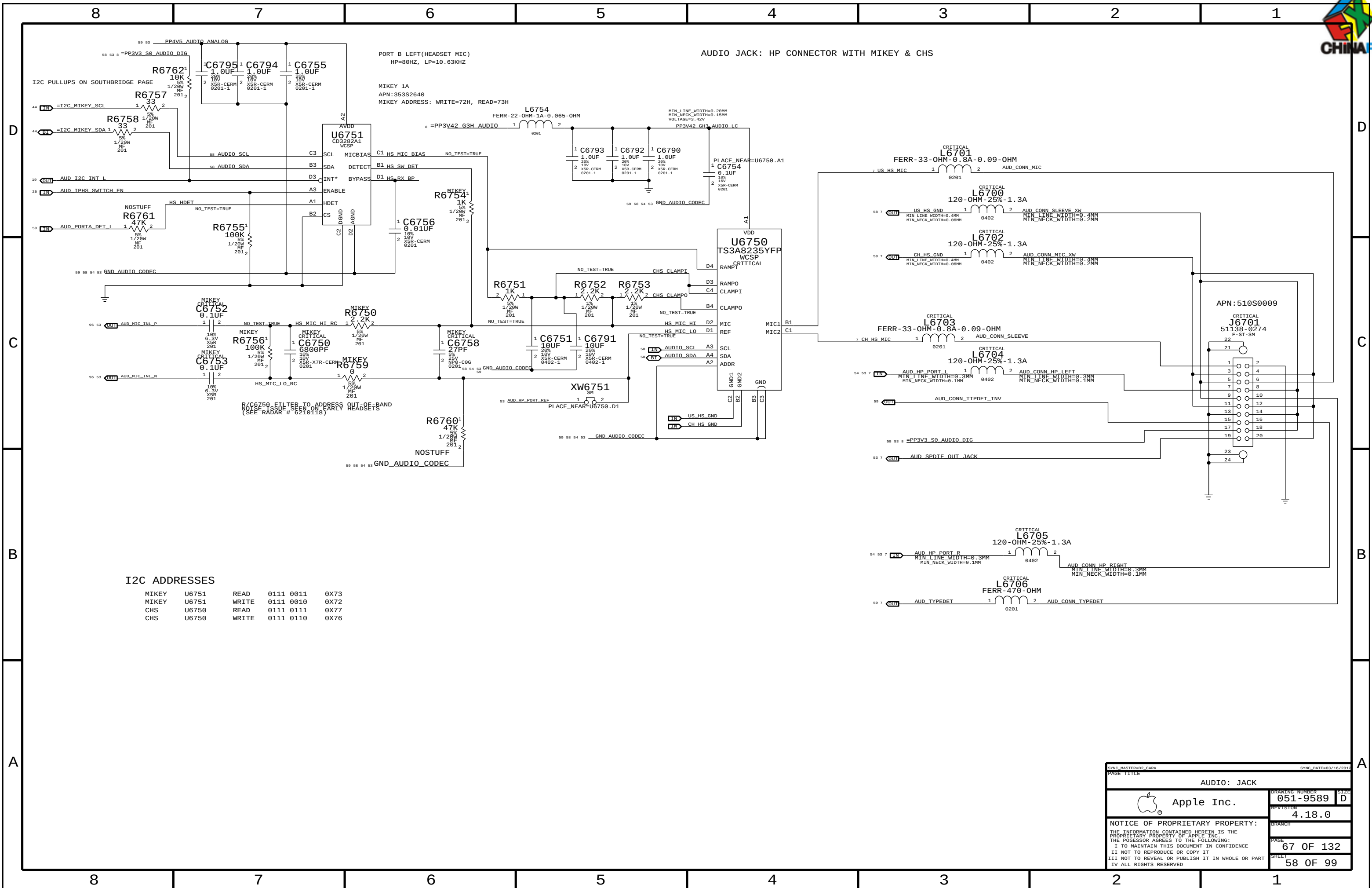
SYNC MASTER=D2 CARA		SYNC DATE=03/16/2012	
PAGE TITLE			
AUDIO: IV SENSE FILTER			
 Apple Inc.	DRAWING NUMBER	051-9589	SIZE D
	REVISION	4.18.0	
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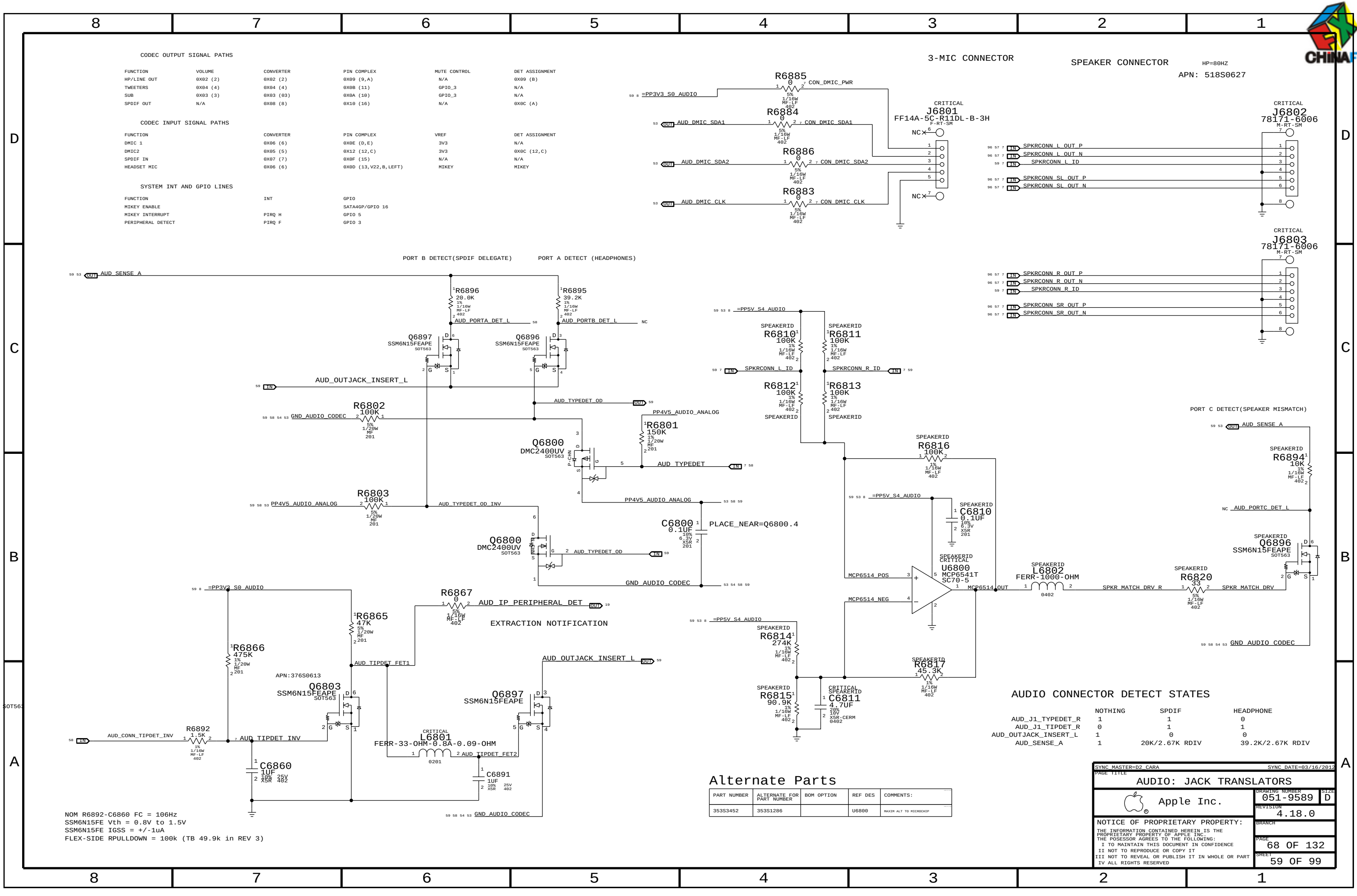


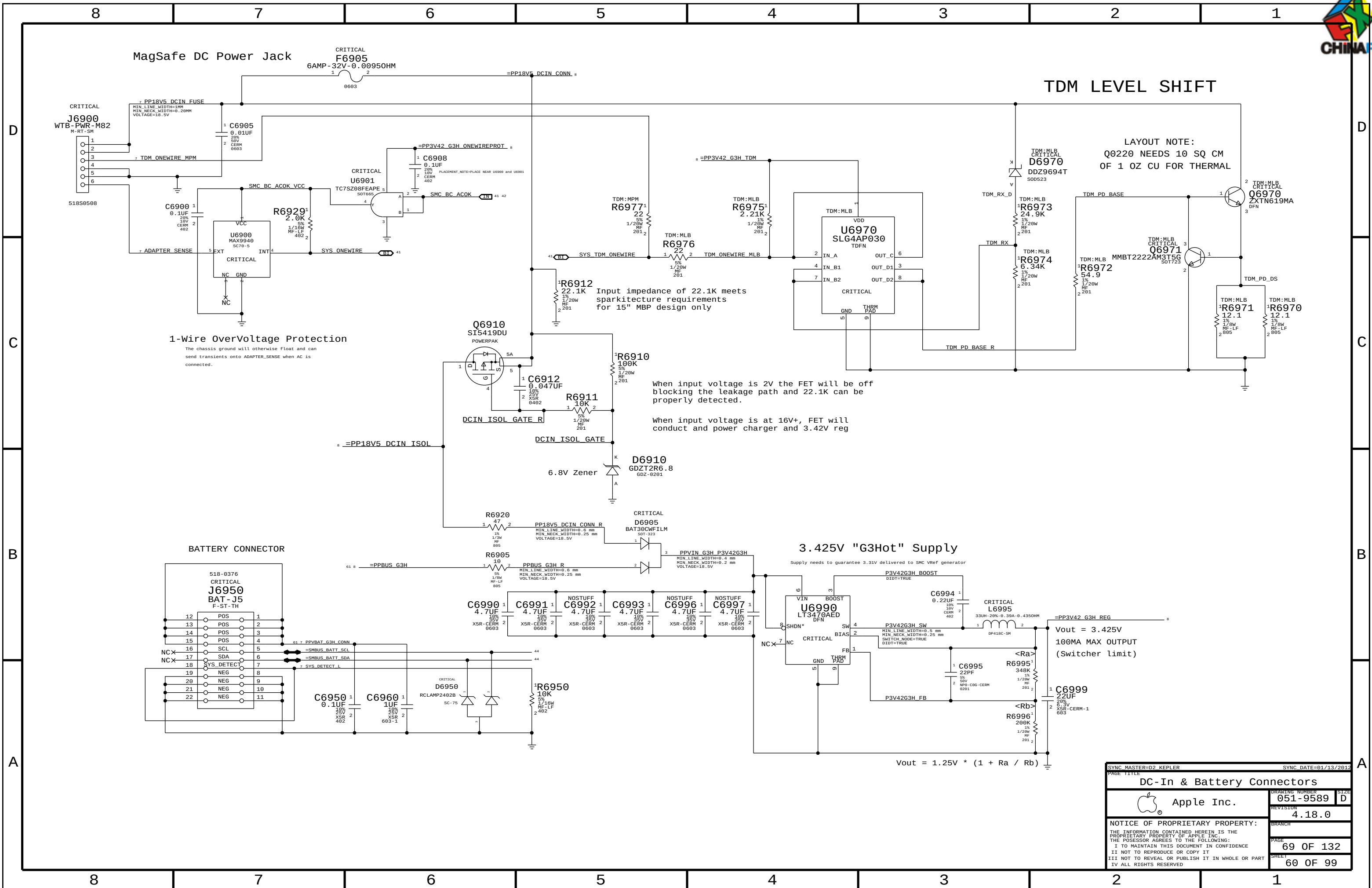
4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)
APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ



SYNC MASTER=D2 CARA		SYNC DATE=83/16/2812	
PAGE TITLE			
AUDIO: SPEAKER AMP			
 Apple Inc.		DRAWING NUMBER	051-9589
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NOTICE OF PROPRIETARY PROPERTY:		PAGE	66 OF 132
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MagSafe DC Power Jack

1-Wire OverVoltage Protection

The chassis ground will otherwise float and can send transients onto ADAPTER_SENSE when AC is connected.

Input impedance of 22.1K meets
sparkitector requirements
for 15" MBP design only

When input voltage is 2V the FET will be off blocking the leakage path and 22.1K can be properly detected.

When input voltage is at 16V+, FET will conduct and power charger and 3.42V reg

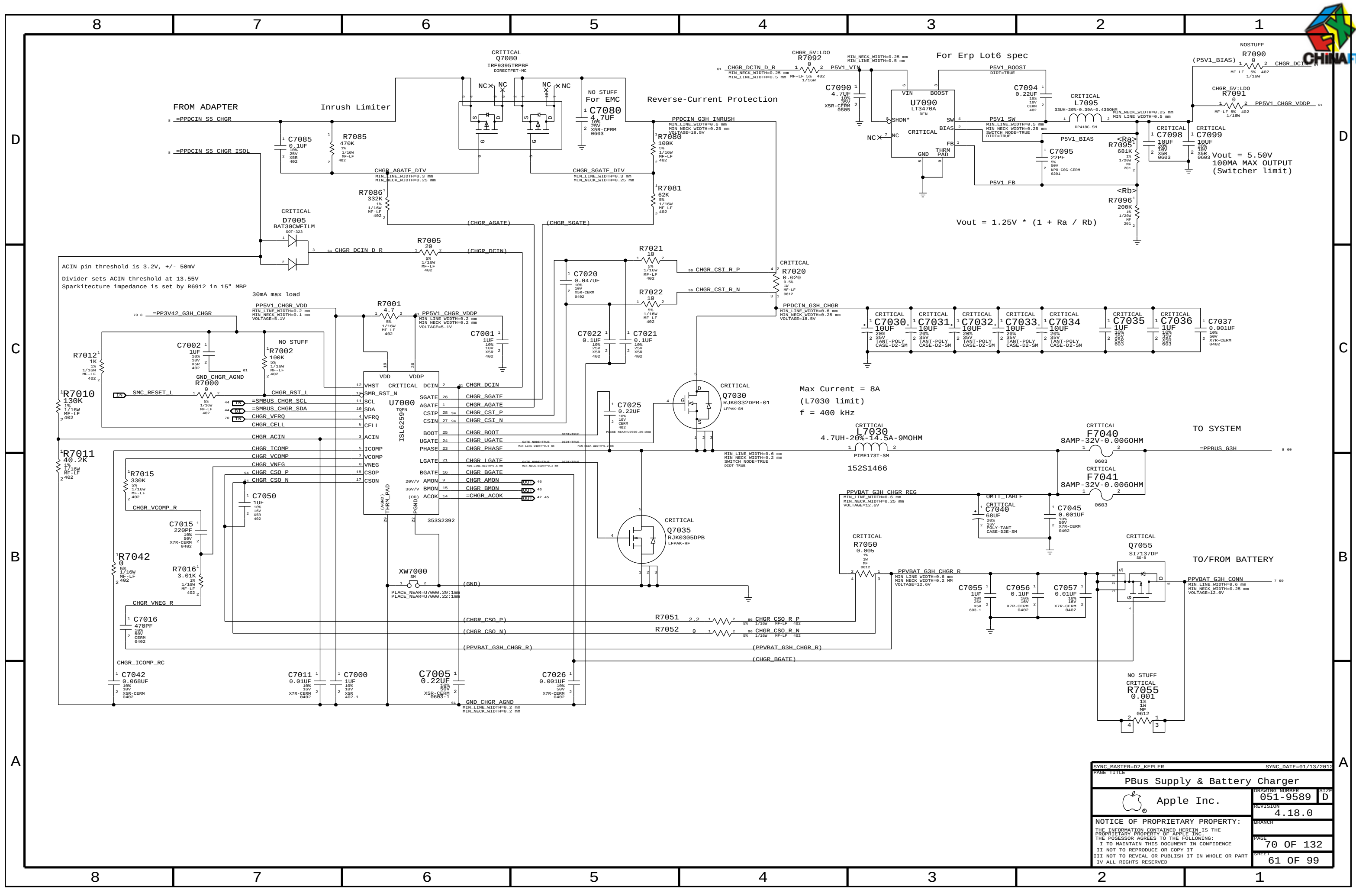
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

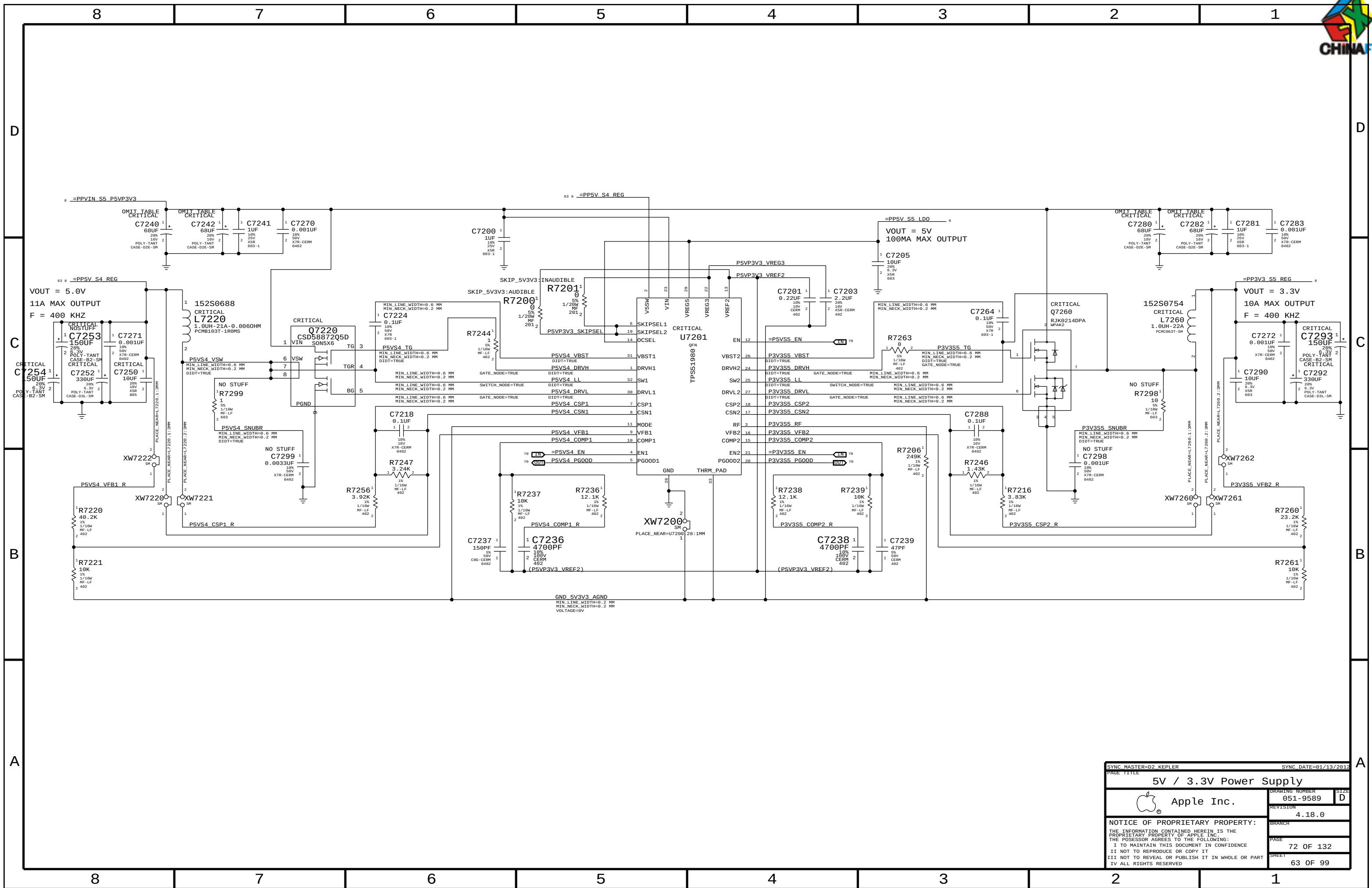
Vout = 3.425V
100MA MAX OUTPUT
(Switcher limit)

$$V_{out} = 1.25V * (1 + R_a / R_b)$$

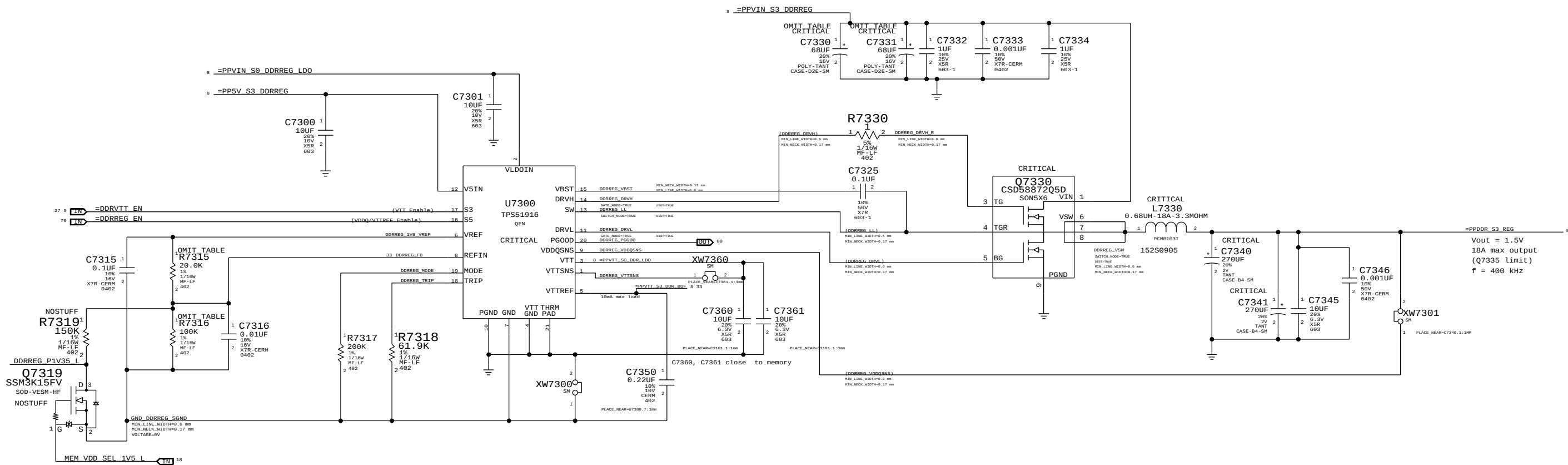
SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
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DC-In & Battery Connectors			
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		69 OF 132	
		SHEET	
		60 OF 99	



SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE			
PBus Supply & Battery Charger			
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		SHEET	61 OF 99



DDR3 (1V5R1V35 S3) REGULATOR



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0343	1	RES, HTL FILM, 1/10W, 20.0K, 1, 0402, SMD, LF	R7315		PPDDR:1V5
114S0342	1	RES, HTL FILM, 1/10W, 20.0K, 1, 0402, SMD, LF	R7315		PPDDR:1V35
114S0411	1	RES, HTL FILM, 1/10W, 100K, 1, 0402, SMD, LF	R7316		PPDDR:1V5
114S0389	1	RES, HTL FILM, 1/10W, 57.0K, 1, 0402, SMD, LF	R7316		PPDDR:1V35



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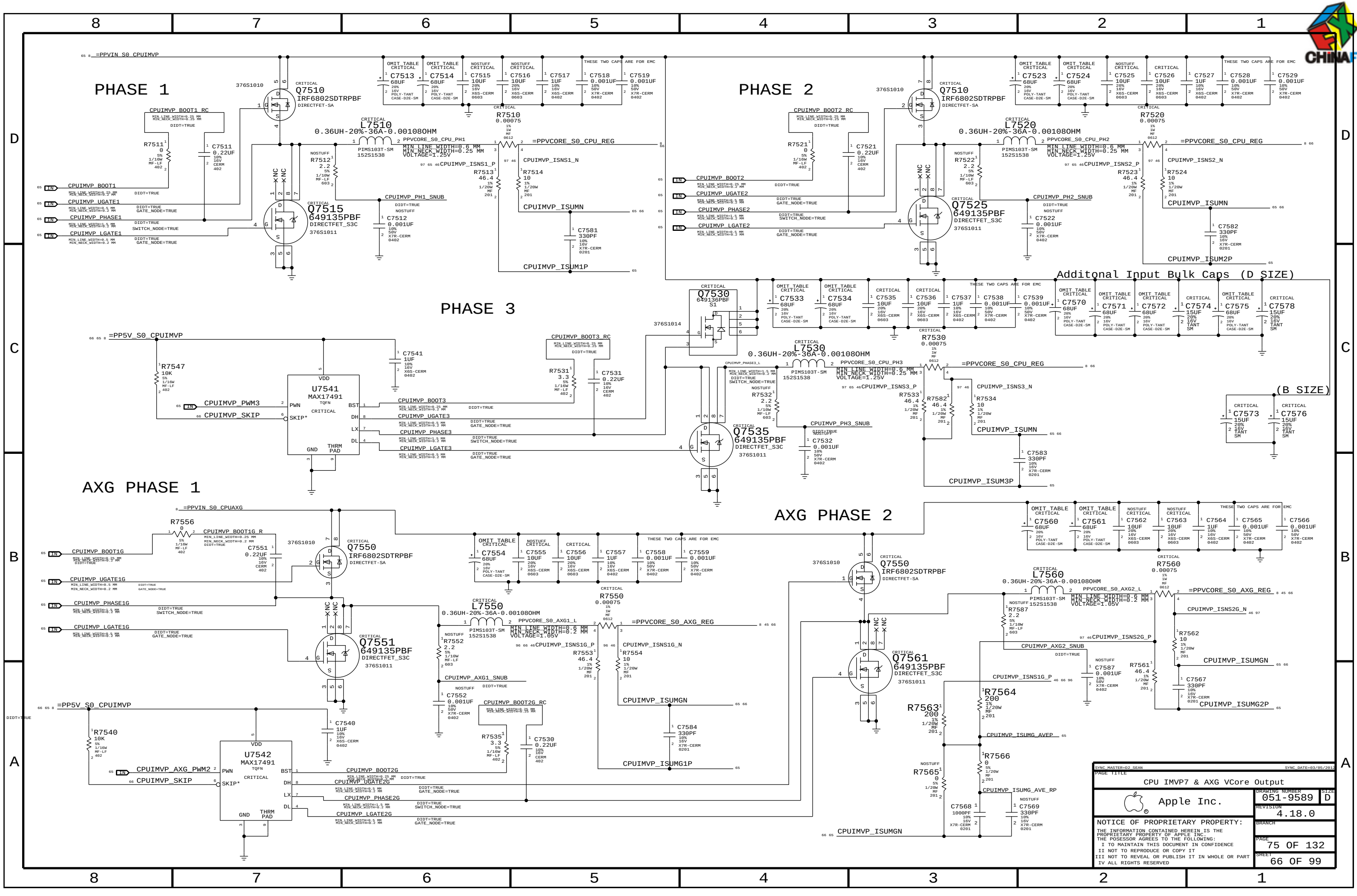


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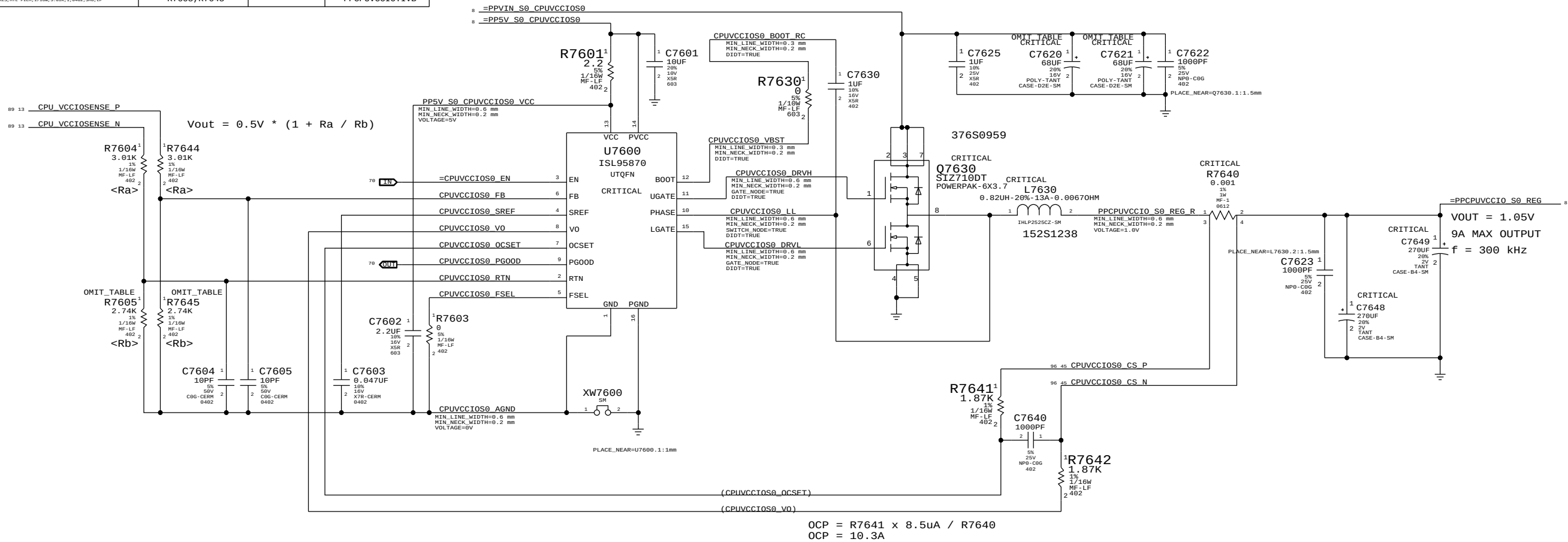
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SYNC MASTER=D2 SEAN		SYNC DATE=03/09/2012	
PAGE TITLE			
CPU IMPV7 & AXG VCore Regulator			
	Apple Inc.		DRAWING NUMBER 051-9589
			SIZE D
		REVISION 4.18.0	
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		SHEET 65 OF 99	



CPU VCCIO (1V0R1V05 S0) REGULATOR

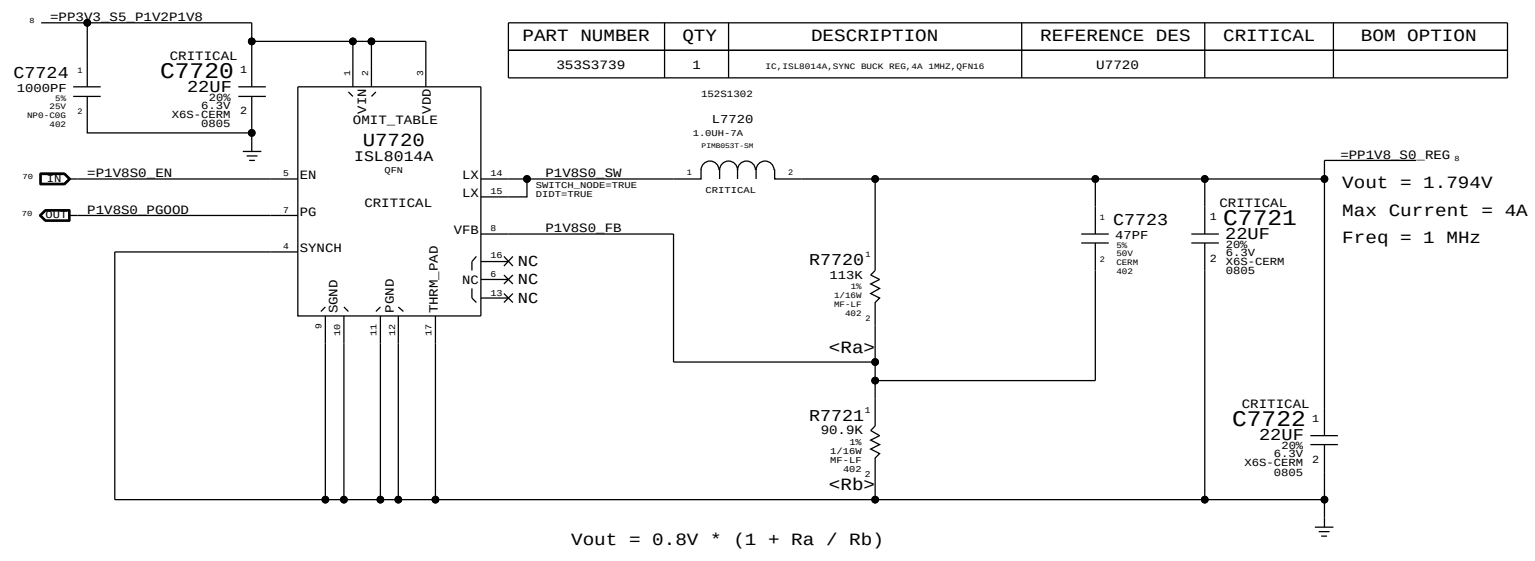
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0260	2	RES,MTL FILM,1/30W,2.74K,1,0402,SMD,LF	R7605,R7645		PPCPUVCCIO:SNB
114S0264	2	RES,MTL FILM,1/30W,3.03K,1,0402,SMD,LF	R7605,R7645		PPCPUVCCIO:IVB



CPU VCCIO (1V0R1V05 S0) POWER SUPPLY	
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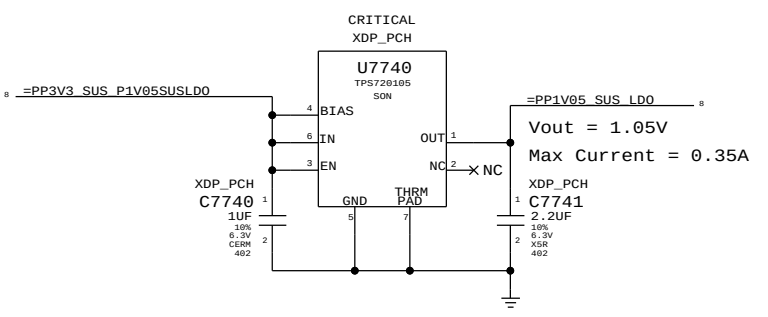


1.8V S0 Regulator

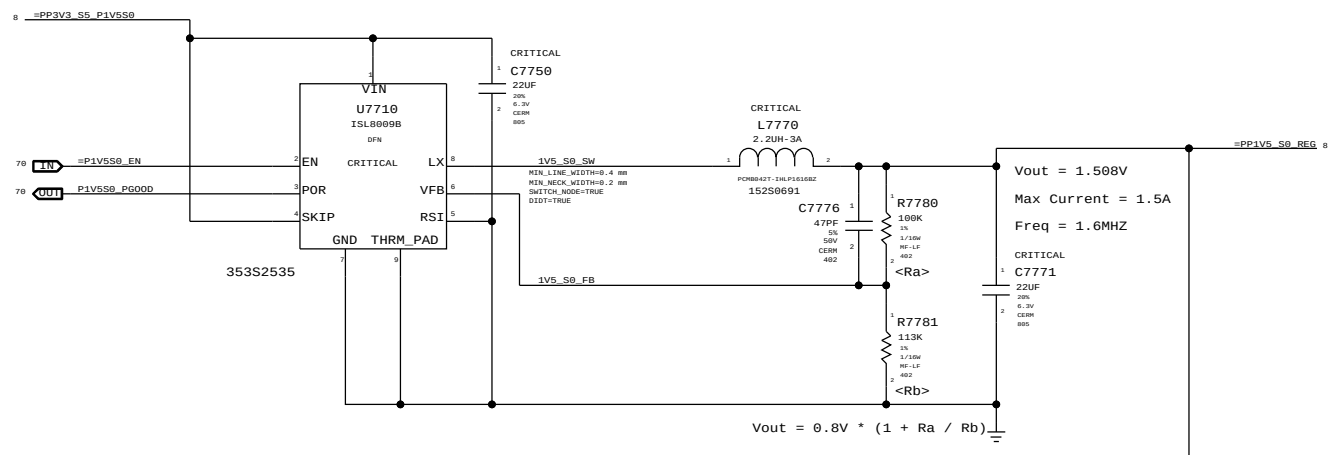


1.05V SUS LDO

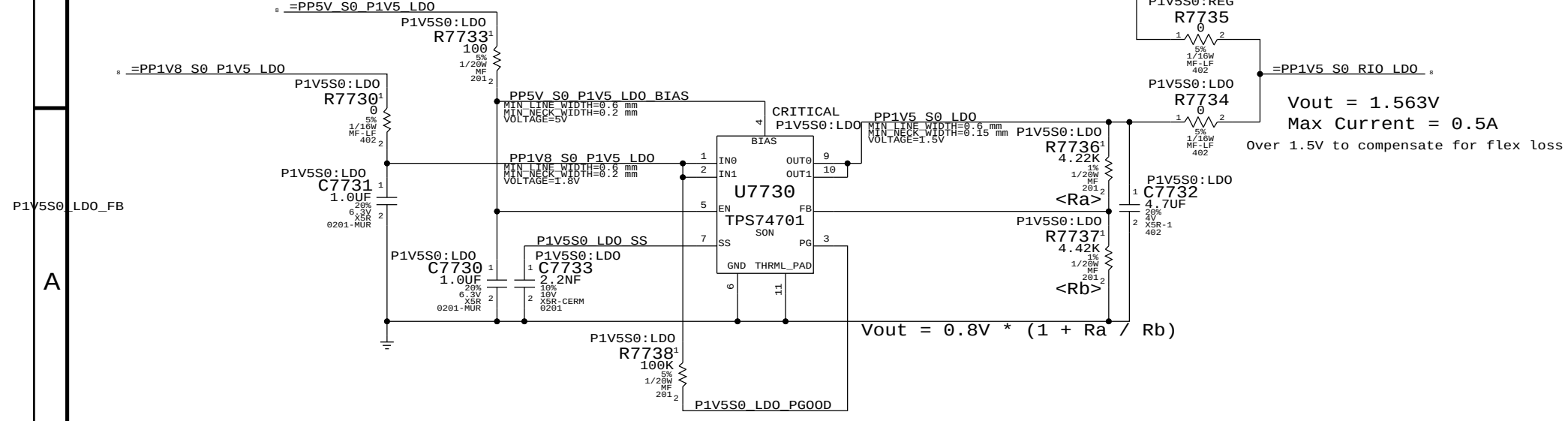
Panther Point-M requires JTAG pull-ups to be powered at 1.05V in Sus. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V Sus, which burns 100mW in all S-states.



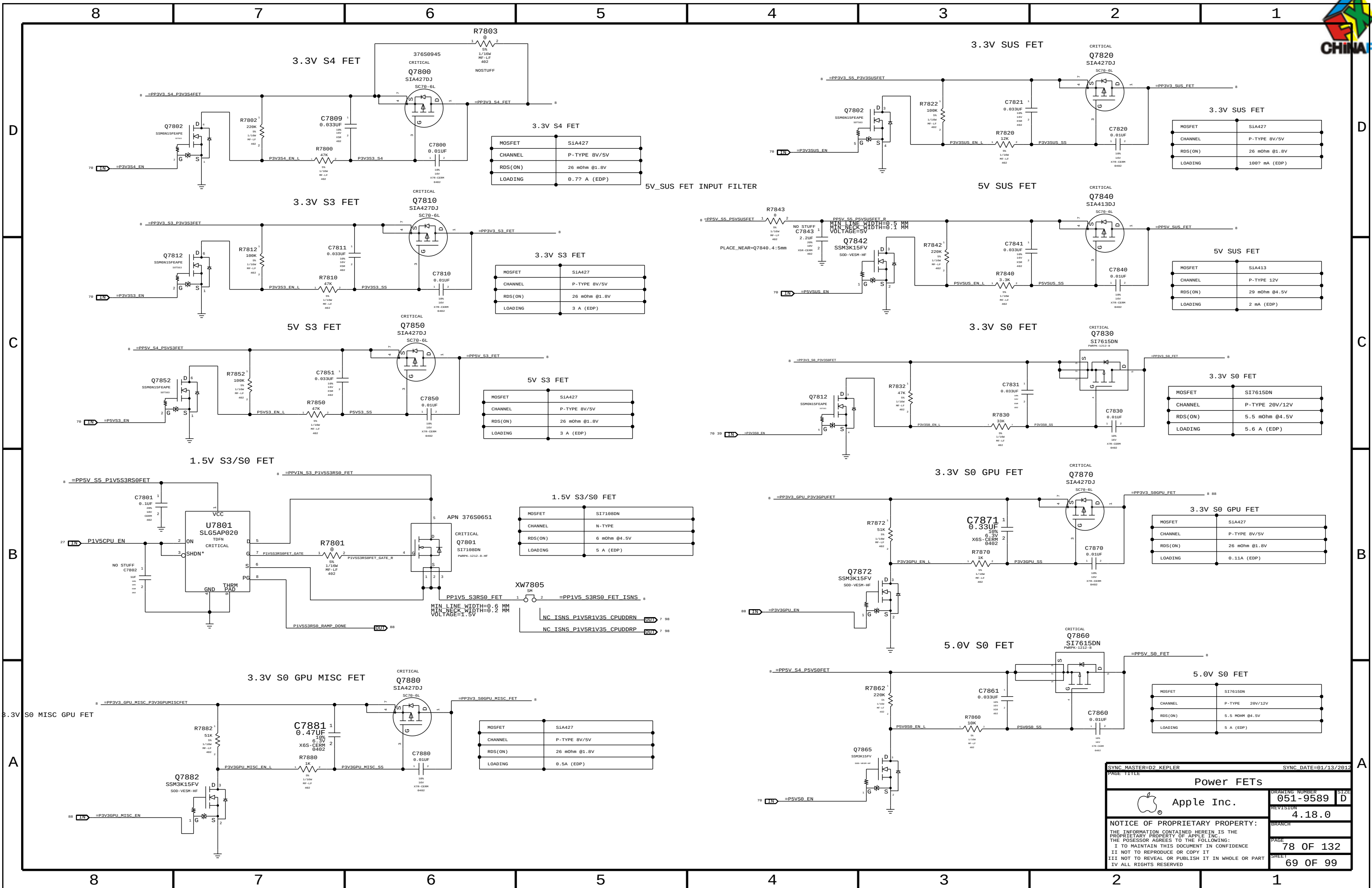
1.5V S0 Regulator

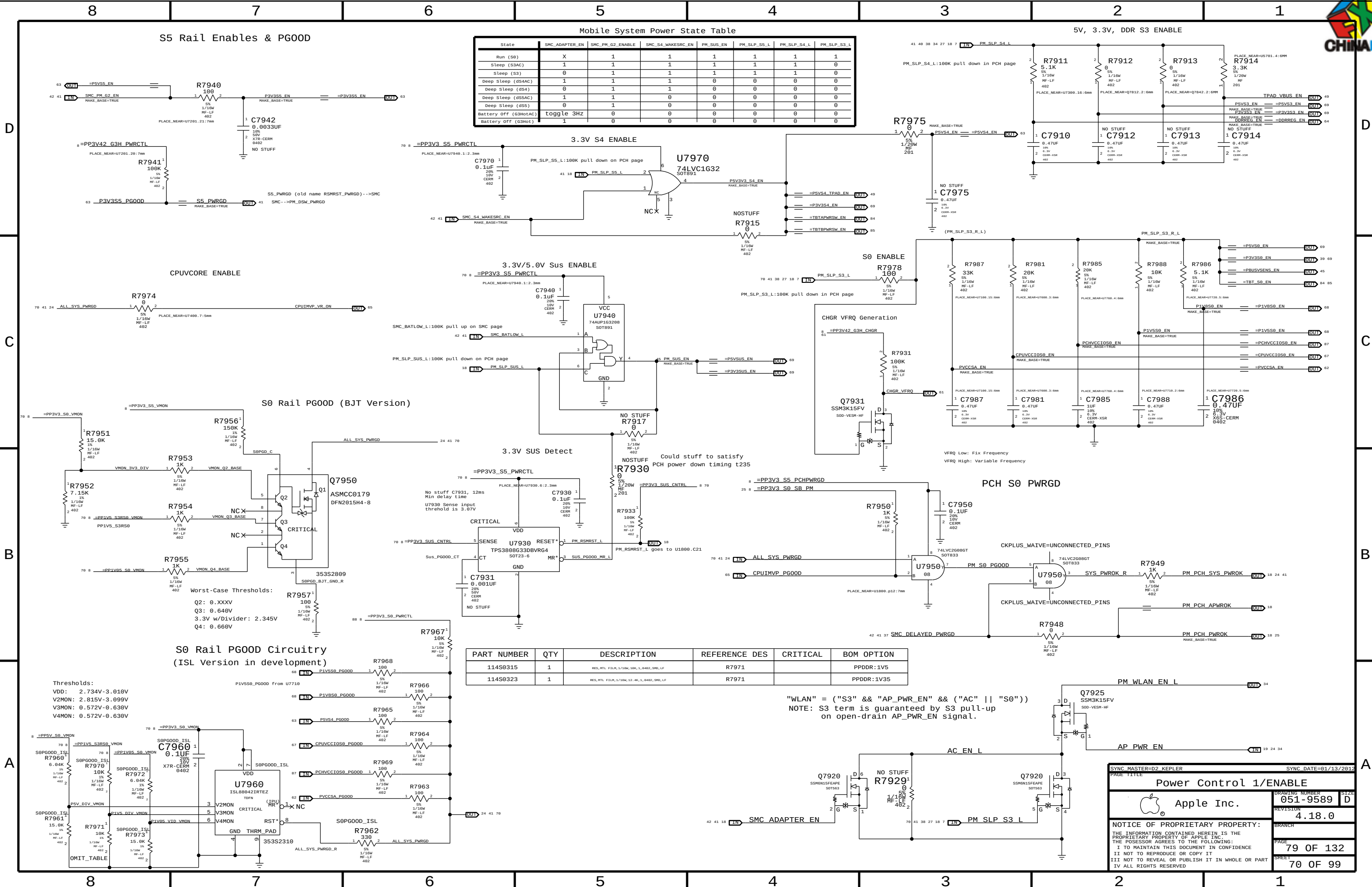


1.5V S0 LDO (RIO)

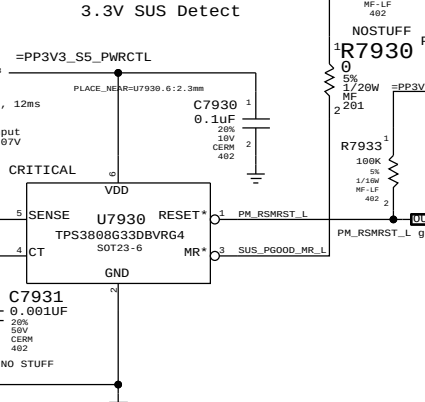
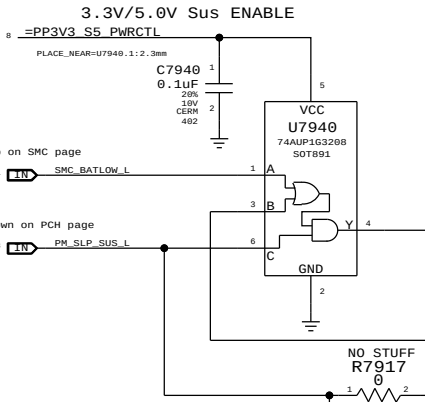
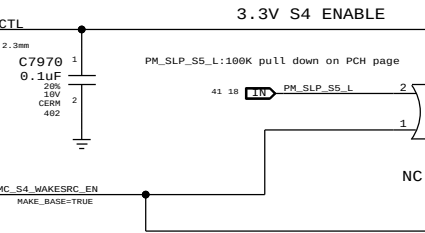


SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE			
Misc Power Supplies			
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		PAGE	77 OF 132
		SHEET	68 OF 99





State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SUS_EN	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Deep Sleep (dS4AC)	0	1	1	0	0	0	0
Deep Sleep (dS4)	0	1	1	0	0	0	0
Deep Sleep (dS5AC)	1	1	0	0	0	0	0
Deep Sleep (dS5)	1	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0315	1	RES, HTL, F15M, 1/10W, 10K, 1, 0402, SMD, LF	R7971		PPDDR:1V5
114S0323	1	RES, HTL, F15M, 1/10W, 12.4K, 1, 0402, SMD, LF	R7971		PPDDR:1V35

"WLAN" = ("S3" & "AP_PWR_EN" & ("AC" || "S0"))
NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.

SYNC MASTER=D2 KEPLER

SYNC DATE=01/13/2012

Power Control 1/ENABLE

Apple Inc.

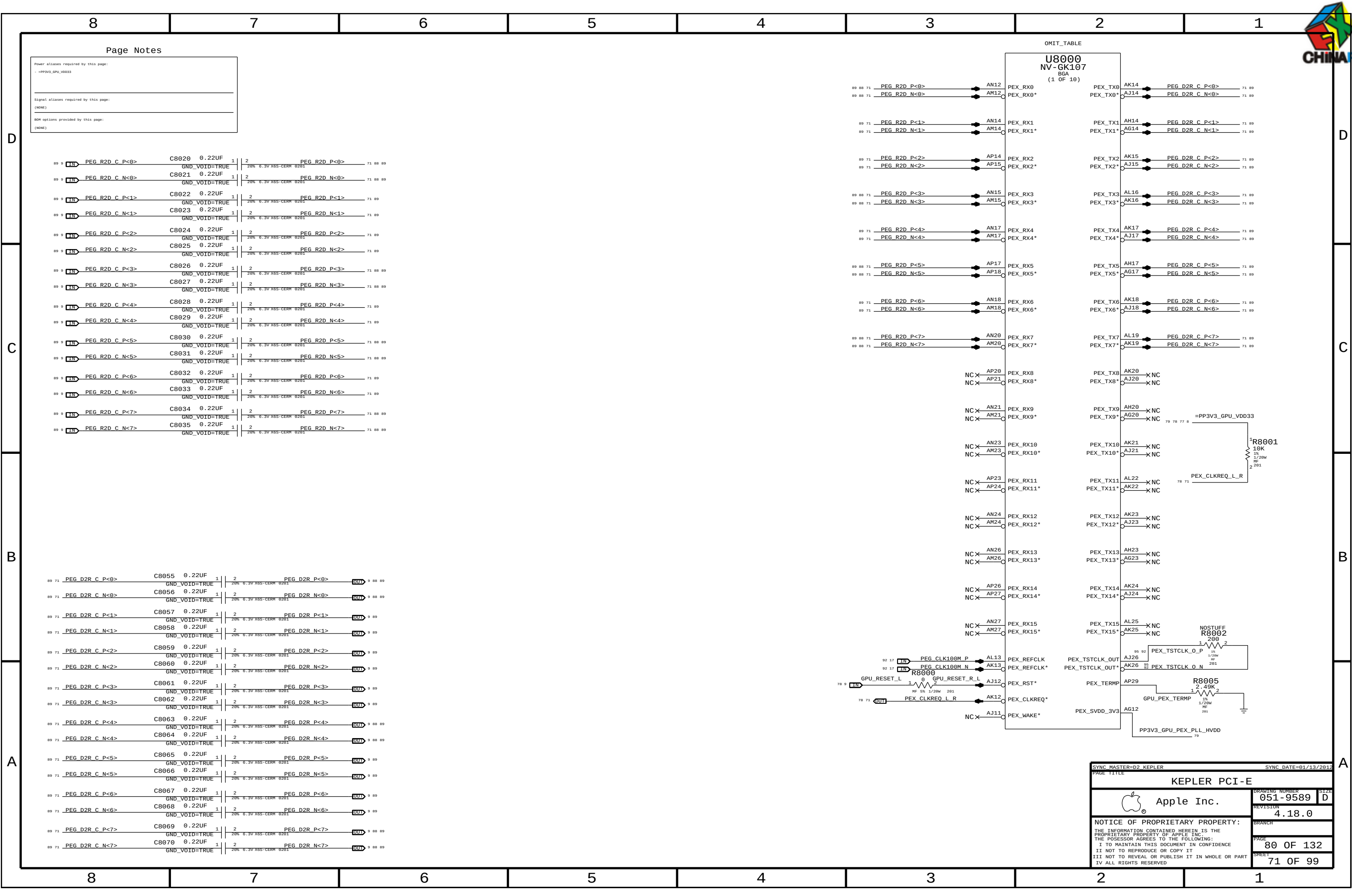
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051-9589

REVISION
4.18.0

PAGE
79 OF 132

SHEET
70 OF 99

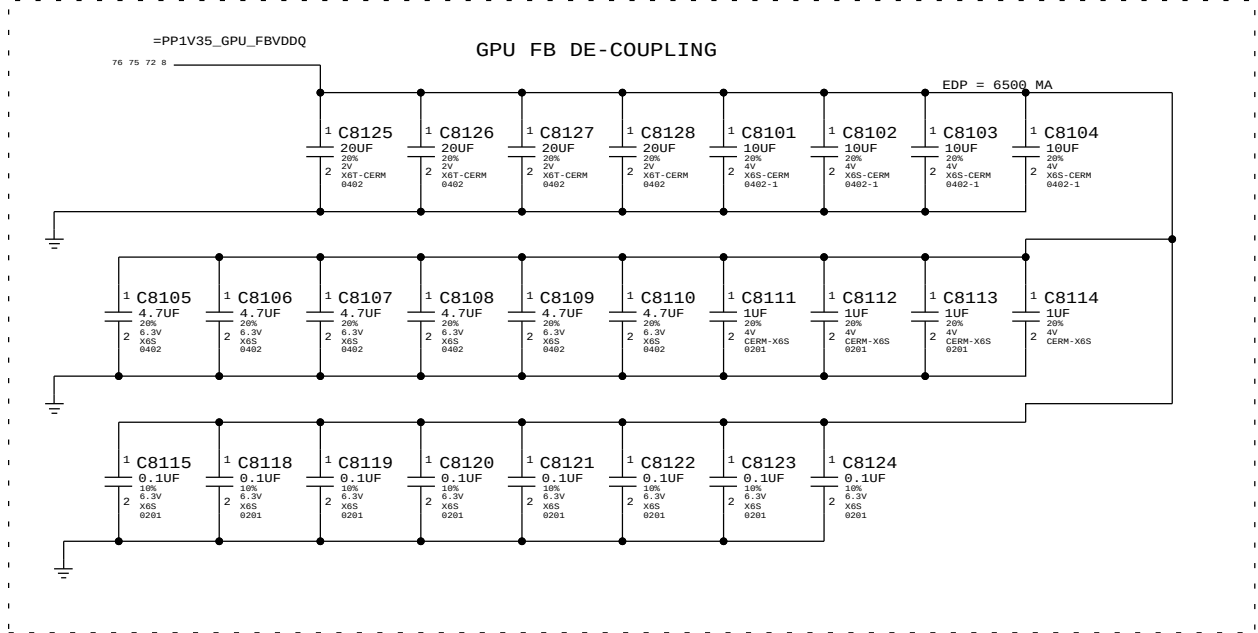
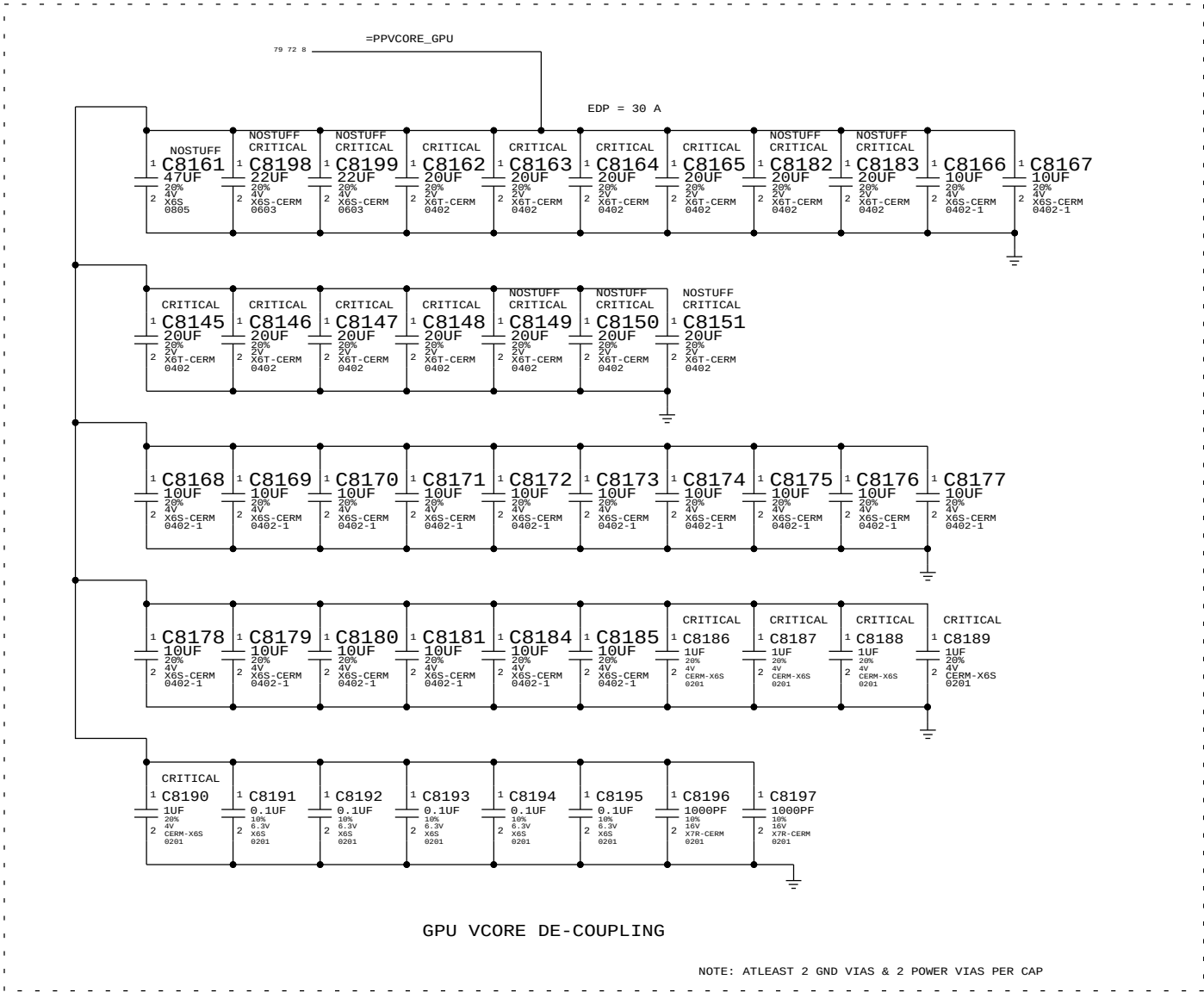




Power aliases required by this page:
- =PPVCORE_GPU
- =PPV35_GPU_FBVDDQ

Signal aliases required by this page:
(NONE)

BOH options provided by this page:
(NONE)



SYNC MASTER=D2 SEAN		SYNC DATE=03/05/2012	
PAGE TITLE			
KEPLER CORE/FB POWER			
 Apple Inc.		DRAWING NUMBER	051-9589
		REVISION	4.18.0
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		SHEET	72 OF 99

D

C

B

A

D

C

B

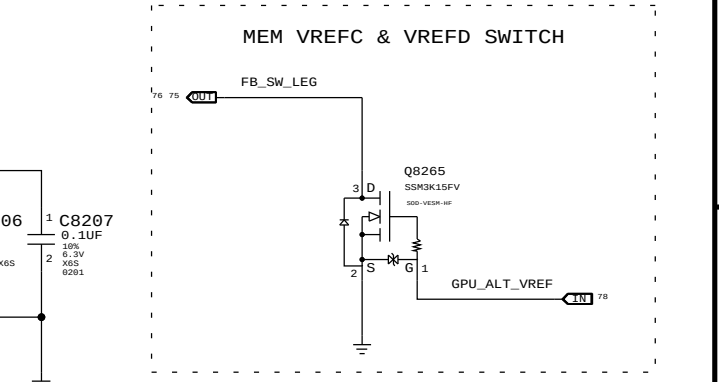
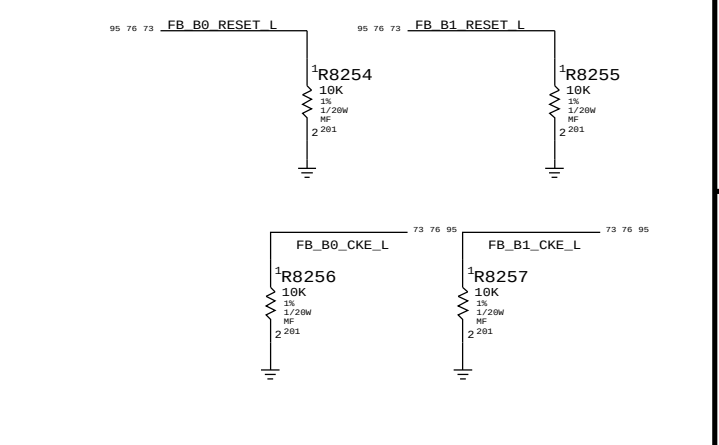
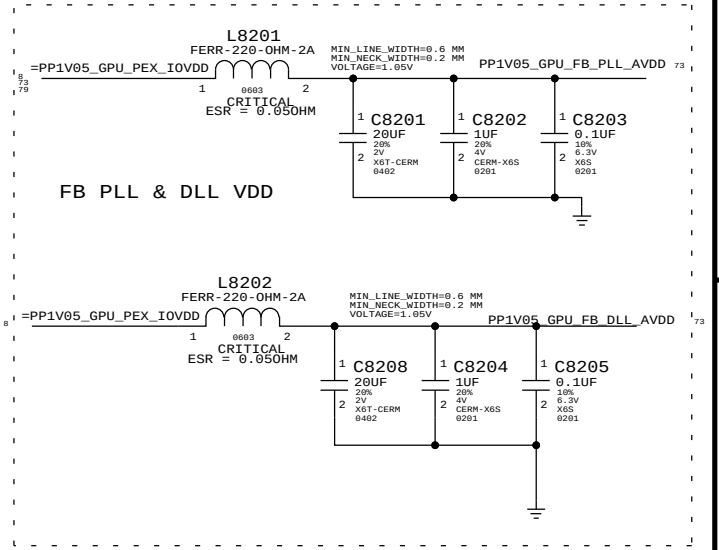
A



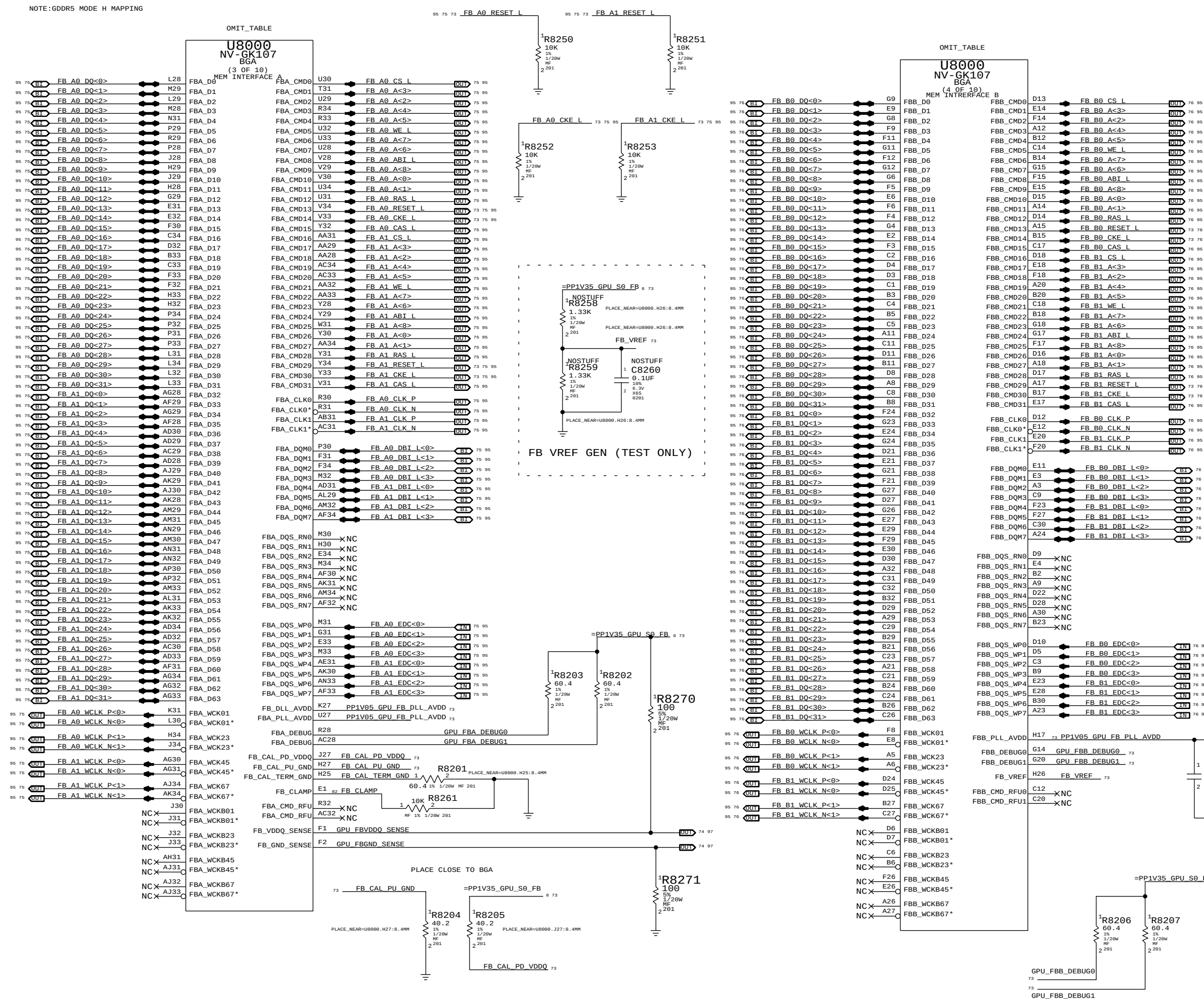
Power aliases required by this page:
-PP1V05_GPU_S0_FB
-PP1V05_GPU_PEX_IOVDD

Signal aliases required by this page:
(NONE)

BOX options provided by this page:
(NONE)

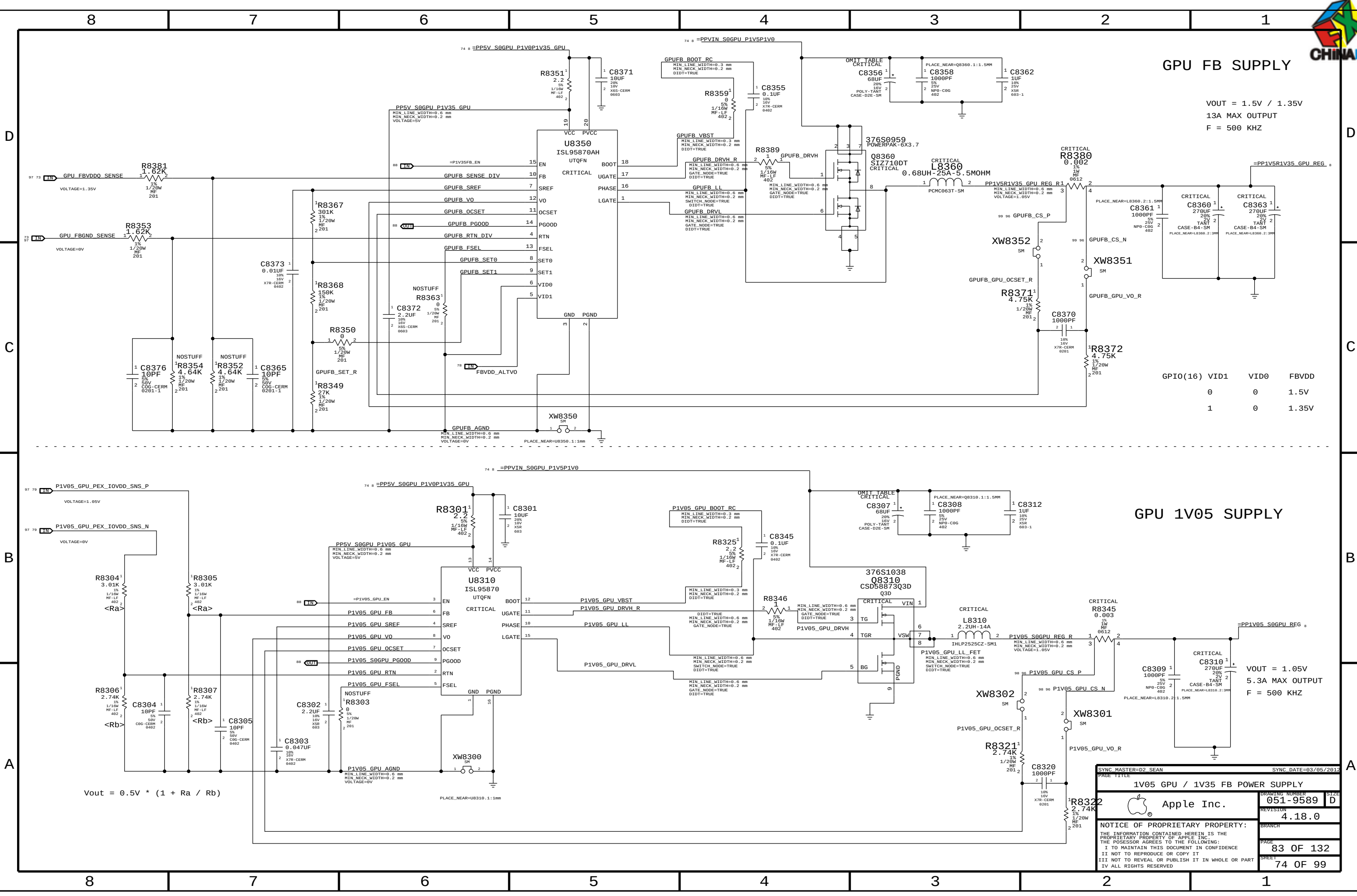


SYNC MASTER=D2 SEAN		SYNC DATE=03/05/2012	
PAGE TITLE		KEPLER FRAME BUFFER I/F	
Apple Inc.		DRAWING NUMBER	051-9589
REVISION		4.18.0	SIZE D
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D
C
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D
C
B
A



$$V_{out} = 0.5V * (1 + R_a / R_b)$$

SYNC MASTER=D2 SEAN		SYNC DATE=03/05/2012	
PAGE TITLE			
1V05 GPU / 1V35 FB POWER SUPPLY			
	DRAWING NUMBER		SIZE
	051-9589		D
Apple Inc.		REVISION	
		4.18.0	
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Page Notes

Power aliases required by this page:

- =PP1V35 GPU FBVDDQ

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

U8400
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C

(1 OF 2)

OMIT_TABLE

U8450
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C

(1 OF 2)

OMIT_TABLE

SYNC MASTER=D2 SEAN SYNC DATE=03/05/2012

GDDR5 Frame Buffer A

Apple Inc.
DRAWING NUMBER: 051-9589
REVISION: 4.18.0

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84 OF 132

75 OF 99



Page Notes

Power aliases required by this page:

- =PP1V35 GPU_FB_VDDQ

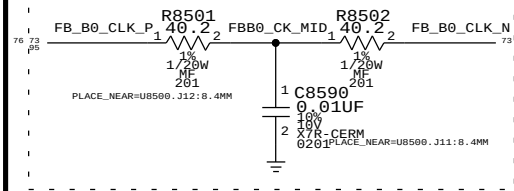
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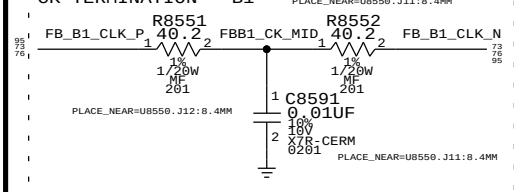
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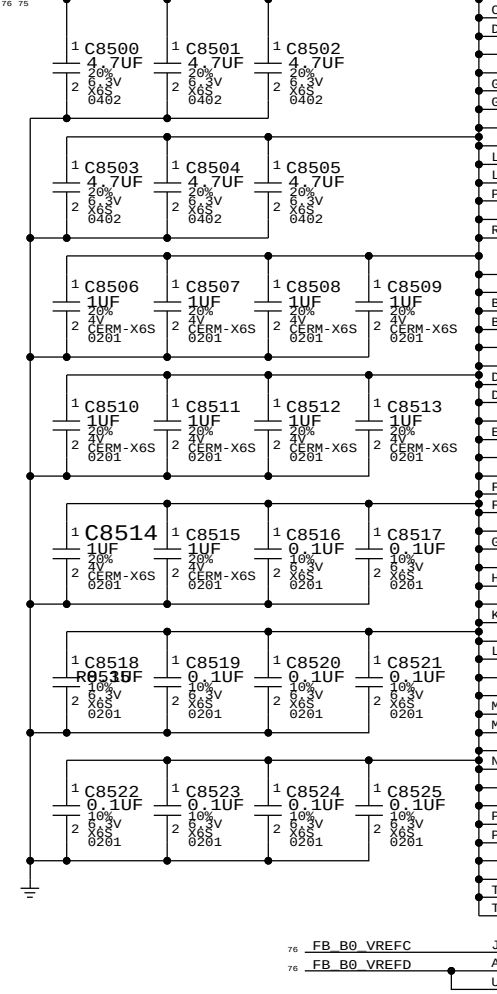
CK TERMINATION - B0



CK TERMINATION - B1



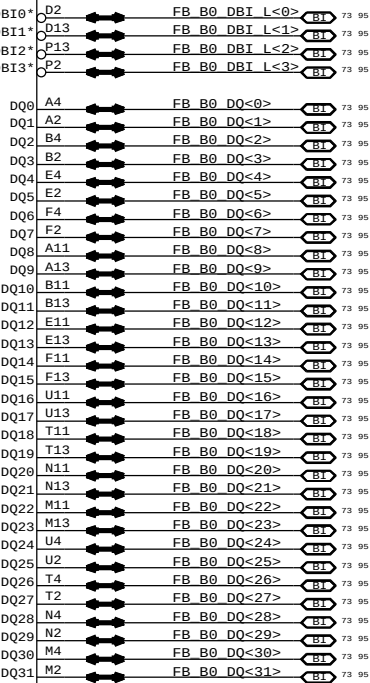
=PP1V35 GPU_FB_VDDQ



U8500
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C

(1 OF 2)

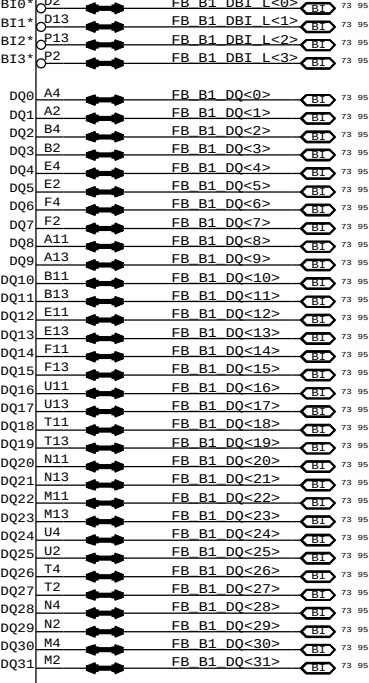
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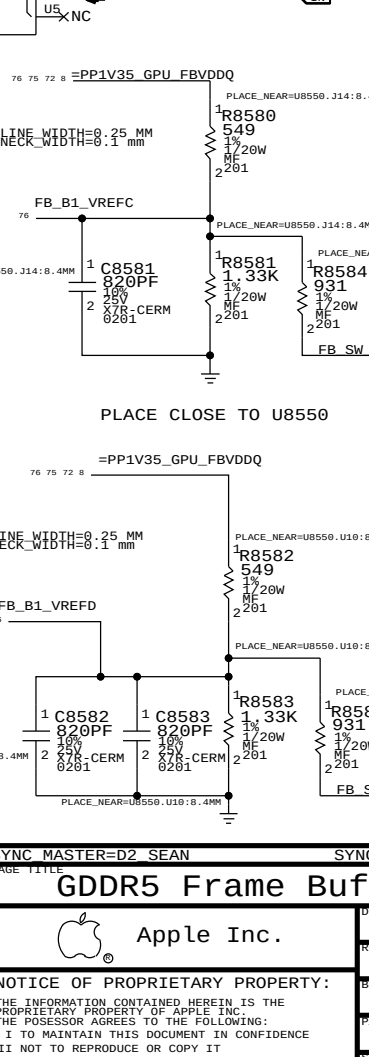
U8550
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C

(1 OF 2)

OMIT_TABLE



=PP1V35 GPU_FB_VDDQ



SYNC MASTER=D2 SEAN		SYNC DATE=03/05/2012	
PAGE TITLE			
GDDR5 Frame Buffer B			
Apple Inc.		DRAWING NUMBER	051-9589
REVISION		4.18.0	
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Page Notes

Power aliases required by this page:

PP3V3_GPU_IFPB_IOVDD
PP3V3_GPU_IFPCD_IOVDD
PP3V3_GPU_IFPAB_PLLVDD
PP3V3_GPU_VDD33
PP1V05_GPU_IPFCD_IOVDD
PP1V05_GPU_IPFAB_PLLVDD
PP1V05_GPU_VID_PLLVDD

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

J31: YES
J32: YES

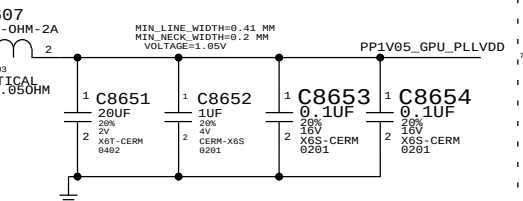
PP1V05_GPU_SP_PLLVDD
PP1V05_GPU_VID_PLLVDD
MAKE_BASE=TRUE

GPU_OSC_27M_XTAL_BUFFOUT_R
GPU_OSC_27M_SSIN
PLACE_NEAR=U8000, J4: 8.4MM
PLACE_NEAR=U8000, H1: 8.4MM

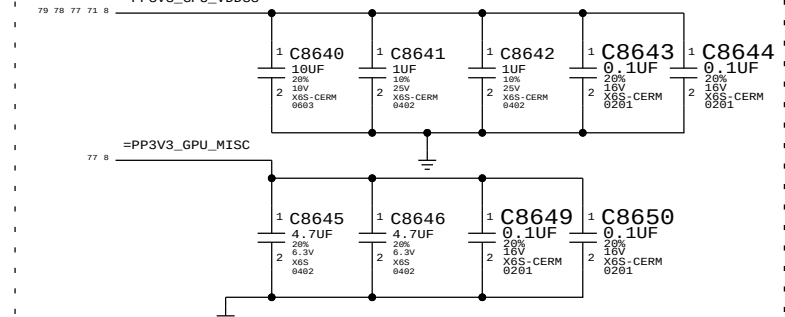
GPIO0 P6 GPU GPIO 0
GPIO1 M3 GPU GPIO 1
GPIO2 L6 GPU GPIO 2
GPIO3 P5 GPU GPIO 3
GPIO4 P7 GPU GPIO 4
GPIO5 L7 GPU GPIO 5
GPIO6 M7 GPU GPIO 6
GPIO7 N8 GPU GPIO 7
GPIO8 M1 GPU GPIO 8
GPIO9 M2 GPU GPIO 9
GPIO10 L1 GPU GPIO 10
GPIO11 M5 GPU GPIO 11
GPIO12 N3 GPU GPIO 12
GPIO13 M4 GPU GPIO 13
GPIO14 N4 GPU GPIO 14
GPIO15 P2 GPU GPIO 15
GPIO16 R8 GPU GPIO 16
GPIO17 M6 GPU GPIO 17
GPIO18 R1 GPU GPIO 18
GPIO19 P3 GPU GPIO 19
GPIO20 P4 GPU GPIO 20
GPIO21 P1 GPU GPIO 21

JTAG_TCK AM10 GPU JTAG TCK
JTAG_TDI AM11 GPU JTAG TDI
JTAG_TDO AP12 GPU JTAG TDO
JTAG_TMS AP11 GPU JTAG TMS
JTAG_TRST AN11 GPU JTAG TRST L
THERMDP K3 GPU TDIODE P
THERMDN K4 GPU TDIODE N

GPU PLL VDD



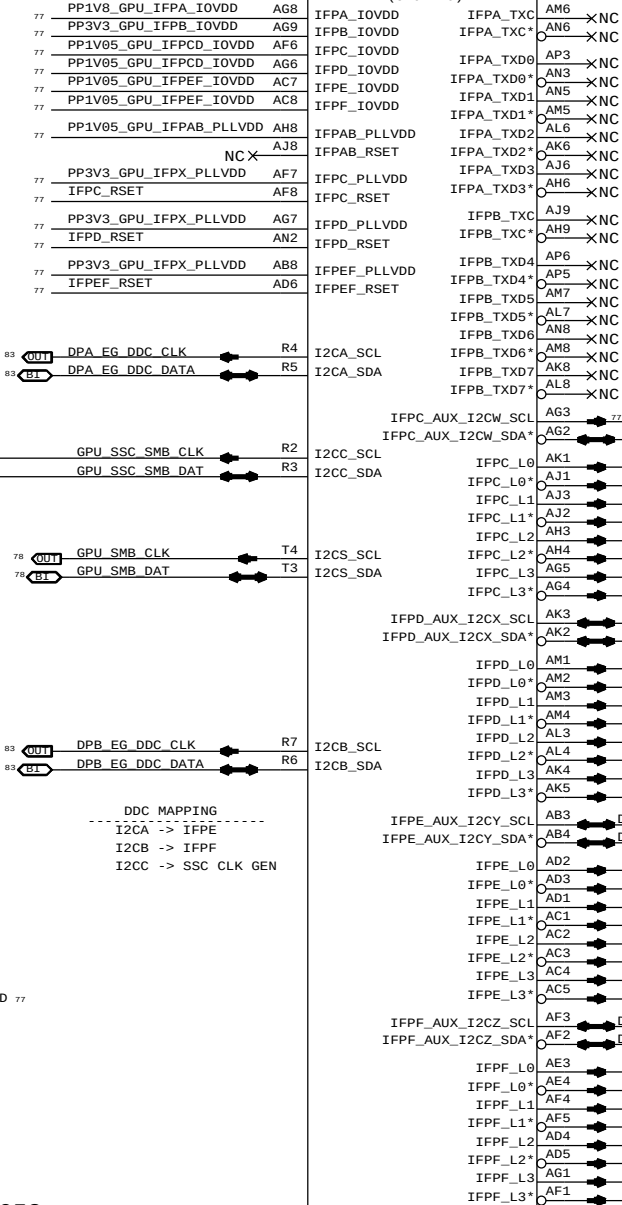
GPU 3V3 VDD



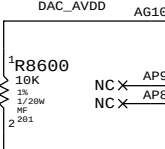
DISABLE PHY A & B FOR 15" MBP

OMIT_TABLE

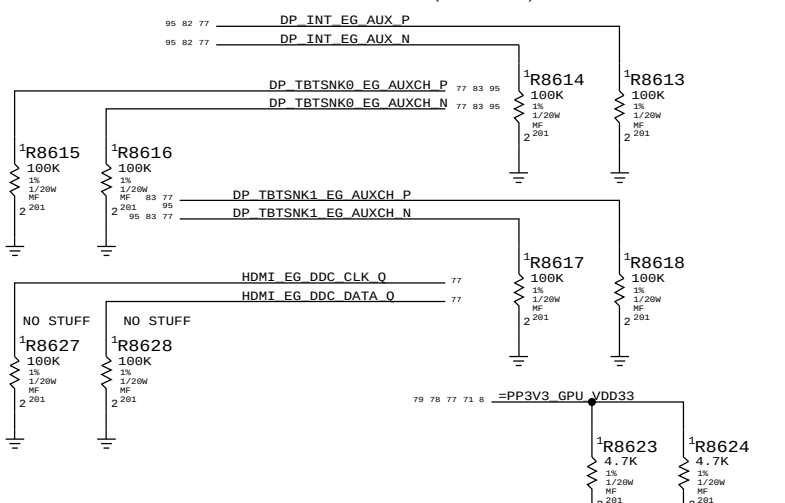
U8000 NV-GK107 (5 OF 10)



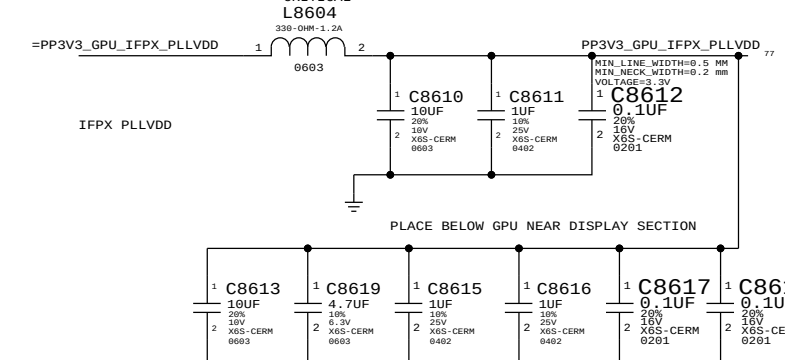
DDC MAPPING
I2CA -> IFPE
I2CB -> IFPF
I2CC -> SSC CLK GEN



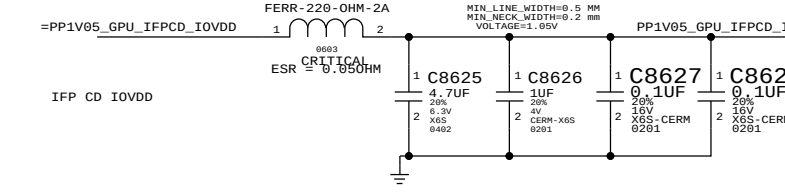
PD FOR AUX CHANNELS (FOR NVIDIA)



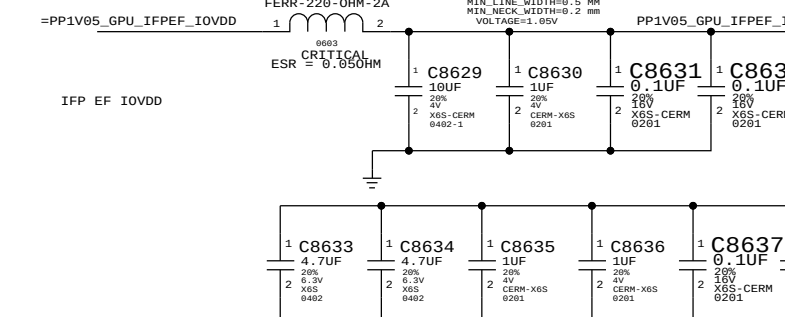
CRITICAL L8604



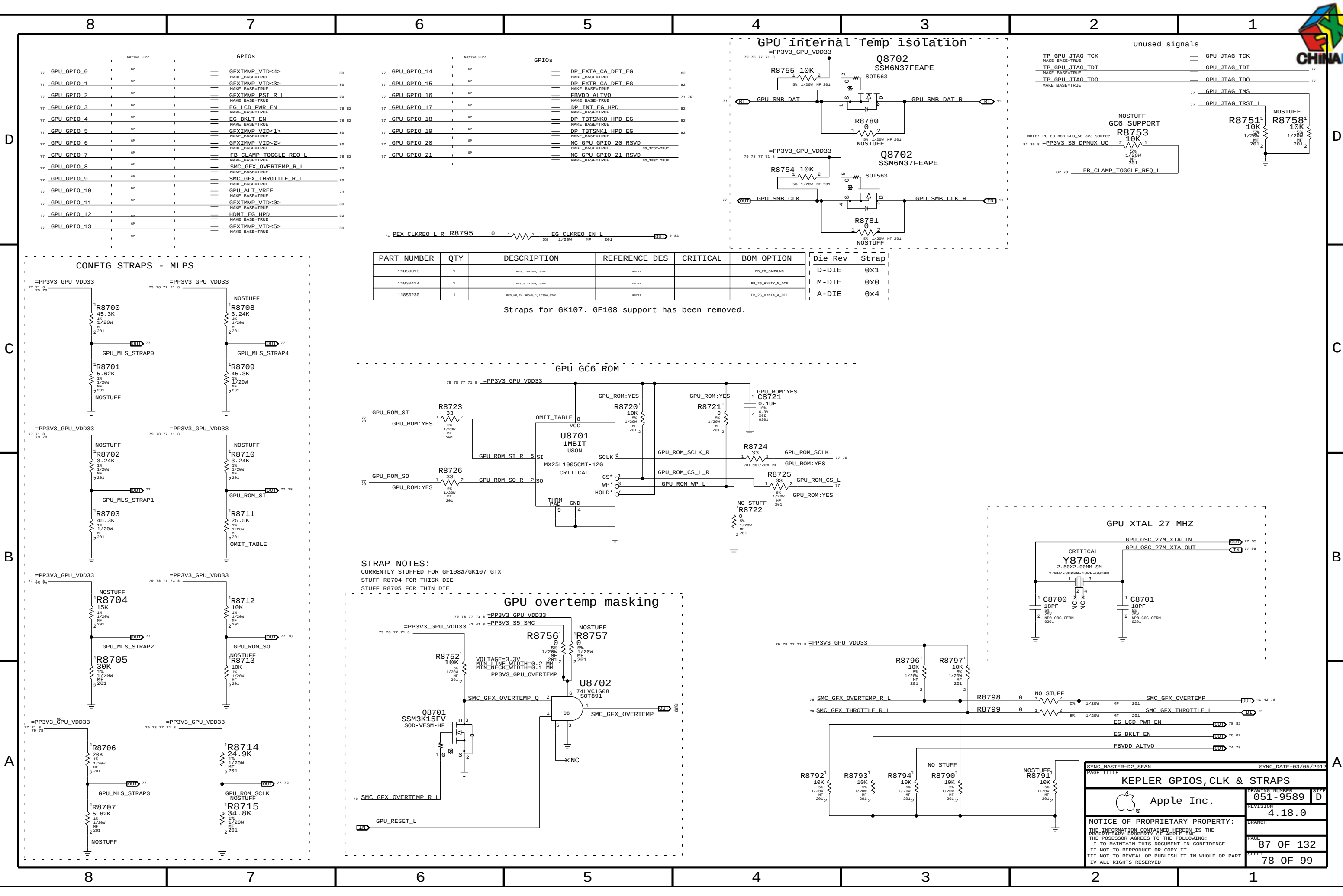
CRITICAL L8605



CRITICAL L8606



SYNC MASTER=D2 SEAN		SYNC DATE=03/05/2012	
PAGE TITLE		KEPLER EDP/DP/GPIO	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0013	1	RES, 180OHM, 0201	R8711		FB_ZG_SAMSUNG
118S0414	1	RES, 5.1KOHM, 0201	R8711		FB_ZG_HYNIX_M_DIE
118S0230	1	RES, 24.5KOHM, 1/20W, 0201	R8711		FB_ZG_HYNIX_A_DIE

Straps for GK107. GF108 support has been removed.

STRAP NOTES:
CURRENTLY STUFFED FOR GF108a/GK107-GTX
STUFF R8704 FOR THICK DIE
STUFF R8705 FOR THIN DIE

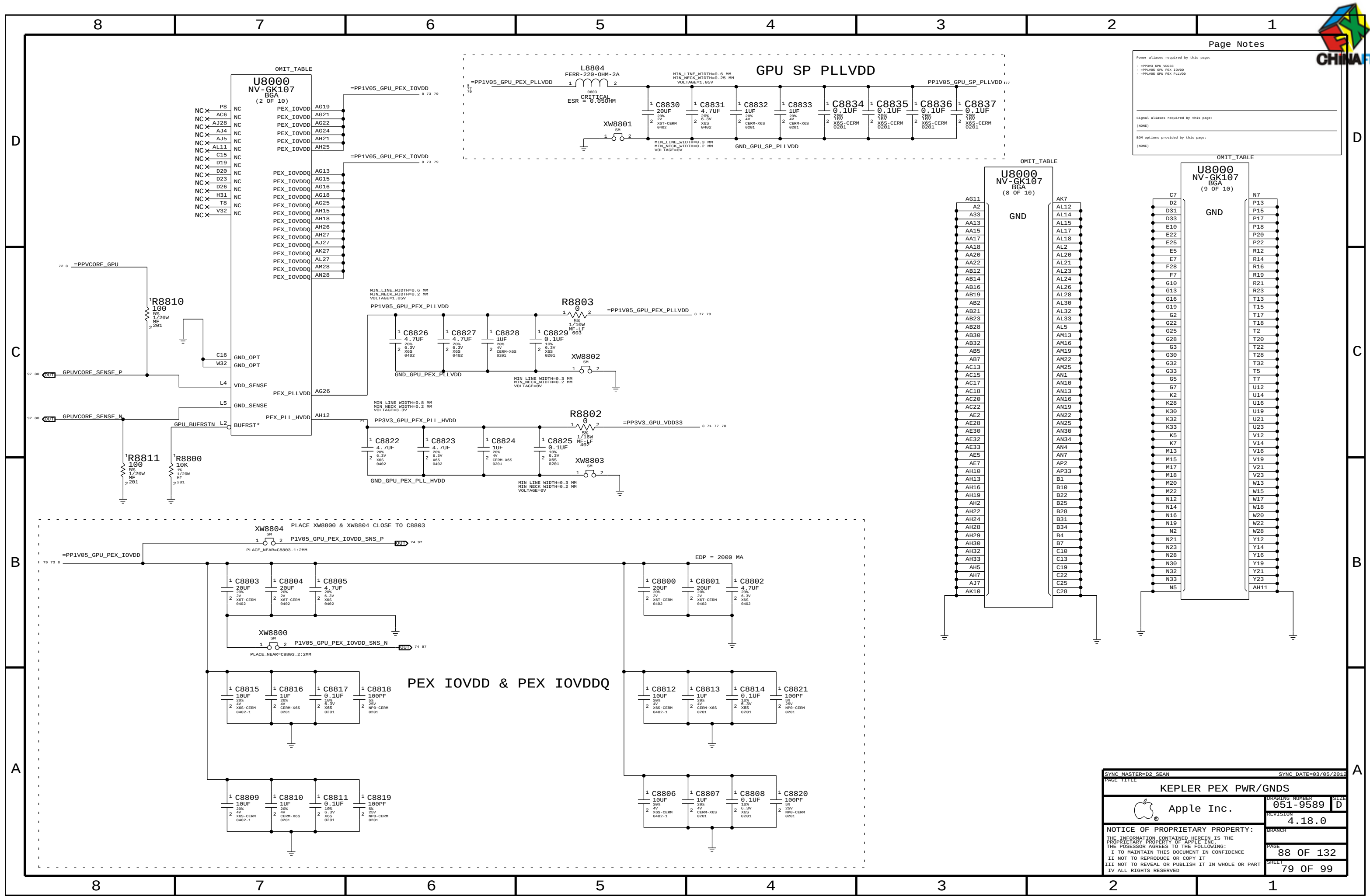
SYNC MASTER=D2 SEAN		SYNC DATE=03/05/2012	
PAGE TITLE		KEPLER GPIOs,CLK & STRAPS	
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		REVISION	4.18.0
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		SHEET	78 OF 99



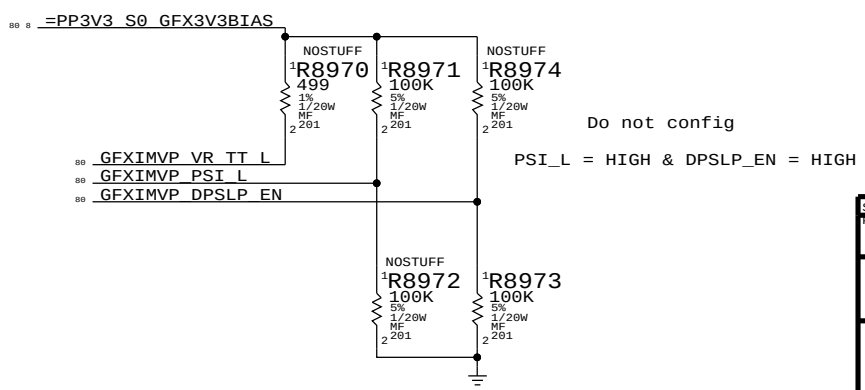
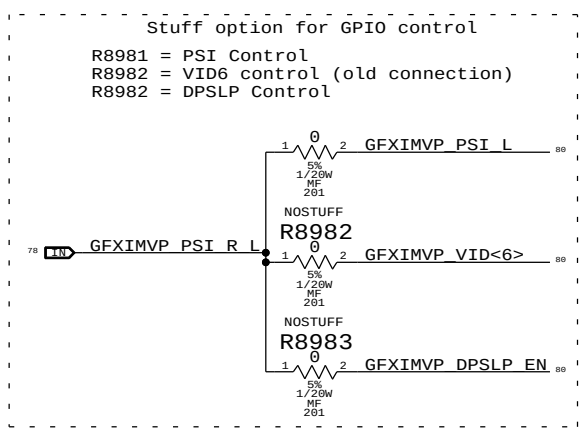
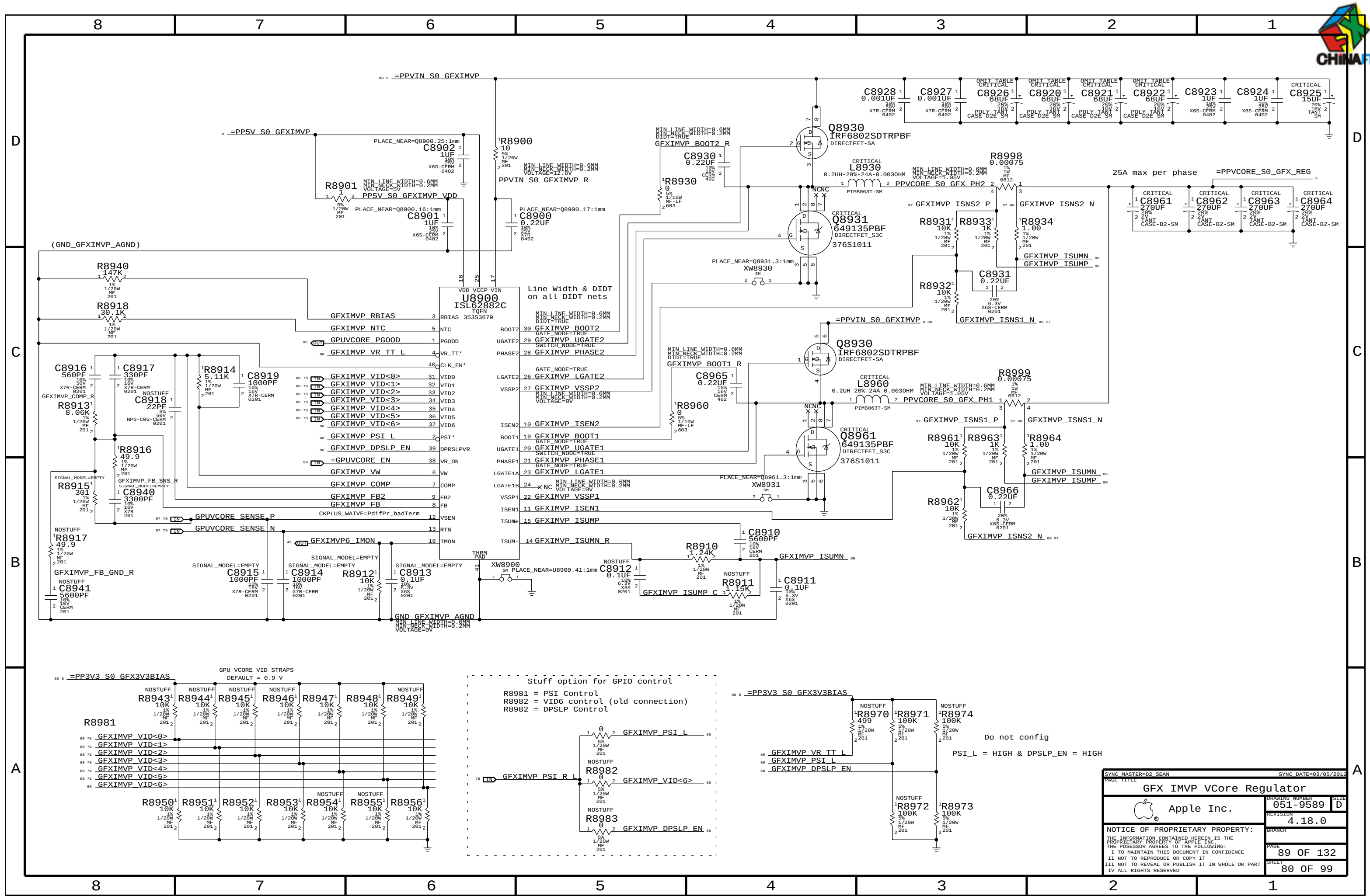
Power aliases required by this page:
- PP3V3_GPU_VDD33
- PP1V05_GPU_PEX_IOVDD
- PP1V05_GPU_PEX_PLLVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC MASTER=D2 SEAN		SYNC DATE=03/05/2012	
PAGE TITLE		KEPLER PEX PWR/GNDS	
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		PAGE	88 OF 132
		SHEET	79 OF 99



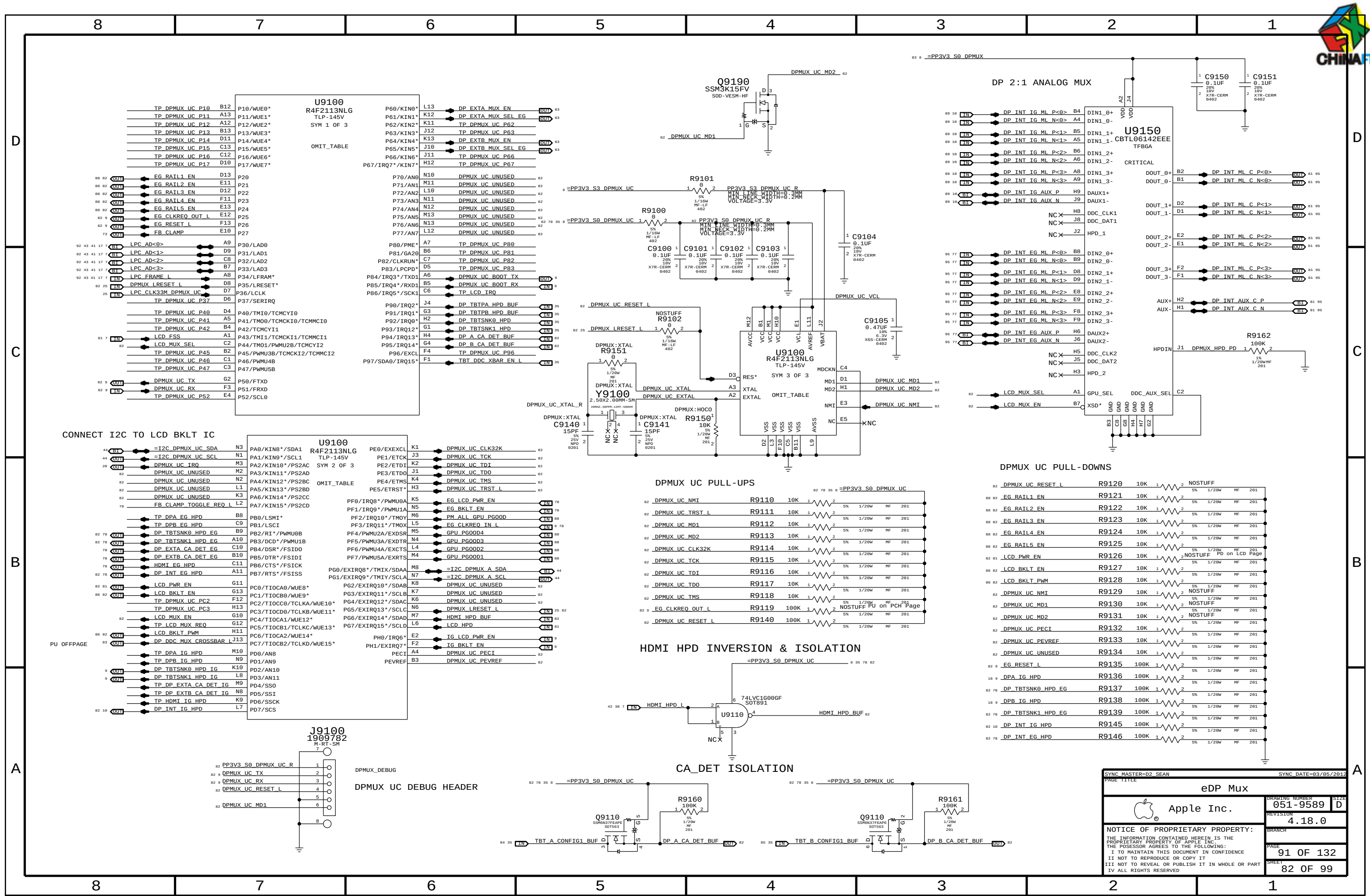
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PAGE TITLE			
GFX IMVP VCore Regulator			
	Apple Inc.	DRAWING NUMBER	SIZE
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		SHEET	80 OF 99



J9000
20525-130E-01
F-RT-SM



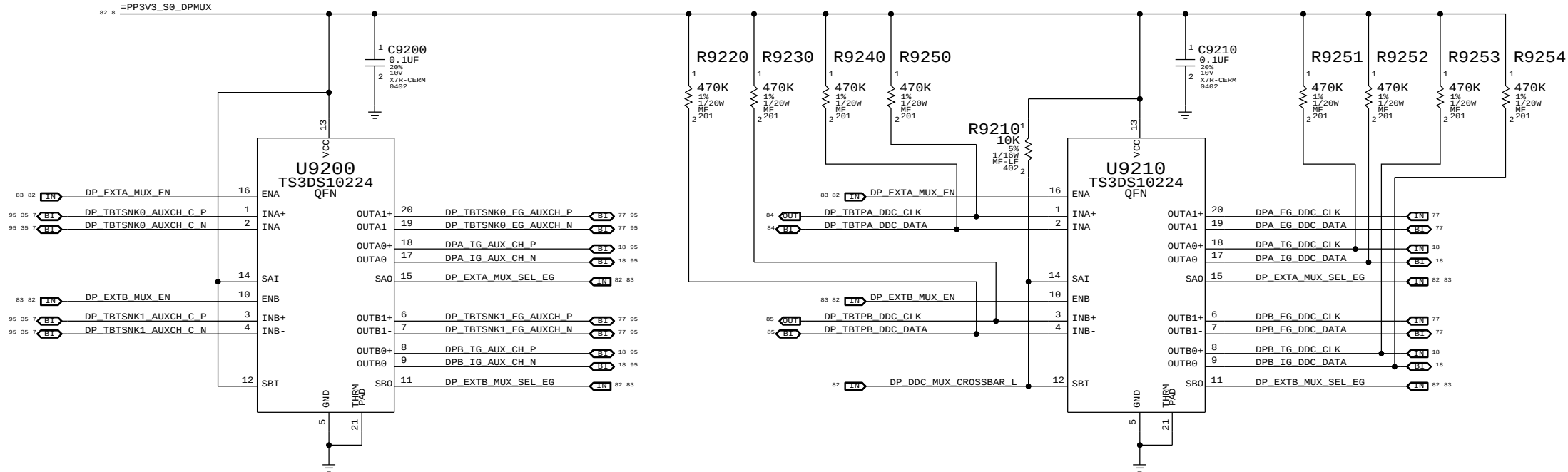
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PAGE TITLE		eDP Mux	
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		REVISION	4.18.0
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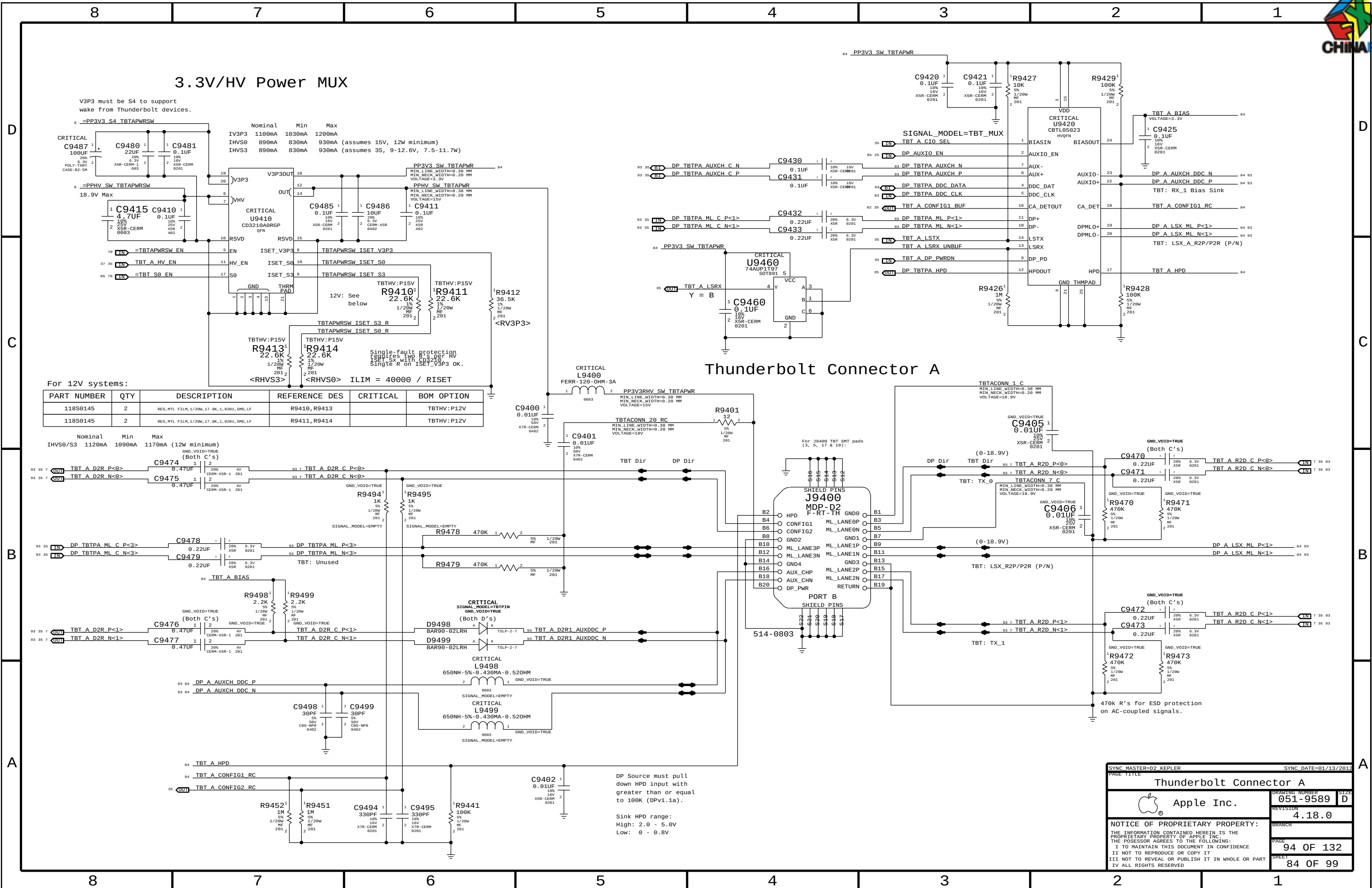
DP A & DP B AUX MUX

DP A & DP B DDC MUX



MUX TRUTH TABLE

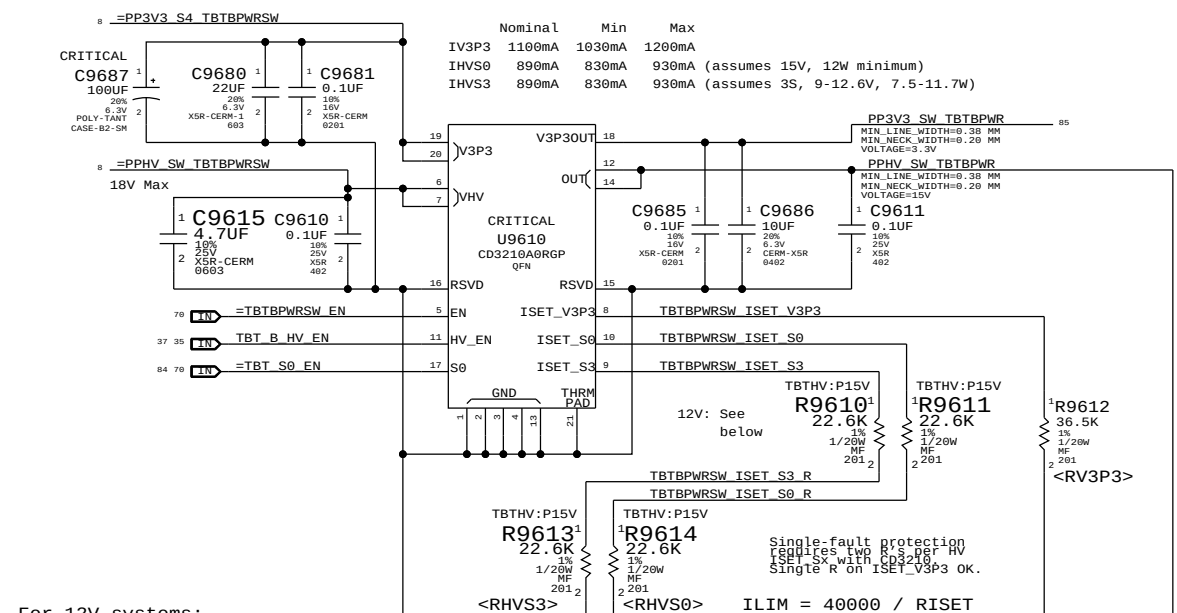
SAI/SBI	SAO	SBO	INA	INB
0	0	0	OUTB0	OUTA0
0	0	1	OUTB1	OUTA0
0	1	0	OUTB0	OUTA1
0	1	1	OUTB1	OUTA1
1	0	0	OUTA0	OUTB0
1	0	1	OUTA0	OUTB1
1	1	0	OUTA1	OUTB0
1	1	1	OUTA1	OUTB1



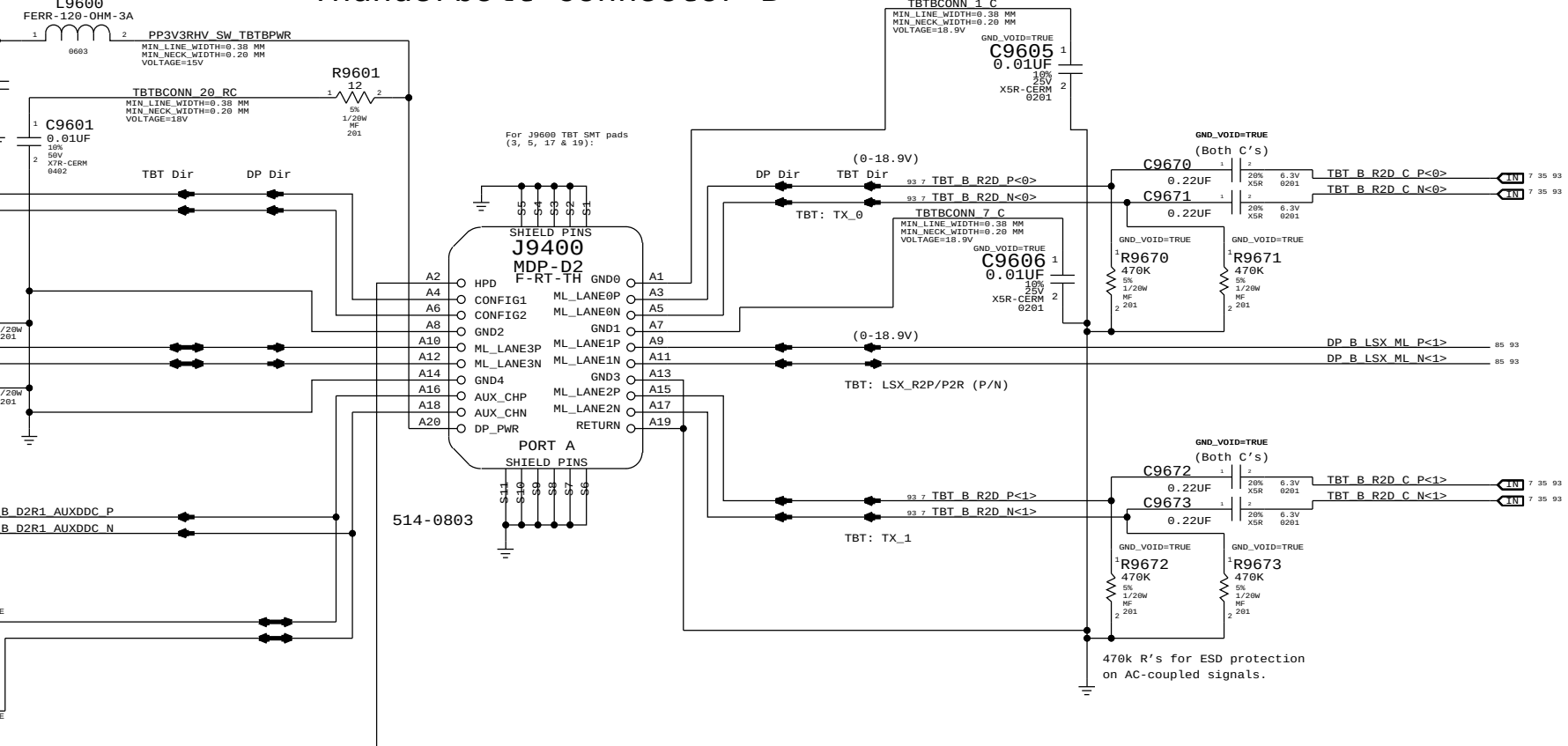


3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.



Thunderbolt Connector B



SYNC MASTER=D2 KEPLER

SYNC DATE=01/13/2012

Thunderbolt Connector B

Apple Inc.

051-9589

4.18.0

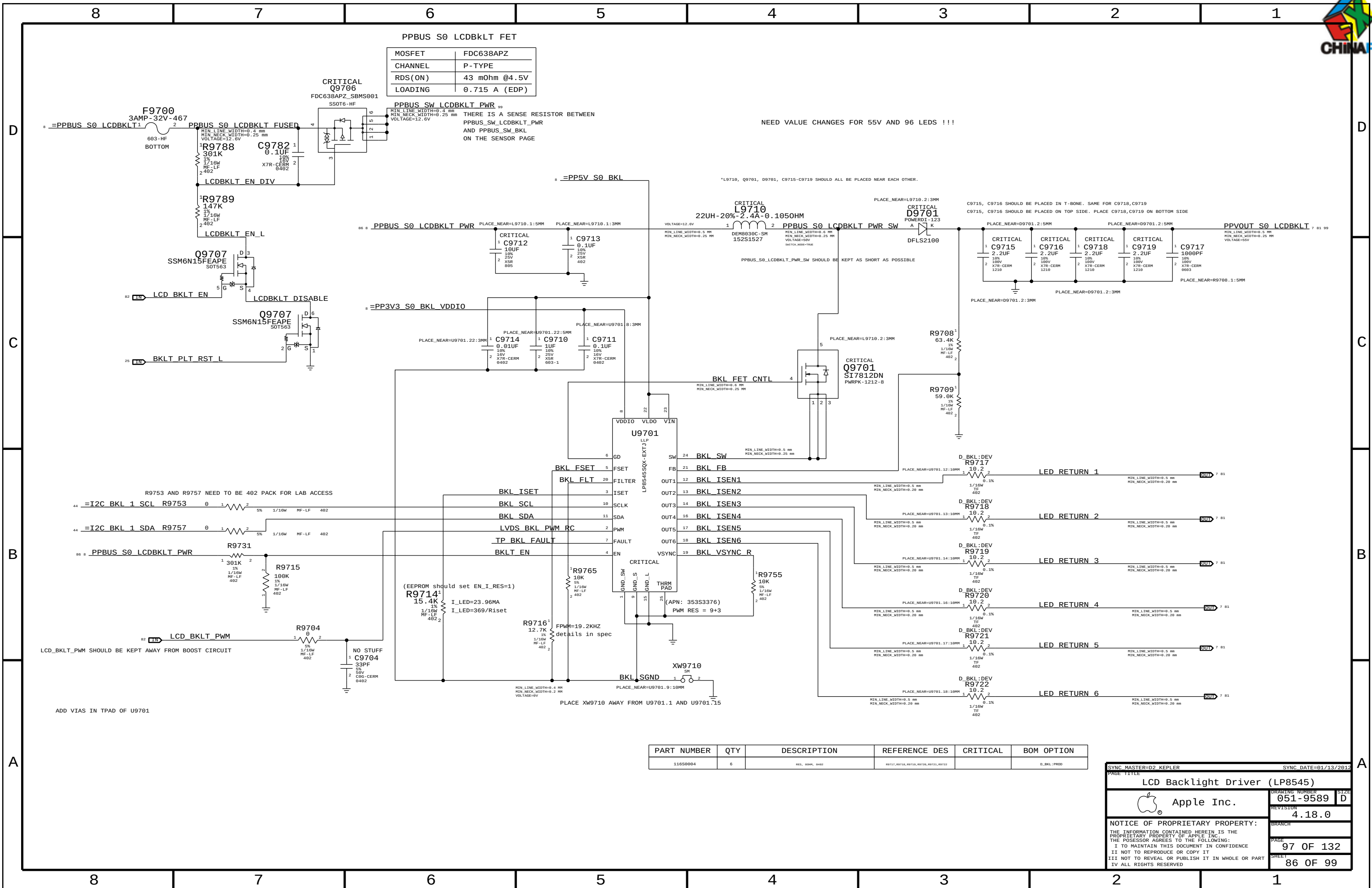
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PAGE 111

96 OF 132

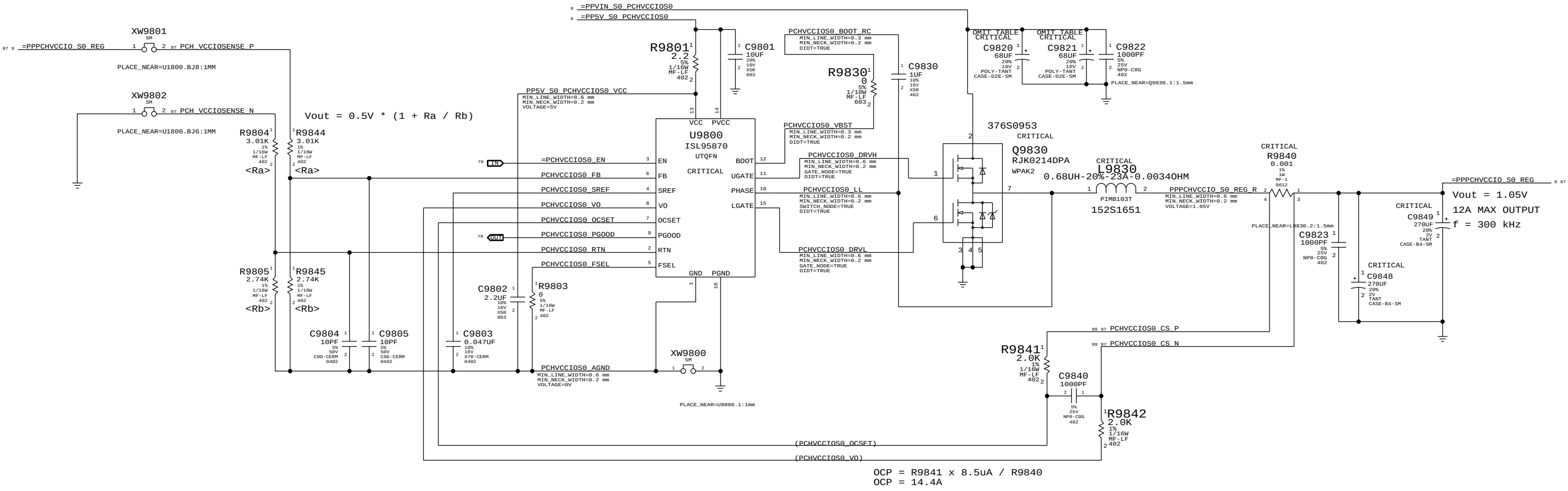
85 OF 99



SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE			
LCD Backlight Driver (LP8545)			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-9589		D
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BRANCH		PAGE	
		97 OF 132	
SHEET		86 OF 99	



PCH VCCIO (1.05V S0) REGULATOR



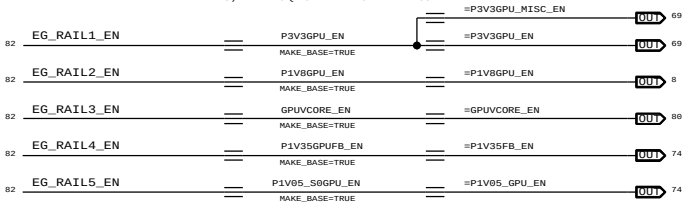
PAGE TITLE		PAGE 1	
PCH VCCIO (1.05V) POWER SUPPLY		DRAWING NUMBER	
Apple Inc.		051-9589	
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GPU Rail Sequencing

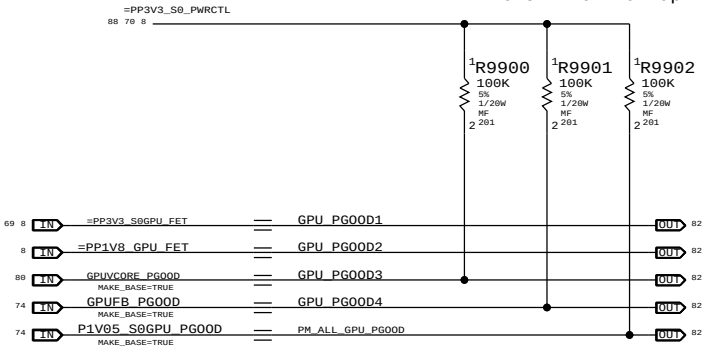
KEPLER GPU REQUIRES RAILS TO COME UP IN THE FOLLOWING ORDER:

- 1) GPU_3.3V
- 2) IFPX IOVDD - 1.8V
- 3) GPUVCORE
- 4) FBVDDQ/GDDR5 1.35V
- 5) PEXVDD/Q OR IFPY IOVDD - 1.85V



NOTE: 1V8 MAY NOT BE REQUIRED FOR KEPLER IF THERE IS NO LVDS

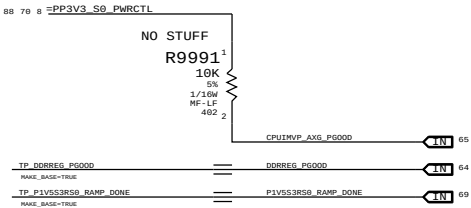
EXT GPU PWRGD Pullup



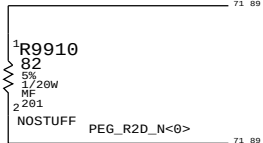
NOTE: NO PU ON 3V3 AND 1V8 PG00DS SINCE THEY ARE SYNTHETIC.

NOTE 2: CHECK IF 1V8 IS READ AS LOGIC HIGH BY GMUX

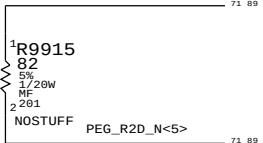
Unused PG00D signal



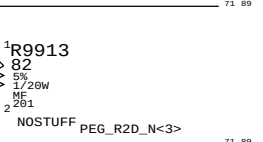
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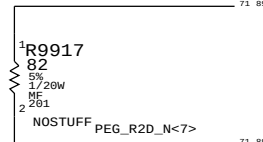
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PEG_R2D_P<3>



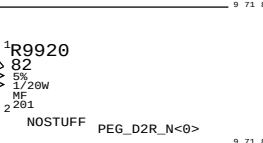
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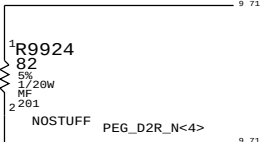
PLACE R9910 - R9917 CLOSE TO U8000

PCIE TEST STRUCTURES (FOR LAB USE)

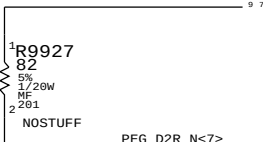
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PEG_D2R_P<4>



PEG_D2R_P<7>



PLACE R9920 - R9927 CLOSE TO U1000

SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE			
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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?	CPU_AGTL	TOP,BOTTOM	=2X_DIELECTRIC	?
CPU_BMIL	*	8 MIL	?	CPU_VID	*	0.457 MM	?
CPU_COMP	*	20 MIL	?	CPU_VREF	*	12 MIL	?
CPU_ITP	*	=2:1_SPACING	?				
CPU_VCCSENSE	*	25 MIL	?				

Most CPU signals with impedance requirements are 50-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: IVB PLATFORM DG , Tables 205-207

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	15 MIL	?	PCIE	TOP,BOTTOM	15 MIL	?
CLK_PCIE	*	20 MIL	?				

PEG

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PEG_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_RXRX	*	=4X_DIELECTRIC	?
PEG_TXTX	*	=4X_DIELECTRIC	?
PEG_TXRX	*	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PEG_D2R	PEG_D2R	*	PEG_RXRX
PEG_R2D	PEG_R2D	*	PEG_TXTX
PEG_D2R	PEG_R2D	*	PEG_TXRX

CPU Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
DMI_S2N	PCIE_85D	PCIE	DMI_S2N P<3:0>	10 18
DMI_S2N	PCIE_85D	PCIE	DMI_S2N N<3:0>	10 18
DMI_N2S	PCIE_85D	PCIE	DMI_N2S P<3:0>	10 18
DMI_N2S	PCIE_85D	PCIE	DMI_N2S N<3:0>	10 18
FDI_DATA	PCIE_85D	PCIE	FDI_DATA P<7:0>	9 10
FDI_DATA	PCIE_85D	PCIE	FDI_DATA N<7:0>	9 10
FDI_FSYNC	CPU_50S	CPU_AGTL	FDI_FSYNC<1..0>	9 10
FDI_LSYNC	CPU_50S	CPU_AGTL	FDI_LSYNC<1..0>	9 10
FDI_INT	CPU_50S	CPU_AGTL	FDI_INT	10 18
DMI_CLK100M	CLK_PCTE_90D	CLK_PCTE	DMI_CLK100M_CPU_P	11 17
DMI_CLK100M	CLK_PCTE_90D	CLK_PCTE	DMI_CLK100M_CPU_N	11 17
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_IG_ML_P<3:0>	10 82
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_IG_ML_N<3:0>	10 82
DP_INT_AUX	DP_85D	DISPLAYPORT	DP_INT_IG_AUX_P	10 82
DP_INT_AUX	DP_85D	DISPLAYPORT	DP_INT_IG_AUX_N	10 82
CPU_EDP_COMP	CPU_27P4S	CPU_COMP	CPU_EDP_COMP	10
CPU_PEG_COMP	CPU_27P4S	CPU_COMP	CPU_PEG_COMP	10
CPU_CFG	CPU_50S	CPU_ITP	CPU_CFG<17..0>	10 24
XDP_CLK_CPU	CLK_PCTE_90D	CLK_PCTE	ITPCPU_CLK100M_P	11 17
XDP_CLK_CPU	CLK_PCTE_90D	CLK_PCTE	ITPCPU_CLK100M_N	11 17
XDP_CLK_PCH	CLK_PCTE_90D	CLK_PCTE	ITPXD_P_CLK100M_P	17 24
XDP_CLK_PCH	CLK_PCTE_90D	CLK_PCTE	ITPXD_P_CLK100M_N	17 24
DPLL_REF_CLK120M	CLK_PCTE_90D	CLK_PCTE	DPLL_REF_CLKP	9 11
DPLL_REF_CLK120M	CLK_PCTE_90D	CLK_PCTE	DPLL_REF_CLKN	9 11
XDP_TDI	CPU_50S	CPU_ITP	XDP_CPU_TDI	11 24
XDP_TDO	CPU_50S	CPU_ITP	XDP_CPU_TDO	11 24
XDP_TMS	CPU_50S	CPU_ITP	XDP_CPU_TMS	11 24
XDP_TCK	CPU_50S	CPU_ITP	XDP_CPU_TCK	11 24
XDP_TRST_I	CPU_50S	CPU_ITP	XDP_CPU_TRST_L	11 24
XDP_BPM	CPU_50S	CPU_ITP	XDP_BPM_L<3..0>	11 24
XDP_BPM_L	CPU_50S	CPU_ITP	XDP_BPM_L<7..4>	11 24
XDP_DBRESET_I	CPU_50S	CPU_ITP	XDP_DBRESET_L	11 24 25
XDP_PRODY_I	CPU_50S	CPU_ITP	XDP_CPU_PRODY_L	11 24
XDP_PREQ_I	CPU_50S	CPU_ITP	XDP_CPU_PREQ_L	11 24
CPU_CATERR_I	CPU_50S	CPU_AGTL	CPU_CATERR_L	11 41
CPU_PROC_SEL_L	CPU_50S	CPU_AGTL	CPU_PROC_SEL_L	11 20
CPU_PECT	CPU_50S	CPU_VID	CPU_PECT	11 20 42
CPU_PROCHOT_I	CPU_50S	CPU_AGTL	CPU_PROCHOT_L	11 41 42 65
XDP_CPU_PWRGD	CPU_50S	CPU_ITP	XDP_CPU_PWRGD	24
PM_THRMTRIP_I	CPU_50S	CPU_BMIL	PM_THRMTRIP_L	11 20 42
PM_SYNC	CPU_50S	CPU_AGTL	PM_SYNC	11 18
PM_MEM_PWRGD	CPU_50S	CPU_AGTL	PM_MEM_PWRGD	11 18 27
CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD	11 20 24
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<2..0>	11
	CPU_50S	CPU_VID	CPU_VIDSOUT	13 65
	CPU_50S	CPU_VID	CPU_VIDSCLK	13 65
	CPU_50S	CPU_VID	CPU_VIDALERT_L	13 65
	CPU_55S	CPU_VID	CPU_VCCSA_VID<1..0>	13 62
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	13 65
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	13 65
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCIOSENSE_P	13 67
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCIOSENSE_N	13 67
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_SENSE_P	13 65
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_SENSE_N	13 65
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_P	13
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_N	13
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_P	13
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_N	13
CPU_VCCSASENSE	CPU_50S	CPU_AGTL	CPU_VCCSASENSE	13 62
CPU_MEM_VREF		CPU_VREF	PPCPU_MEM_VREFDQ_A	10 33
CPU_MEM_VREF		CPU_VREF	PPCPU_MEM_VREFDQ_B	10 33
CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFDQ_A	28 29 33
CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFDQ_B	30 31 33
CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFCA_A	28 29 33
CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFCA_B	30 31 33
XDP_CLK_TTP	CLK_PCTE_90D	CLK_PCTE	XDP_CPU_CLK100M_P	24
XDP_CLK_TTP	CLK_PCTE_90D	CLK_PCTE	XDP_CPU_CLK100M_N	24
PEG_R2D	PEG_80D	PEG_R2D	PEG_R2D_P<7..0>	71 88
PEG_R2D	PEG_80D	PEG_R2D	PEG_R2D_N<7..0>	71 88
PEG_R2D	PEG_80D	PEG_R2D	PEG_R2D_C_P<7..0>	9 71
PEG_R2D	PEG_80D	PEG_R2D	PEG_R2D_C_N<7..0>	9 71
PEG_D2R	PEG_80D	PEG_D2R	PEG_D2R_P<7..0>	9 71 88
PEG_D2R	PEG_80D	PEG_D2R	PEG_D2R_N<7..0>	9 71 88
PEG_D2R	PEG_80D	PEG_D2R	PEG_D2R_C_P<7..0>	71
PEG_D2R	PEG_80D	PEG_D2R	PEG_D2R_C_N<7..0>	71

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CPU Constraints			
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		PAGE	100 OF 132
		SHEET	89 OF 99



Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=3:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MILS	?
MEM_DQBL2BL	*	16 MILS	?
MEM_DQCH2CH	*	25 MILS	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_*	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_*	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_*	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQ_BYTE*	MEM_*	*	MEM_DATA2MEM
MEM_*_DQ_BYTE*	=SAME	*	MEM_DATA2DATA
MEM_A_DQ_BYTE*	MEM_A_DQ_BYTE*	*	MEM_DQBL2BL
MEM_B_DQ_BYTE*	MEM_B_DQ_BYTE*	*	MEM_DQBL2BL
MEM_A_DQ_BYTE*	MEM_B_DQ_BYTE*	*	MEM_DQCH2CH

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_*	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*	*	*	MEM_20THER

DDR3 (Memory Down):

DQ signals should be matched within 0.508mm of associated DQS pair

DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.

DQS to clock matching should be within [CLK-139.73mm] and [CLK-30.48mm].

CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.

CONTROL signals should be matched within [CLK-2.54mm] to [CLK+0mm] of CLK pairs.

A/BA/CMD signals should be matched within [CLK-2.54mm] to [CLK+2.54mm] of CLK pairs.

DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.

Maximum length of any signal from die pad to first DRAM device is 139.7mm max, to last DRAM device is 194.31mm max.

SOURCE: Chief River SFF Platform DG, Rev 0.7 (#460452), Section 2.6.3

Memory Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<5..0> 12 28 29 32
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<5..0> 12 28 29 32
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A CKE<3..0> 12 28 29 32
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A CS L<3..2> 12 28 29 32
MEM_A_CNT11	MEM_37S	MEM_CTRL	MEM A CS L<1> 12 29 32
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A CS L<0> 12 28 32
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A ODT<3..2> 12 29 32
MEM_A_CNT11	MEM_37S	MEM_CTRL	MEM A ODT<1> 12 29 32
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A ODT<0> 12 28 32
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0> 12 28 29 32
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0> 12 28 29 32
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L 12 28 29 32
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L 12 28 29 32
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L 12 28 29 32
MEM_A_DQ_BYTE0	MEM_50S	MEM_A_DQ_BYTE0	MEM A DQ<7..0> 12 28 29
MEM_A_DQ_BYTE1	MEM_50S	MEM_A_DQ_BYTE1	MEM A DQ<15..8> 12 28 29
MEM_A_DQ_BYTE2	MEM_50S	MEM_A_DQ_BYTE2	MEM A DQ<23..16> 12 28 29
MEM_A_DQ_BYTE3	MEM_50S	MEM_A_DQ_BYTE3	MEM A DQ<31..24> 12 28 29
MEM_A_DQ_BYTE4	MEM_50S	MEM_A_DQ_BYTE4	MEM A DQ<39..32> 12 28 29
MEM_A_DQ_BYTE5	MEM_50S	MEM_A_DQ_BYTE5	MEM A DQ<47..40> 12 28 29
MEM_A_DQ_BYTE6	MEM_50S	MEM_A_DQ_BYTE6	MEM A DQ<55..48> 12 28 29
MEM_A_DQ_BYTE7	MEM_50S	MEM_A_DQ_BYTE7	MEM A DQ<63..56> 12 28 29
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS P<0> 12 28 29
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS N<0> 12 28 29
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS P<1> 12 28 29
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS N<1> 12 28 29
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS P<2> 12 28 29
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS N<2> 12 28 29
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS P<3> 12 28 29
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS N<3> 12 28 29
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS P<4> 12 28 29
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS N<4> 12 28 29
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS P<5> 12 28 29
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS N<5> 12 28 29
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS P<6> 12 28 29
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS N<6> 12 28 29
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS P<7> 12 28 29
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS N<7> 12 28 29
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<5..0> 12 30 31 32
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<5..0> 12 30 31 32
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B CKE<3..2> 12 31 32
MEM_B_CNT11	MEM_37S	MEM_CTRL	MEM B CKE<1> 12 31 32
MEM_B_CNT10	MEM_37S	MEM_CTRL	MEM B CKE<0> 12 30 32
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B CS L<3..0> 12 30 31 32
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B ODT<3..1> 12 31 32
MEM_B_CNT10	MEM_37S	MEM_CTRL	MEM B ODT<0> 12 30 32
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..7> 12 30 31 32
MEM_B_CMD6	MEM_40S	MEM_CMD	MEM B A<6> 12 30 31 32
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<5..0> 12 30 31 32
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0> 12 30 31 32
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L 12 30 31 32
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L 12 30 31 32
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L 12 30 31 32
MEM_B_DQ_BYTE0	MEM_50S	MEM_B_DQ_BYTE0	MEM B DQ<7..0> 12 30 31
MEM_B_DQ_BYTE1	MEM_50S	MEM_B_DQ_BYTE1	MEM B DQ<15..8> 12 30 31
MEM_B_DQ_BYTE2	MEM_50S	MEM_B_DQ_BYTE2	MEM B DQ<23..16> 12 30 31
MEM_B_DQ_BYTE3	MEM_50S	MEM_B_DQ_BYTE3	MEM B DQ<31..24> 12 30 31
MEM_B_DQ_BYTE4	MEM_50S	MEM_B_DQ_BYTE4	MEM B DQ<39..32> 12 30 31
MEM_B_DQ_BYTE5	MEM_50S	MEM_B_DQ_BYTE5	MEM B DQ<47..40> 12 30 31
MEM_B_DQ_BYTE6	MEM_50S	MEM_B_DQ_BYTE6	MEM B DQ<55..48> 12 30 31
MEM_B_DQ_BYTE7	MEM_50S	MEM_B_DQ_BYTE7	MEM B DQ<63..56> 12 30 31
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS P<0> 12 30 31
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS N<0> 12 30 31
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS P<1> 12 30 31
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS N<1> 12 30 31
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS P<2> 12 30 31
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS N<2> 12 30 31
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS P<3> 12 30 31
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS N<3> 12 30 31
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS P<4> 12 30 31
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS N<4> 12 30 31
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS P<5> 12 30 31
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS N<5> 12 30 31
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS P<6> 12 30 31
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS N<6> 12 30 31
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS P<7> 12 30 31
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS N<7> 12 30 31

SYNC MASTER=D2 KEPLER		SYNC DATE=01/13/2012	
PAGE TITLE			
Memory Constraints			
 Apple Inc.		DRAWING NUMBER	051-9589
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		PAGE	101 OF 132
		SHEET	90 OF 99



Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_DP_99D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
LVDS_85D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_DISPLAYPORT	T0L3, T0L4, T0L9, T0L10	=4:1_SPACING	?	PCH_DISPLAYPORT	TOP,BOTTOM	=4:1_SPACING	?
LVDS	T0L3, T0L4, T0L9, T0L10	=4:1_SPACING	?	LVDS	TOP,BOTTOM	=4:1_SPACING	?

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_98D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
SATA_37SE	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE
SATA_50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	T0L3, T0L4, T0L9, T0L10	=5:1_SPACING	?	SATA	TOP,BOTTOM	=5:1_SPACING	?
SATA_ICOMP	*	15 MIL	?				

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RB1AS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	T0L3, T0L4, T0L9, T0L10	=4:1_SPACING	?	USB	TOP,BOTTOM	=4:1_SPACING	?
USB_RB1AS	*	15 MIL	?				

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

USB 3.0 INTERFACE CONSTRAINTS

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3	T0L3, T0L4, T0L9, T0L10	=5:1_SPACING	?	USB3	TOP,BOTTOM	=5:1_SPACING	?

SOURCE: CR SFF PLATFORM DESIGN GUIDE V0.7, TABLE 4-211, 1X1+

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?				
CLK_25M	*	=5x_DIELECTRIC	?				

NOTE: 25MHz system clocks very sensitive to noise.

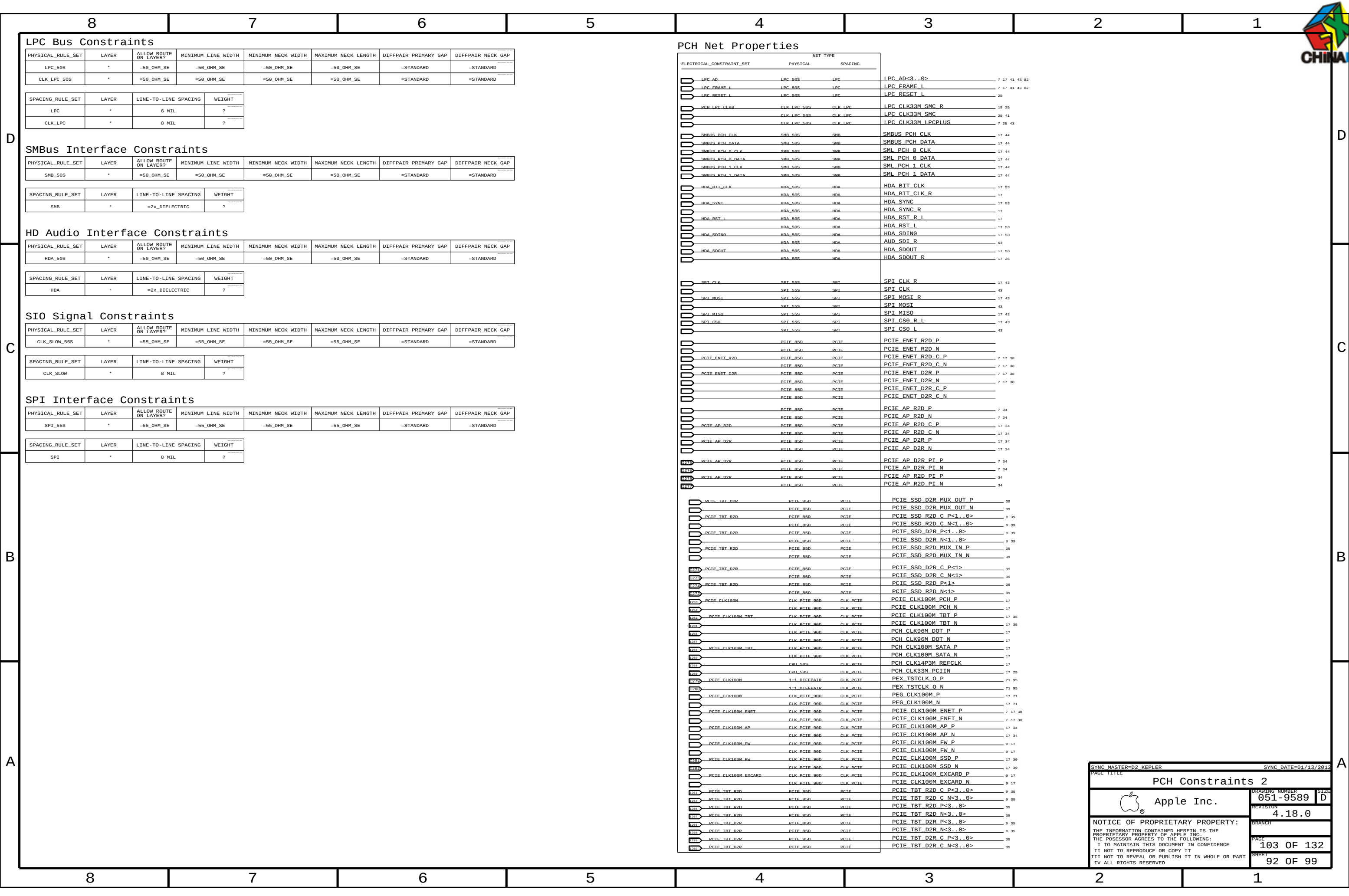
PCH Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
LVDS_TG_A_CLK	LVDS_85D	LVDS	LVDS IG A CLK P 9 18
LVDS_TG_A_CLK	LVDS_85D	LVDS	LVDS IG A CLK N 9 18
LVDS_TG_A_DATA	LVDS_85D	LVDS	LVDS IG A DATA P<2..0> 9 18
LVDS_TG_A_DATA	LVDS_85D	LVDS	LVDS IG A DATA N<2..0> 9 18
LVDS_TG_A_DATA3	LVDS_85D	LVDS	LVDS IG A DATA P<3> 9 18
LVDS_TG_A_DATA3	LVDS_85D	LVDS	LVDS IG A DATA N<3> 9 18
LVDS_TG_B_DATA	LVDS_85D	LVDS	LVDS IG B DATA P<2..0> 9 18
LVDS_TG_B_DATA	LVDS_85D	LVDS	LVDS IG B DATA N<2..0> 9 18
SATA_HDD_R2D	SATA_90D	SATA	SATA HDD R2D C P 17 39
SATA_HDD_R2D	SATA_90D	SATA	SATA HDD R2D C N 17 39
SATA_HDD_D2R	SATA_90D	SATA	SATA HDD D2R P 17 39
SATA_HDD_D2R	SATA_90D	SATA	SATA HDD D2R N 17 39
SATA_HDD_D2R	SATA_90D	SATA	SATA SSD D2R MUX_OUT_P 39
SATA_HDD_D2R	SATA_90D	SATA	SATA SSD D2R MUX_OUT_N 39
SATA_HDD_R2D	SATA_90D	SATA	SATA SSD R2D MUX_IN_P 39
SATA_HDD_R2D	SATA_90D	SATA	SATA SSD R2D MUX_IN_N 39
SATA_HDD_D2R	SATA_90D	SATA	SATA SSD D2R P 39
SATA_HDD_D2R	SATA_90D	SATA	SATA SSD D2R N 39
SATA_HDD_R2D	SATA_90D	SATA	SATA SSD R2D P 39
SATA_HDD_R2D	SATA_90D	SATA	SATA SSD R2D N 39
SATA_HDD_R2D	SATA_90D	SATA	SATA HDD R2D UF_P 39
SATA_HDD_R2D	SATA_90D	SATA	SATA HDD R2D UF_N 39
SATA_ODD_R2D	SATA_90D	SATA	SATA ODD R2D C P 9 17
SATA_ODD_R2D	SATA_90D	SATA	SATA ODD R2D C N 9 17
SATA_ODD_R2D	SATA_90D	SATA	SATA ODD R2D P 9 17
SATA_ODD_R2D	SATA_90D	SATA	SATA ODD R2D N 9 17
SATA_ODD_D2R	SATA_90D	SATA	SATA ODD D2R P 9 17
SATA_ODD_D2R	SATA_90D	SATA	SATA ODD D2R N 9 17
SATA_ODD_D2R	SATA_90D	SATA	SATA ODD D2R UF_P 9 17
SATA_ODD_D2R	SATA_90D	SATA	SATA ODD D2R UF_N 9 17
PCH_SATA3_TCOMP	SATA_50SE	SATA_TCOMP	PCH SATA3COMP 17
PCH_SATA_TCOMP	SATA_37SE	SATA_TCOMP	PCH SATA1COMP 17
USB_HUB1_UP	USB_85D	USB	USB EXT_B_XHCI_P 19 26
USB_HUB1_UP	USB_85D	USB	USB EXT_B_XHCI_N 19 26
USB_HUB1_UP	USB_85D	USB	USB EXT_B_EHCI_P 19 26
USB_HUB1_UP	USB_85D	USB	USB EXT_B_EHCI_N 19 26
USB_HUB2_UP	USB_85D	USB	USB HUB UP_P 19 26
USB_HUB2_UP	USB_85D	USB	USB HUB UP_N 19 26
USB_EXT_A	USB_85D	USB	USB EXT_A_P 19 40
USB_EXT_A	USB_85D	USB	USB EXT_A_N 19 40
USB_EXT_B	USB_85D	USB	USB EXT_B_P 7 26 38
USB_EXT_B	USB_85D	USB	USB EXT_B_N 7 26 38
USB_EXT_C	USB_85D	USB	USB EXT_C_P 9 19
USB_EXT_C	USB_85D	USB	USB EXT_C_N 9 19
USB_CAMERA	USB_85D	USB	USB CAMERA_CONN_P 7 34
USB_CAMERA	USB_85D	USB	USB CAMERA_CONN_N 7 34
USB_BT	USB_85D	USB	USB_BT_P 9 34
USB_BT	USB_85D	USB	USB_BT_N 9 34
USB_BT	USB_85D	USB	USB_BT_CONN_P 7 34
USB_BT	USB_85D	USB	USB_BT_CONN_N 7 34
USB_BT	USB_85D	USB	USB_BT_WAKE_P 34
USB_BT	USB_85D	USB	USB_BT_WAKE_N 34
USB_TPAD	USB_85D	USB	USB_TPAD_P 9 49
USB_TPAD	USB_85D	USB	USB_TPAD_N 9 49
USB_IR	USB_85D	USB	USB_SMC_P 9 41
USB_IR	USB_85D	USB	USB_SMC_N 9 41
PCH_USB_RB1AS	PCH_USB_RB1AS	USB_RB1AS	PCH USB RB1AS 19
USB_T29A	USB_85D	USB	USB_EXTD_XHCI_P 19 26
USB_T29A	USB_85D	USB	USB_EXTD_XHCI_N 19 26
USB_EXT_A	USB_85D	USB	USB_EXT_A_MUXED_P 40
USB_EXT_A	USB_85D	USB	USB_EXT_A_MUXED_N 40
USB_CAMERA	USB_85D	USB	USB_CAMERA_P 19 34
USB_CAMERA	USB_85D	USB	USB_CAMERA_N 19 34
USB_EXT_A	USB_85D	USB	USB_LT1_P 40
USB_EXT_A	USB_85D	USB	USB_LT1_N 40
USB3_EXTB_TX	USB_85D	USB3	USB3_EXTB_TX_P 19 38
USB3_EXTB_TX	USB_85D	USB3	USB3_EXTB_TX_N 19 38
USB3_EXTB_RX	USB_85D	USB3	USB3_EXTB_RX_P 7 19 38
USB3_EXTB_RX	USB_85D	USB3	USB3_EXTB_RX_N 7 19 38
USB3_EXTC_TX	USB_85D	USB3	USB3_EXTC_TX_P 9 19
USB3_EXTC_TX	USB_85D	USB3	USB3_EXTC_TX_N 9 19
USB3_EXTC_RX	USB_85D	USB3	USB3_EXTC_RX_P 9 19
USB3_EXTC_RX	USB_85D	USB3	USB3_EXTC_RX_N 9 19
USB3_EXT_A_TX	USB_85D	USB3	USB3_EXT_A_TX_P 19 40
USB3_EXT_A_TX	USB_85D	USB3	USB3_EXT_A_TX_N 19 40
USB3_EXT_A_RX	USB_85D	USB3	USB3_EXT_A_RX_P 19 40
USB3_EXT_A_RX	USB_85D	USB3	USB3_EXT_A_RX_N 19 40

Clock Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
SYSCLK_CLK32K_RTC	CLK_SLOW_55S	CLK_SLOW	SYSCLK CLK32K RTC 17 25
SYSCLK_CLK25M_SB	CLK_25M_55S	CLK_25M	SYSCLK CLK25M_SB 17 25
SYSCLK_CLK25M_ENET	CLK_25M_55S	CLK_25M	SYSCLK CLK25M_ENET 17
SYSCLK_CLK25M_TBT	CLK_25M_55S	CLK_25M	SYSCLK CLK25M_TBT 25 35
SYSCLK_CLK25M_TBT	CLK_25M_55S	CLK_25M	SYSCLK CLK25M_TBT_R 35

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PAGE TITLE		PCH Constraints 1	
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DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
TBT_SPI	*	=2X_DIELECTRIC	?

Thunderbolt/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
TBTDp_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
TBTDp_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
TBTDp_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
TBTDp	*	=5X_DIELECTRIC	?	TBTDp	TOP,BOTTOM	=7X_DIELECTRIC	?

NOTE: Thunderbolt high-speed nets are NOT directly assigned to TBTDp.*D physical rules.

TABLE_PHYSICAL_ASSIGNMENT symbols must be used to create the assignments.

Proper differential impedance depends on mDP connector used.

For 514-0637: R2D nets (SMT pins) = 80D, D2R nets (TH pins) = 100D

SOURCE: Bill Cornelius's Thunderbolt Routing Notes

Thunderbolt/DP Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
TBT_A_R2D	TBTDp_85D	TBTDp	TBT_A_R2D_C_P<1..0>	7 35 84
TBT_A_R2D	TBTDp_85D	TBTDp	TBT_A_R2D_C_N<1..0>	7 35 84
	TBTDp_85D	TBTDp	TBT_A_R2D_P<1..0>	7 84
	TBTDp_85D	TBTDp	TBT_A_R2D_N<1..0>	7 84
DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3..1:2>	35 84
DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3..1:2>	35 84
	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3..1:2>	84
	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3..1:2>	84
	DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>	84
	DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>	84
	TBTDp_85D	TBTDp	TBT_A_D2R_C_P<1..0>	7 84
	TBTDp_85D	TBTDp	TBT_A_D2R_C_N<1..0>	7 84
TBT_A_D2R	TBTDp_85D	TBTDp	TBT_A_D2R_P<1..0>	7 35 84
TBT_A_D2R	TBTDp_85D	TBTDp	TBT_A_D2R_N<1..0>	7 35 84
TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_P	35 84
TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_C_N	35 84
	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_P	84
	DP_85D	DISPLAYPORT	DP_TBTPA_AUXCH_N	84
	DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_P	84
	DP_85D	DISPLAYPORT	DP_A_AUXCH_DDC_N	84
	TBTDp_85D	TBTDp	TBT_A_D2R1_AUXDDC_P	84
	TBTDp_85D	TBTDp	TBT_A_D2R1_AUXDDC_N	84
TBT_B_R2D	TBTDp_85D	TBTDp	TBT_B_R2D_C_P<1..0>	7 35 85
TBT_B_R2D	TBTDp_85D	TBTDp	TBT_B_R2D_C_N<1..0>	7 35 85
	TBTDp_85D	TBTDp	TBT_B_R2D_P<1..0>	7 85
	TBTDp_85D	TBTDp	TBT_B_R2D_N<1..0>	7 85
DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3..1:2>	35 85
DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3..1:2>	35 85
	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3..1:2>	85
	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3..1:2>	85
	DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	85
	DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	85
	TBTDp_85D	TBTDp	TBT_B_D2R_C_P<1..0>	7 85
	TBTDp_85D	TBTDp	TBT_B_D2R_C_N<1..0>	7 85
TBT_B_D2R	TBTDp_85D	TBTDp	TBT_B_D2R_P<1..0>	7 35 85
TBT_B_D2R	TBTDp_85D	TBTDp	TBT_B_D2R_N<1..0>	7 35 85
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_P	35 85
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_C_N	35 85
	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_P	85
	DP_85D	DISPLAYPORT	DP_TBTPB_AUXCH_N	85
	DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_P	85
	DP_85D	DISPLAYPORT	DP_B_AUXCH_DDC_N	85
TBT_B_D2R	TBTDp_85D	TBTDp	TBT_B_D2R1_AUXDDC_P	85
	TBTDp_85D	TBTDp	TBT_B_D2R1_AUXDDC_N	85

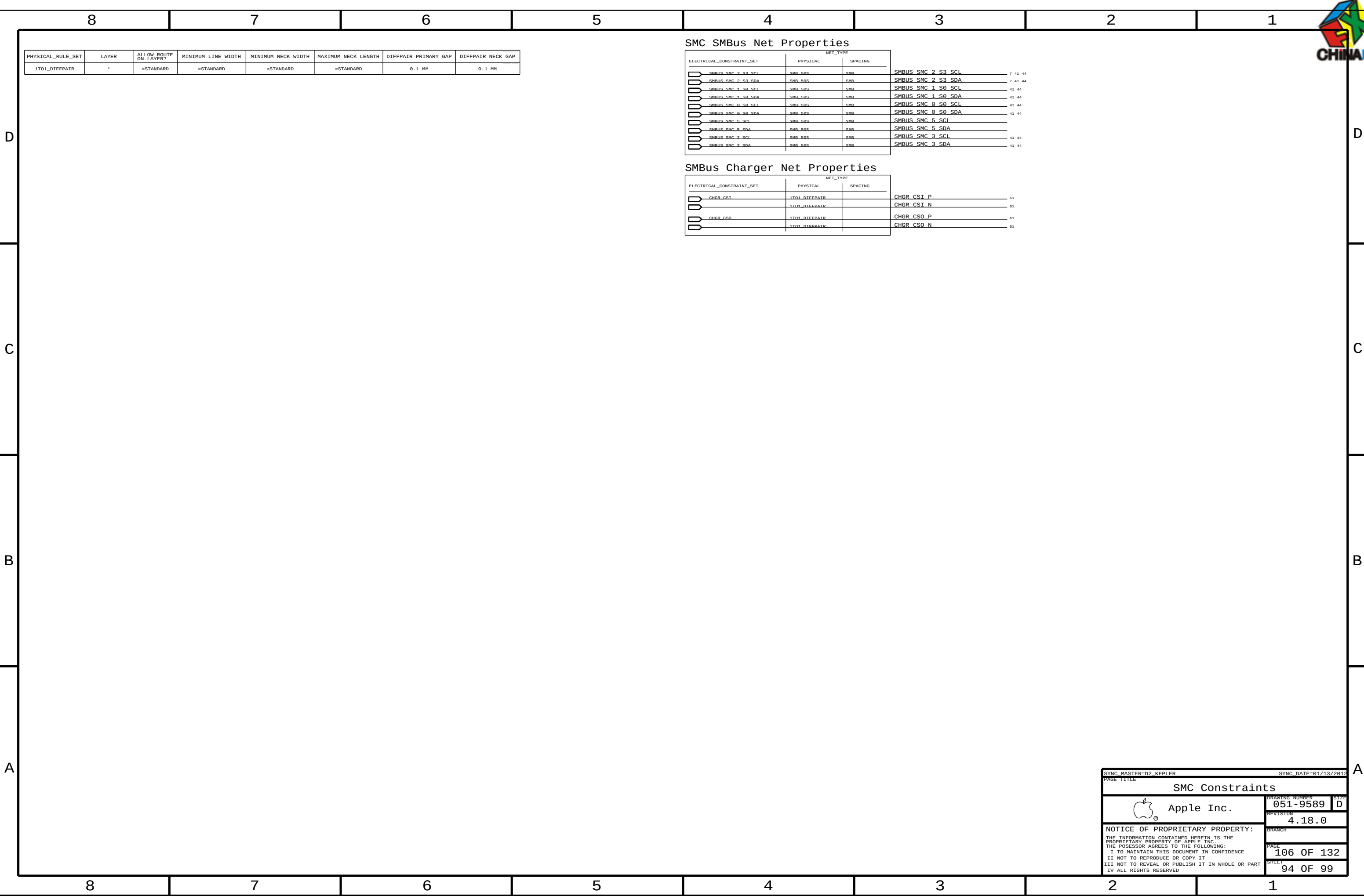
Only used on dual-port hosts.

Thunderbolt IC Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_P	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_N	
TBT_SPI_CLK	TBT_SPI_55S	TBT_SPI	TBT_SPI_CLK	35
TBT_SPI_MOSI	TBT_SPI_55S	TBT_SPI	TBT_SPI_MOSI	35
TBT_SPI_MISO	TBT_SPI_55S	TBT_SPI	TBT_SPI_MISO	35
TBT_SPI_CS_L	TBT_SPI_55S	TBT_SPI	TBT_SPI_CS_L	35

Only used on hosts supporting Thunderbolt video-in

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	4.18.0		
PAGE		105 OF 132	
SHEET		93 OF 99	



PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
1T01_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

NET_TYPE				
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
 SMBUS_SMC_2_S3_SCL	SMB_SBS	SMB	SMBUS_SMC_2_S3_SCL	7 41 44
 SMBUS_SMC_2_S3_SDA	SMB_SBS	SMB	SMBUS_SMC_2_S3_SDA	7 41 44
 SMBUS_SMC_1_S0_SCL	SMB_SBS	SMB	SMBUS_SMC_1_S0_SCL	41 44
 SMBUS_SMC_1_S0_SDA	SMB_SBS	SMB	SMBUS_SMC_1_S0_SDA	41 44
 SMBUS_SMC_0_S0_SCL	SMB_SBS	SMB	SMBUS_SMC_0_S0_SCL	41 44
 SMBUS_SMC_0_S0_SDA	SMB_SBS	SMB	SMBUS_SMC_0_S0_SDA	41 44
 SMBUS_SMC_5_SCL	SMB_SBS	SMB	SMBUS_SMC_5_SCL	
 SMBUS_SMC_5_SDA	SMB_SBS	SMB	SMBUS_SMC_5_SDA	
 SMBUS_SMC_3_SCL	SMB_SBS	SMB	SMBUS_SMC_3_SCL	41 44
 SMBUS_SMC_3_SDA	SMB_SBS	SMB	SMBUS_SMC_3_SDA	41 44

SMBus Charger Net Properties

NET_TYPE				
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
 CHGR_CST	1T01_DIEFPATR		CHGR_CST_P	61
 CHGR_CST	1T01_DIEFPATR		CHGR_CST_N	61
 CHGR_CSO	1T01_DIEFPATR		CHGR_CSO_P	61
 CHGR_CSO	1T01_DIEFPATR		CHGR_CSO_N	61

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PAGE_TITLE

SMC Constraints

 Apple Inc.

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PAGE

106 OF 132

SHEET

94 OF 99



GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD
GDDR5_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR5_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?	GDDR5_CLK	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR5_CMD	*	=3x_DIELECTRIC	?	GDDR5_CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?	GDDR5_DATA	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR5_EDC	*	=5x_DIELECTRIC	?	GDDR5_EDC	TOP,BOTTOM	=5x_DIELECTRIC	?

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
HDMI_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?	DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
HDMI	*	=3x_DIELECTRIC	?	HDMI	TOP,BOTTOM	=4x_DIELECTRIC	?

DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.

DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.

MAX LENGTH OF DISPLAYPORT/TMDS TRACES: 13 INCHES.

SOURCE: Calpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.

GDDR5 FB A Net Properties

ELECTRICAL_CONSTRAINT_SET	NET TYPE		
	PHYSICAL	SPACING	
FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB A0 CLK P 73 76
FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB A0 CLK N 73 76
FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB A1 CLK P 73 76
FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB A1 CLK N 73 76
FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB A0 A<8..0> 73 76
FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB A1 A<8..0> 73 76
FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB A0 ABI L 73 76
FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB A1 ABI L 73 76
FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB A0 RAS L 73 76
FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB A1 RAS L 73 76
FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB A0 CAS L 73 76
FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB A1 CAS L 73 76
FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB A0 WE L 73 76
FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB A1 WE L 73 76
FB_A0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB A0 CKE L 73 76
FB_A1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB A1 CKE L 73 76
FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB A0 CS L 73 76
FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB A1 CS L 73 76
FB_A0_EDC0	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<0> 73 76
FB_A0_EDC1	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<1> 73 76
FB_A0_EDC2	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<2> 73 76
FB_A0_EDC3	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<3> 73 76
FB_A1_EDC0	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<0> 73 76
FB_A1_EDC1	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<1> 73 76
FB_A1_EDC2	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<2> 73 76
FB_A1_EDC3	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<3> 73 76
FB_A0_DBI_I_0	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<0> 73 76
FB_A0_DBI_I_1	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<1> 73 76
FB_A0_DBI_I_2	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<2> 73 76
FB_A0_DBI_I_3	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<3> 73 76
FB_A1_DBI_I_0	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<0> 73 76
FB_A1_DBI_I_1	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<1> 73 76
FB_A1_DBI_I_2	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<2> 73 76
FB_A1_DBI_I_3	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<3> 73 76
FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB A0 WCLK P<0> 73 76
FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB A0 WCLK N<0> 73 76
FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB A0 WCLK P<1> 73 76
FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB A0 WCLK N<1> 73 76
FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB A1 WCLK P<0> 73 76
FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB A1 WCLK N<0> 73 76
FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB A1 WCLK P<1> 73 76
FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB A1 WCLK N<1> 73 76
FB_A0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<7..0> 73 76
FB_A0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<15..8> 73 76
FB_A0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<23..16> 73 76
FB_A0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<31..24> 73 76
FB_A1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<7..0> 73 76
FB_A1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<15..8> 73 76
FB_A1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<23..16> 73 76
FB_A1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<31..24> 73 76
FB_A0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB A0 RESET L 73 76
FB_A1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB A1 RESET L 73 76

GDDR5 FB B Net Properties

ELECTRICAL_CONSTRAINT_SET	NET TYPE		
	PHYSICAL	SPACING	
FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB B0 CLK P 73 76
FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB B0 CLK N 73 76
FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB B1 CLK P 73 76
FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB B1 CLK N 73 76
FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 A<8..0> 73 76
FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 A<8..0> 73 76
FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 ABI L 73 76
FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 ABI L 73 76
FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 RAS L 73 76
FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 RAS L 73 76
FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 CAS L 73 76
FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 CAS L 73 76
FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 WE L 73 76
FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 WE L 73 76
FB_B0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB B0 CKE L 73 76
FB_B1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB B1 CKE L 73 76
FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 CS L 73 76
FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 CS L 73 76
FB_B0_EDC0	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<0> 73 76
FB_B0_EDC1	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<1> 73 76
FB_B0_EDC2	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<2> 73 76
FB_B0_EDC3	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<3> 73 76
FB_B1_EDC0	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<0> 73 76
FB_B1_EDC1	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<1> 73 76
FB_B1_EDC2	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<2> 73 76
FB_B1_EDC3	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<3> 73 76
FB_B0_DBI_I_0	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<0> 73 76
FB_B0_DBI_I_1	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<1> 73 76
FB_B0_DBI_I_2	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<2> 73 76
FB_B0_DBI_I_3	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<3> 73 76
FB_B1_DBI_I_0	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<0> 73 76
FB_B1_DBI_I_1	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<1> 73 76
FB_B1_DBI_I_2	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<2> 73 76
FB_B1_DBI_I_3	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<3> 73 76
FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB B0 WCLK P<0> 73 76
FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB B0 WCLK N<0> 73 76
FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB B0 WCLK P<1> 73 76
FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB B0 WCLK N<1> 73 76
FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB B1 WCLK P<0> 73 76
FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB B1 WCLK N<0> 73 76
FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB B1 WCLK P<1> 73 76
FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB B1 WCLK N<1> 73 76
FB_B0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<7..0> 73 76
FB_B0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<15..8> 73 76
FB_B0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<23..16> 73 76
FB_B0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<31..24> 73 76
FB_B1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<7..0> 73 76
FB_B1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<15..8> 73 76
FB_B1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<23..16> 73 76
FB_B1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<31..24> 73 76
FB_B0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB B0 RESET L 73 76
FB_B1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB B1 RESET L 73 76

MUXGFX & DP AUX MUX NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET TYPE		
	PHYSICAL	SPACING	
DP_TINT_ML	DP_85D	DISPLAYPORT	DP INT ML C P<3..0> 81 82
DP_TINT_ML	DP_85D	DISPLAYPORT	DP INT ML C N<3..0> 81 82
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP INT AUX C P 81 82
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP INT AUX C N 81 82
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP INT AUX P 7 81
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP INT EG AUX P 77 82
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP INT EG AUX N 77 82
DP_TINT_ML	DP_85D	DISPLAYPORT	DP INT ML P<3..0> 7 81
DP_TINT_ML	DP_85D	DISPLAYPORT	DP INT ML N<3..0> 7 81
DP_TINT_ML	DP_85D	DISPLAYPORT	DP INT ML F P<3..0> 81
DP_TINT_ML	DP_85D	DISPLAYPORT	DP INT ML F N<3..0> 81
DP_TINT_ML	DP_85D	DISPLAYPORT	DP INT EG ML P<3..0> 77 82
DP_TINT_ML	DP_85D	DISPLAYPORT	DP INT EG ML N<3..0> 77 82
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DPA IG AUX CH P 18 83
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DPA IG AUX CH N 18 83
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DPB IG AUX CH P 18 83
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DPB IG AUX CH N 18 83
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK0 EG AUXCH P 77 83
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK0 EG AUXCH N 77 83
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK1 EG AUXCH P 77 83
DP_TINT_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK1 EG AUXCH N 77 83
TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK0 AUXCH C P 7 35 83
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK0 AUXCH C N 7 35 83
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK1 AUXCH C P 7 35 83
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK1 AUXCH C N 7 35 83
DP_TINT_ML	DP_85D	DISPLAYPORT	DP TBTSNK0 ML C P<3..0> 7 35 77
DP_TINT_ML	DP_85D	DISPLAYPORT	DP TBTSNK0 ML C N<3..0> 7 35 77
DP_TINT_ML	DP_85D	DISPLAYPORT	DP TBTSNK1 ML C P<3..0> 7 35 77
DP_TINT_ML	DP_85D	DISPLAYPORT	DP TBTSNK1 ML C N<3..0> 7 35 77
TBT_A_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK1 ML C P<3..0> 7 35
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK0 AUXCH P 7 35
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK0 AUXCH N 7 35
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK1 AUXCH P 7 35
TBT_B_AUXCH	DP_85D	DISPLAYPORT	DP TBTSNK1 AUXCH N 7 35
DP_TINT_ML	DP_85D	DISPLAYPORT	DP TBTSNK0 ML P<3..0> 7 35
DP_TINT_ML	DP_85D	DISPLAYPORT	DP TBTSNK0 ML N<3..0> 7 35
DP_TINT_ML	DP_85D	DISPLAYPORT	DP TBTSNK1 ML P<3..0> 7 35
DP_TINT_ML	DP_85D	DISPLAYPORT	DP TBTSNK1 ML N<3..0> 7 35

Kepler Net Properties

ELECTRICAL_CONSTRAINT_SET	NET TYPE		
	PHYSICAL	SPACING	
GPU_CLK27M	CLK_SLOW_55S	CLK_SLOW	GPU_OSC_27M_XTALIN 77 78
GPU_CLK27M	CLK_SLOW_55S	CLK_SLOW	GPU_OSC_27M_XTALOUT 77 78
GPU_CLK27M	CLK_SLOW_55S	CLK_SLOW	GPU_OSC_27M_XTAL_BUFFOUT 77 78
GPU_CLK27M	CLK_SLOW_55S	CLK_SLOW	GPU_OSC_27M_SSIN 77
1:1 DIFPAIR			PEX_TSTCLK_0_P 71 92
1:1 DIFPAIR			PEX_TSTCLK_0_N 71 92
HDMI_DATA	HDMI_90D	HDMI	HDMI EG DATA C P<2..0> 7 38 77
HDMI_90D	HDMI		HDMI EG DATA C N<2..0> 7 38 77
HDMI_CLK	HDMI_90D	HDMI	HDMI EG CLK C P 7 38 77
HDMI_90D	HDMI		HDMI EG CLK C N 7 38 77

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PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SENSE_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIO000FFF	*	=1:1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
THERM_55S_CPU1WPI5NS1	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	0.2 MM	0.2 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_CLK	*	GND_P2MM
GND	MEM_CMD	*	GND_P2MM
GND	MEM_CTRL	*	GND_P2MM
GND	MEM_*_DQ_BYTE*	*	GND_P2MM
GND	MEM_DQS	*	GND_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_40S_OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D_OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_37S_OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D_OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D_OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D_OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S_OVERRIDE	BOTTOM OVERRIDE	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE

Graphics ,SATA Constraint Relaxations

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
LVDS_85D	BGA	LVDS_85D
DP_85D	BGA	100_DIFF_BGA
SATA_80D	BGA	100_DIFF_BGA
CLK_PCIE_90D	BGA	100_DIFF_BGA

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

D2 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
SENSE_DIEFPATR	THERM_1T01_55S	THERM	CPU1THMSNS_D2_P
SENSE_DIEFPATR	THERM_1T01_55S	THERM	CPU1THMSNS_D2_N
SENSE_DIEFPATR	THERM_1T01_55S	THERM	DDR3THMSNS_D1_P
SENSE_DIEFPATR	THERM_1T01_55S	THERM	DDR3THMSNS_D1_N
SENSE_DIEFPATR	THERM_1T01_55S	THERM	GPU1THMSNS_D_P
SENSE_DIEFPATR	THERM_1T01_55S	THERM	GPU1THMSNS_D_N
SENSE_DIEFPATR	THERM_1T01_55S	THERM	GPU_TDIODE_P
SENSE_DIEFPATR	THERM_1T01_55S	THERM	GPU_TDIODE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	VCCSA50_CS_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	VCCSA50_CS_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	VCCSA1SNS_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	VCCSA1SNS_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_1V5_MEM_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_1V5_MEM_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPUVCC10S0_CS_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPUVCC10S0_CS_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPUVCC10SNS_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPUVCC10SNS_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	GPU1SENS_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	GPU1SENS_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_1V5_MEM_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_1V5_MEM_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_AIRPORT_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_AIRPORT_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_AIRPORT_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_AIRPORT_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	GPUFB_CS_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	GPUFB_CS_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_PP1V8_S0GPU_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_PP1V8_S0GPU_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_PP1V8_S0GPU_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_PP1V8_S0GPU_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_PP1V8_S0GPU_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_PP1V8_S0GPU_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	PIV85_GPU_CS_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	PIV85_GPU_CS_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_PP1V5_S0GPU_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_PP1V5_S0GPU_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPU1MVP_ISNS16_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPU1MVP_ISNS16_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPU1MVP_ISNS16_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPU1MVP_ISNS16_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_HS_OTHER_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_HS_OTHER_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_HS_GPU_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_HS_GPU_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_HS_COMPUTING_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_HS_COMPUTING_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPU1MVP_ISNS_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPU1MVP_ISNS_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ADC1_VSENSE_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ADC1_VSENSE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ADC2_VSENSE_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ADC2_VSENSE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ADC2_ISENSE_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ADC2_ISENSE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ADC2_ISENSE_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ADC2_ISENSE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	SPKR_R_RSENSE_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	SPKR_R_RSENSE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	SPKR_L_RSENSE_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	SPKR_L_RSENSE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_LO1_L_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_LO1_L_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_LO1_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_LO1_R_N
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SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_LO2_L_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_LO2_R_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_LO2_R_N
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SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_MIC_INL_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_SPKRAMP_LIN_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_SPKRAMP_LIN_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_SPKRAMP_RIN_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_SPKRAMP_RIN_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_SPKRAMP_LSUBIN_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_SPKRAMP_LSUBIN_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_SPKRAMP_RSUBIN_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	AUD_SPKRAMP_RSUBIN_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	LSPKR_INTIV_RSENSE_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	LSPKR_INTIV_RSENSE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	RSPKR_INTIV_RSENSE_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	RSPKR_INTIV_RSENSE_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	LSPKR_INTIV_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	LSPKR_INTIV_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	RSPKR_INTIV_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	RSPKR_INTIV_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_TBT_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_TBT_P
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_TBT_R_N
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_TBT_R_P

D2 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
PCIE_CLK100M_AP	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N
	1T01_DIEFPATR		CHGR_CS1_R_P
	1T01_DIEFPATR		CHGR_CS1_R_N
	1T01_DIEFPATR		CHGR_CS0_R_P
	1T01_DIEFPATR		CHGR_CS0_R_N
(USB_EXTN)	USB_85D	USB	USB2_EXTN_MIXED_P
(USB_EXTN)	USB_85D	USB	USB2_EXTN_MIXED_N
(USB_EXTN)	USB_85D	USB	USB2_LT1_P
(USB_EXTN)	USB_85D	USB	USB2_LT1_N
	USB_85D	USB	CONN_USB2_BT_P
	USB_85D	USB	CONN_USB2_BT_N
	USB_85D	USB	USB_LT2_P
	USB_85D	USB	USB_LT2_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	SPKRAMP_LIN_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	SPKRAMP_LIN_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	SPKRAMP_RIN_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	SPKRAMP_RIN_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	SSM2375SL_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	SSM2375SL_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	SSM2375SR_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	SSM2375SR_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_SL_OUT_P_R
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_SL_OUT_N_R
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_SL_OUT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_SL_OUT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	LSPKR_VSENSE_FILT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	LSPKR_VSENSE_FILT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	RSPKR_VSENSE_FILT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	RSPKR_VSENSE_FILT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_SR_OUT_P_R
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_SR_OUT_N_R
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_SR_OUT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_SR_OUT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_L_OUT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_L_OUT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_R_OUT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_R_OUT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_S_OUT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SPKRCONN_S_OUT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	LSPKR_ISENSE_FILT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	LSPKR_ISENSE_FILT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	RSPKR_ISENSE_FILT_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	RSPKR_ISENSE_FILT_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	RSUBIN_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	RSUBIN_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	LSUBIN_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	LSUBIN_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SSM4321SR_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SSM4321SR_N
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SSM4321SL_P
AUDIO_DIEFPATR	DIEFPATR	AUDIO	SSM4321SL_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	LSPKR_VSENSE_IN_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	LSPKR_VSENSE_IN_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	RSPKR_VSENSE_IN_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	RSPKR_VSENSE_IN_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	LSPKR_ISENSE_RDIVIDE_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	LSPKR_ISENSE_RDIVIDE_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	RSPKR_ISENSE_RDIVIDE_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	RSPKR_ISENSE_RDIVIDE_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	LSPKR_VSENSE_RDIVIDE_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	LSPKR_VSENSE_RDIVIDE_N
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	RSPKR_VSENSE_RDIVIDE_P
AUDIO_DIEFPATR	AUDIO000FFF	AUDIO	RSPKR_VSENSE_RDIVIDE_N
AUDIO_DIEFPATR	USB_85D	USB	USB_TPAD_R_P
AUDIO_DIEFPATR	USB_85D	USB	USB_TPAD_R_N
	SB_POWER	SB_POWER	PP3V3_5S
	SB_POWER	SB_POWER	PP3V3_SB
	SB_POWER	SB_POWER	PP1V5_S3RS0_CPU0DR
	GND	GND	GND

D

C

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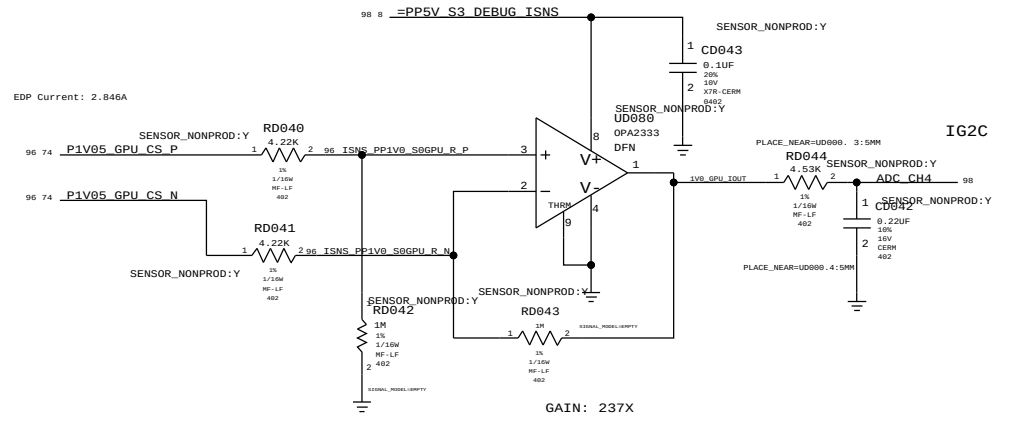
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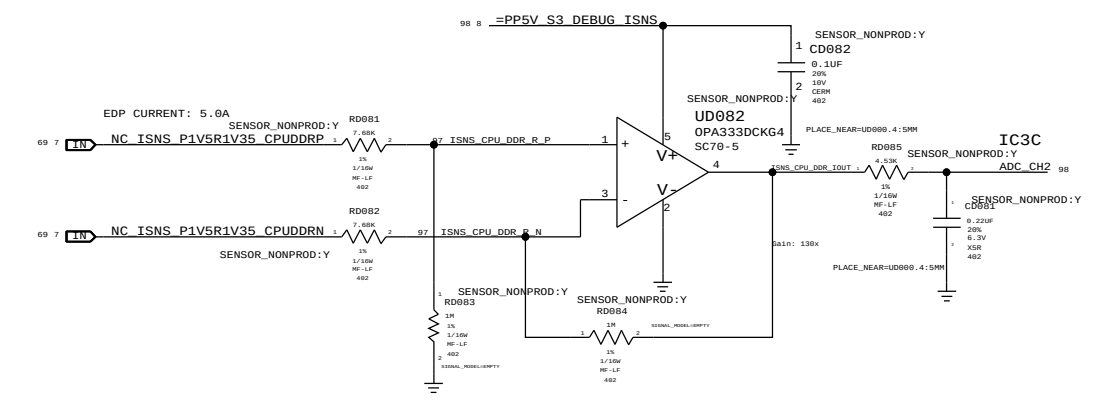
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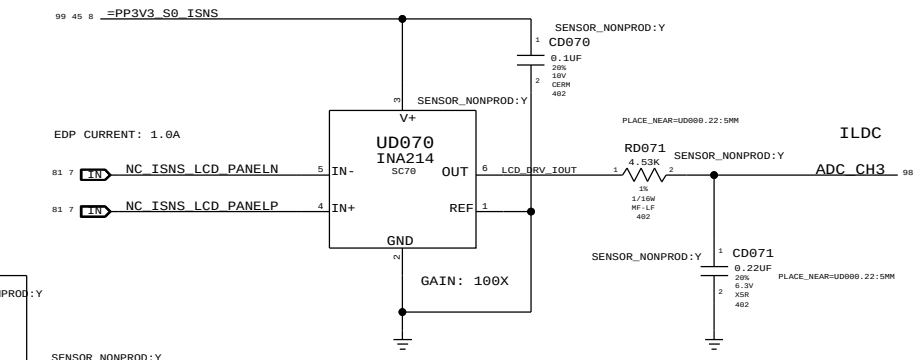
GPU 1.0V CURRENT SENSE



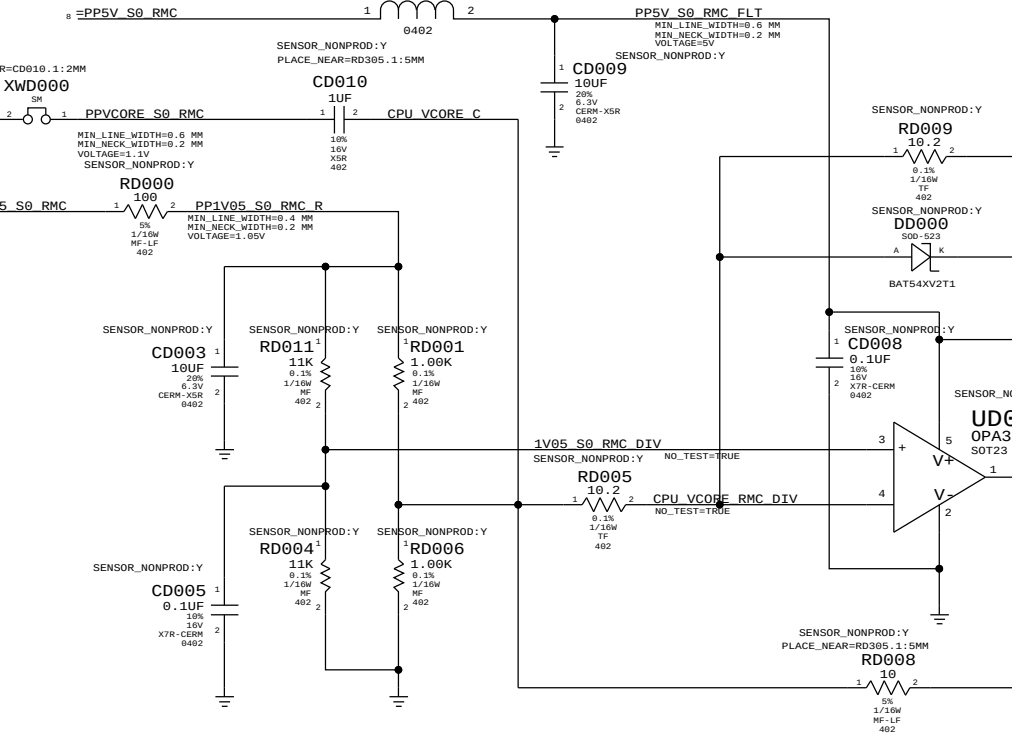
CPU DDR CURRENT SENSE



LCD PANEL CURRENT SENSE



LD000 1200HM-0.3A



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DEBUG SENSORS AND ADC		051-9589		D
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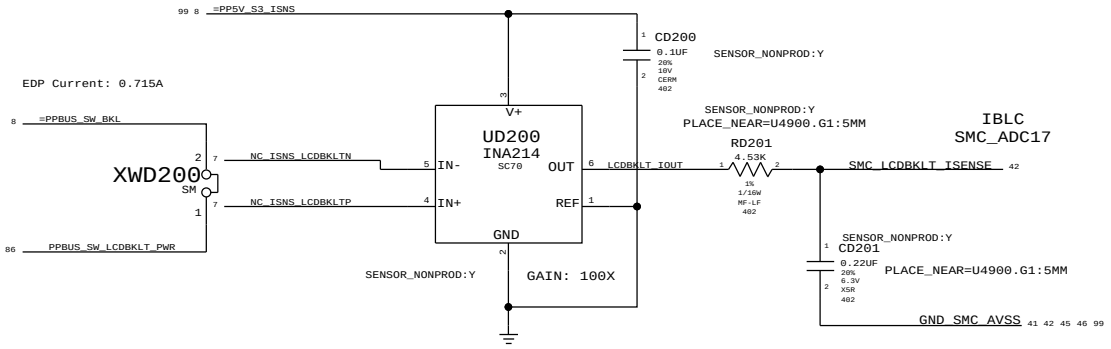
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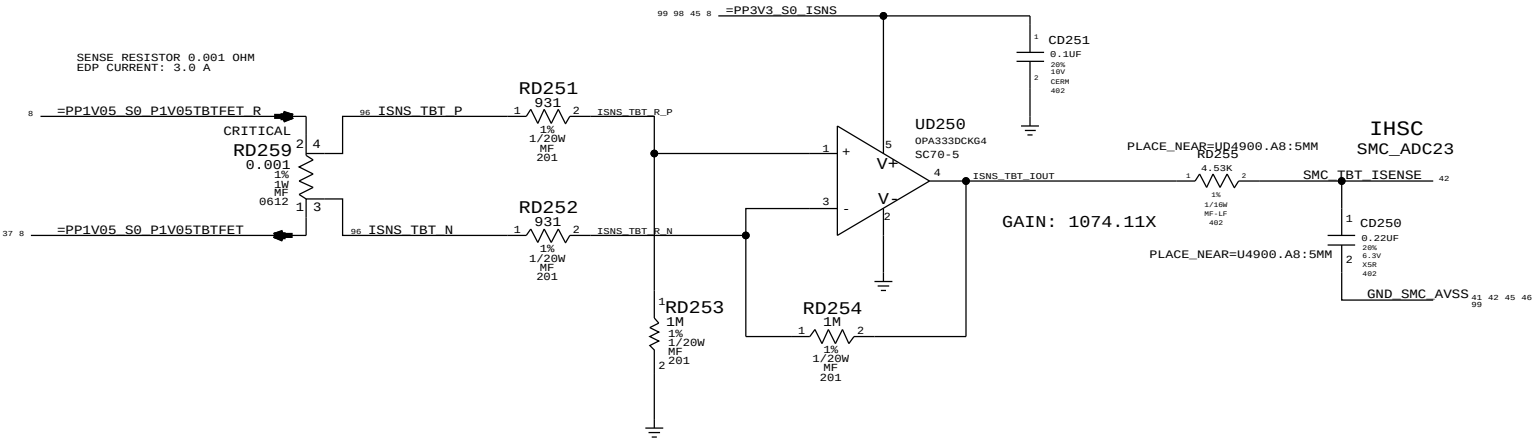
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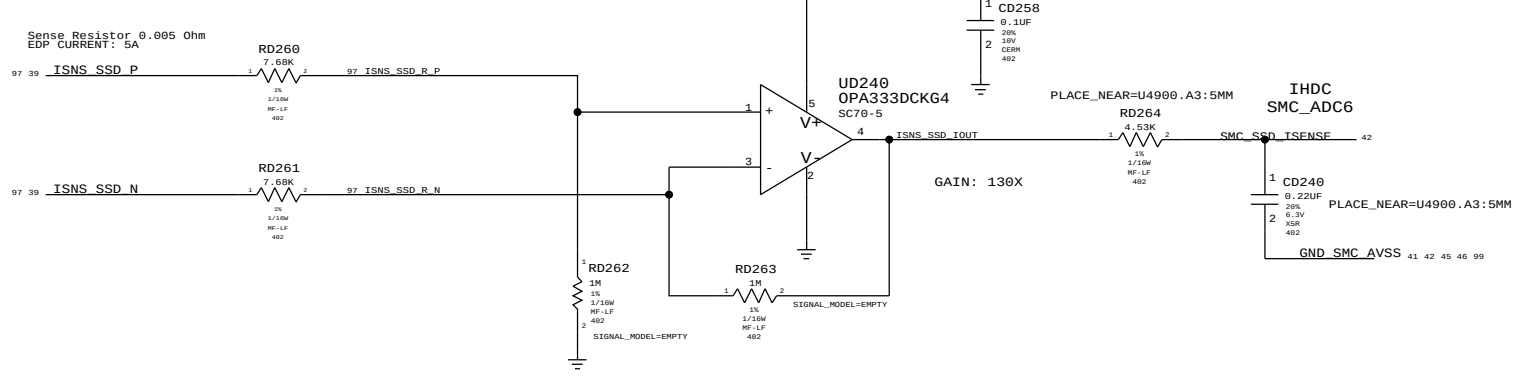
LCD BKLT Current Sense



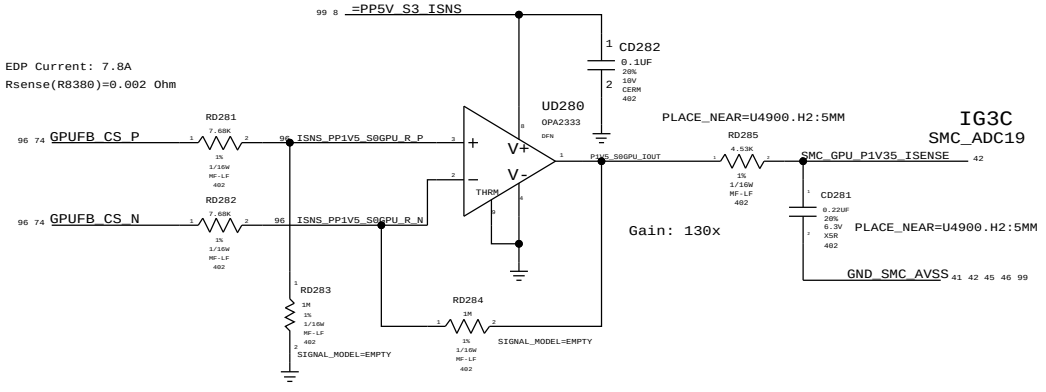
TBT (T29) CURRENT SENSE



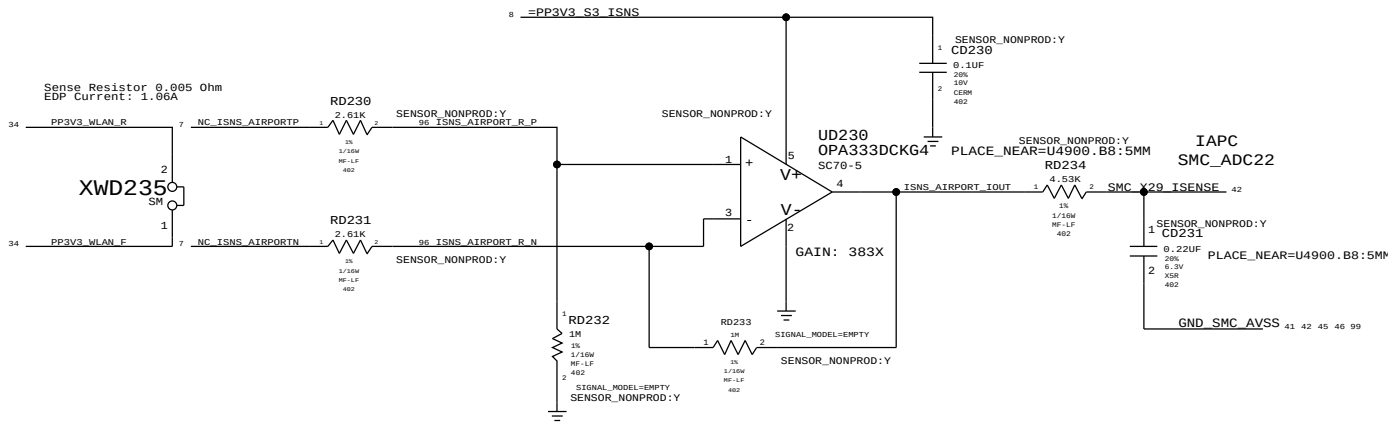
SSD CURRENT SENSE



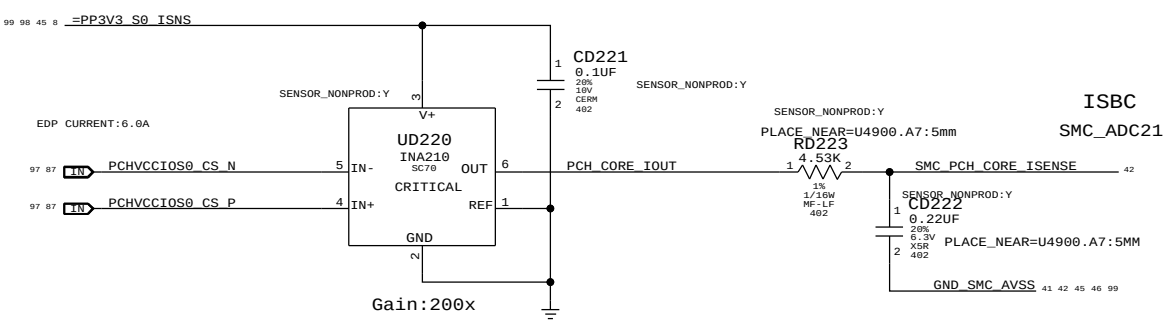
GPU FB (1.35V/1.5V) CURRENT SENSE



X29 AIRPORT CURRENT SENSE

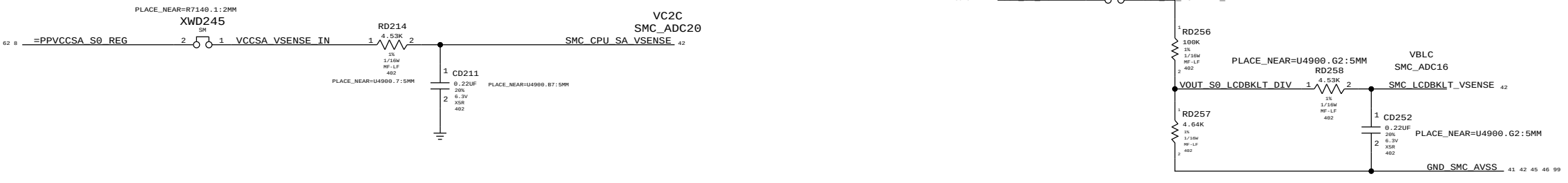


PCH VCORE CURRENT SENSE



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
11650114	4	RES, HTL, F12H, 100K, 5, 1/16W, 0402, SMD, LF	C0001, C0022, C0031		SENSOR_NONPROD: N

LCD BKLT Voltage Sense



SYNC MASTER=02 KEPLER

SYNC DATE=01/13/2015

PAGE TITLE

SMC12 SENSORS EXTENDED

Apple Inc.

DRAWING NUMBER
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REVISION
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PAGE
132 OF 132

SHEET
99 OF 99